Repeaterless Low-Swing On-Chip Interconnects

A thesis submitted in partial fulfillment of
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by

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To my parents and teachers ...

Thesis Approval

This thesis entitled

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Declaration

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Abstract

CMOS technology, with its continuous technology scaling, has been able to improve the performance of digital processing systems in every new generation. Performance of transistors improves as technology scales and this, coupled with smaller device dimensions and larger die dimensions, allows very sophisticated systems to be built. However, the performance of interconnects has not been scaling equally well. Long interconnects in particular have now become the performance bottlenecks in these systems. Repeater insertion is widely used to alleviate this problem, but it comes with high power consumption, routing complexity and increase in the active area of interconnects.

These limitations have stimulated extensive research in low swing repeaterless interconnects that use equalizers to achieve high performance with low power. The promise shown by these schemes has raised a lot of interest. However, it has also led to reservations about their adoption in mainstream digital system design because these interconnects use mixed signal and low swing circuits. The high latency of interconnects necessitates clock synchronization. Since the latency of long interconnects is high, and can vary with environmental changes, measuring this latency and its variation with time accurately is important for understanding and designing systems. The testability of these interconnect schemes in the established digital design flow has also raised concerns.

In this thesis, we have tried to address these concerns. The thesis reports circuits for performing equalization at the transmitter and receiver. We have designed a two tap capacitively coupled equalizer for transmitter end equalization. A method has been outlined for selecting the parameters for this transmitter using worst case sequences. We have also designed a low latency Decision Feedback Equalizer (DFE) circuit, which includes offset compensation, using the sense amplifier comparator. We then report a clock retiming circuit suitable for repeaterless low swing interconnects. The circuit uses a coarse+fine type controller for generating a retimed sampling clock. A Delay Locked Loop (DLL) is used to generate multiple phases of the clock

which are used for performing coarse phase correction and the fine correction is performed using a Voltage Controlled Delay Line (VCDL). Using the multiple phases of the DLL, a clock domain transfer circuit then transfers the sampled data to the receiver clock domain with a maximum latency of three clock cycles. This effectively makes the interconnect a transparent digital interconnect. We then demonstrate that with the addition of a little circuitry, these mixed signal systems can be tested along with the rest of the digital system using established digital test methods like scan chains and Built In Self Test (BIST) with digital I/O's.

Finally, we study the settling time of mesochronous clock retiming circuits in the presence of timing jitter. We show how timing jitter can result in large, and possibly indefinite, increase in the settling time. By modeling the system as a Markov chain, we analyze the effects of different types of jitter, which include data dependent jitter, random jitter and also simultaneous data dependent and random jitter. Using the insights provided by the model, different synchronizers with reduced settling time are then proposed and verified with circuit simulations.

Contents

List of Tables

List of Figures

Abbreviations

DFE Decision Feedback Equalizer.

DLL Delay Locked Loop.

VCDL Voltage Controlled Delay Line.

BIST Built In Self Test.

DVFS Dynamic Voltage and Frequency Scaling.

DAC Digital to Analog Converter.

BER Bit Error Rate.

ISI Inter Symbol Interference.

MTBF Mean Time Between Failures.

CAD Computer Aided Design.

PLL Phase Locked Loop.

FIR Finite Impulse Response.

FFE Feed Forward Equalizer.

PRBS Pseudo Random Binary Sequence Generator.

SPICE Simulation Program with Integrated Circuit Emphasis.

ADC Analog to Digital Converter.

CMRR Common Mode Rejection Ratio.

LPF Low Pass Filter.

FSM Finite State Machine.

UI Unit Interval.

RMS Root Mean Squared.

TDC Time to Digital Converter.

DTC Digital to Time Converter.

FPGA Field Programmable Gate Array.

OSR Over Sampling Ratio.

ENOB Effective Number Of Bits.

CDF Cumulative Distribution Function.

List of Symbols

$\mathcal{W}$ Window of susceptibility to extended settling time.

ϕ_{Tx} Phase of the transmitter clock.

ϕ_{Rx} Phase of the receiver clock.

ϕ_d Phase of the sampling clock.

SS Slow N and Slow P corner.

TT Typical process parameters.

FF Fast N and Fast P corner.

SNFP Slow N and Fast P corner.

FNSP Fast N and Slow P corner.

V_{th} Threshold voltage of the transistor.

$\Delta\phi$ Initial phase error between the data and the sampling clock.

α Data activity factor.

ϕ_d^0 Initial value of the sampling clock phase.

$T_{\mathcal{W}}$ Width of the window of susceptibility $\mathcal{W}$.

Chapter 1

Introduction

CMOS technology has been scaling down closely following Moore's prediction of exponential growth in packing density [1], made nearly half a century ago. This growth has been sustained because of continuous improvements and innovations in the fabrication process for CMOS devices. Continuous improvements in fabrication processes have also allowed systems of increasing sophistication to be built, by allowing for larger die sizes. However, interconnect performance has not shared the benefits of scaling. In fact, performance of interconnects has actually been worsening with the down-scaling of the feature size and the up-scaling of the die size [2]. Delay of interconnects is known to increase with the square of the distance [3]. Hence, this problem is exacerbated all the more for long interconnects which are used for communicating between different sub-modules in a chip. These are interconnects which run into millimeters in length. Repeater insertion is widely used to reduce the latency of long interconnects, which makes the latency of interconnects proportional to the distance [3, 4]. Minimum latency is achieved when the delay introduced by the repeater is equal to the interconnect segment wire delay. Since gate delays reduce with scaling down of technology, the number of repeaters required for a given interconnect length increases. This results in increased routing complexity, increase in the active area required for interconnects, and high power consumption. This has resulted in interconnects being the highest contributor to the total system power consumption [5]. While multi-core processors have allowed continued improvement in performance of systems without aggressive frequency scaling, it does not present a scalable solution. This is due to the well known Amdahl's law which states that eventually a single core's performance will limit the multi-core system's performance, due to limitations in parallelization of software programs [6].

The interconnect problem was predicted more than a decade ago [7, 8], and has driven efforts at technology, circuit and system levels towards finding solutions to alleviate this problem. At the technology level, cladded copper allowed integration of copper interconnects in standard CMOS [9]. This provided a quantum of improvement and is now widely deployed. Further technological advances like the use of optical interconnect, nanotubes etc. are however in their nascent stages. Among the circuit solutions, repeater insertion is widely deployed, but ideas like repeaterless interconnects with equalization at the transmitter and receiver are still in research. Preliminary investigations on repeaterless interconnects have shown promising results in terms of power as well as performance.

Repeaterless interconnects have been shown to outperform repeater inserted links in terms of throughput, power and routing complexity by big margins. However, their practical use has not seen the light of the day, due to some justifiable reservations expressed by designers. Repeaterless interconnects still have latencies that are quite high. Hence, appropriate synchronization circuits are required at the receivers. The receiver has to sample the low swing data at the right time, convert it to CMOS levels and then hand it over to the receiver's clock domain. Synchronizers for repeaterless low swing interconnects have not been well explored in the literature. Further, digital technologies have to keep pace with technology nodes which change pretty quickly. Whether repeaterless interconnects, which use mixed signal circuits, will be able keep up with the pace has long been questioned by system designers. Building standard cell digital libraries for new digital processes is not very difficult, but the same cannot be said about the design of repeaterless interconnects.

Another important aspect of high volume digital designs is their testability. All the reported repeaterless interconnect architectures use mixed signal circuits whose testability has not been sufficiently explored. A useful interconnect solution must be end-to-end, i.e. it must include all components needed to handover digital CMOS level data from transmitter to the receiver. It should have low power consumption and must be testable with a high degree of fault coverage. The search for such a solution forms the motivation of this work.

1.1 Motivation

It has been established that repeaterless interconnects can achieve high speed data transmission at significantly low power. Without repeaters, these techniques also simplify layout design and reduce required active silicon area substantially. With these substantial advantages, repeaterless interconnects have opened up new challenges in designing digital systems with these interconnect schemes. These challenges, which are faced in practical use of these interconnect schemes, are listed in the following and addressing these makes up the primary motivation of this work.

Synchronization: Repeaterless interconnects have unknown latencies which can be as high as multiple cycles. Due to the use of low swing and due to the presence of Inter Symbol Interference (ISI), the region of optimal sampling is narrow [10, 11]. Hence, an active clock synchronizing circuit is required in order to ensure the data is sampled at the correct time instant. Further, the synchronizing circuit must be automatic and be able to track changes in latencies due to temperature and voltage variations during operation.

Clock domain transfer: Once sampled with an appropriate sampling clock, data should be handed over to the receiver's clock domain with minimum latency while ensuring that no flip-flop becomes metastable. Serial synchronizers used for transferring data between mesochronous islands trade latency for Mean Time Between Failures (MTBF) and need additional handshaking signals [12].

Testability: High volume digital designs must be testable in order to screen out faulty chips early in the manufacturing line [13]. The cost of a design depends very strongly on the testability of the design. Testability of pure digital circuits has been extensively studied. Standard techniques have been developed to add test infrastructure to designs in order to make them testable and these have been integrated into Computer Aided Design (CAD) tools. Repeaterless interconnects use mixed signal circuits. Hence, in order to be a viable solution, these mixed signal circuits should not hamper the testability of the system. While testability of some mixed signal circuits like Phase Locked Loop (PLL)'s and Delay Locked Loop (DLL)'s have been investigated [14–18], testability of repeaterless interconnects has not been widely investigated.

Hence, while repeaterless interconnects have shown promising results, further work is required to make these schemes usable in practical digital designs. The eventual aim of this work is to design a repeaterless interconnect link that can replace a long repeater inserted link without having the system designer worry about the linear circuits involved for low swing transmission and synchronization.

1.2 Problem Definition

In order for a repeaterless interconnect scheme to be a convenient alternative to repeater inserted links, it must blend with the digital design flow in all aspects, while still offering significant improvements over repeater inserted links. Hence, the list of desirable features of a repeaterless interconnect scheme can be listed as follows:

The interconnect should

- Feature low power consumption, while allowing communication at high data rates.
- Contain minimal and simple analog components to allow easy scaling.
- Include synchronization and clock domain transfer with low latency, for end-to-end reliable communication.
- Feature testability with high fault coverage using standard digital test infrastructure.

The main aim of this work is to design an end-to-end repeaterless interconnect that meets all the above requirements.

1.3 Thesis Contribution

In this work, we have designed low swing interconnects with equalization at the transmitter and at the receiver. Clock synchronizing circuits, that provide a retimed sampling clock, to sample the low swing data at optimum time instant and convert to CMOS levels were designed. Clock domain transfer to the receiver clock domain is included in the core synchronizer circuit. Having designed the low swing interconnects with equalization and clock synchronization, we investigated the circuits for their testability using standard digital test methodologies, and have proposed appropriate amendments to the circuits in order to make them testable with high fault coverage. Further, the settling time of mesochronous clock retiming circuits was investigated and new clock synchronizers with reduced settling time were proposed.

The major contributions of this thesis can be summarized as follows

- A low swing interconnect link with a capacitively coupled feed-forward equalizer was designed [19]. A design method based on worst case sequences was formulated with which this circuit can be designed quickly. A comparator with reduced kickback noise was designed which was used as the receiver circuit. The circuit and the worst case sequence design method were tested for their effectiveness by designing interconnects of different geometries and in different technologies like UMC 130 nm CMOS technology, UMC 90 nm CMOS technology and in TSMC 65 nm CMOS technology.

- A Decision Feedback Equalizer (DFE) circuit was designed for receiver equalization. The circuit features reduced loop latency, which allows high speed operation. Further, the circuit allowed compensation of the offset of the comparator without any additional circuitry. The circuit was designed in TSMC 65 nm CMOS technology and UMC 130 nm CMOS technology. The circuit was also tested practically in UMC 130 nm CMOS technology.

- A mesochronous clock synchronizer circuit for low swing interconnects was designed to sample the incoming low swing data at optimal time [20, 21]. The circuit is adaptive with background tracking. The clock control feedback loop uses a coarse digital correction along with a fine analog correction for accurate clock positioning. A low latency clock domain transfer to the receiver clock domain is included in the design. The proposed retiming circuit was fabricated and tested in UMC 130 nm CMOS technology. Measuring clock skew between remote nodes, and its variation with time, is useful in optimizing the design of synchronizers. A technique of measuring skew between remote nodes using an all-digital sigma delta modulator has been proposed [22].

- It is shown that with the addition of a little extra circuitry, repeaterless interconnects with the proposed clock synchronizers can be tested using standard digital test methodologies like scan test and Built In Self Test (BIST) with digital test interfaces [23]. The fault coverage was quantified and found to be satisfactory. UMC 130 nm CMOS technology was used for the study.

- The dependence of settling time of mesochronous clock retiming circuits on the jitter in the incoming data was demonstrated with simulations as well as experimentally [24, 25]. It was shown that timing jitter in the incoming data can extend the settling time of repeaterless interconnects indefinitely. The system was modeled as a Markov chain and analyzed for different types of timing jitter. These included data dependent jitter (ISI induced), random noise as well as timing jitter induced simultaneously by ISI and random noise. The settling time was estimated by analyzing the model, and its predictions were verified with behavioural simulations.

- Using the insights provided by synchronizer circuit's model, three different fast settling synchronizers were designed and their settling time was quantified. The improvement in the settling time of these fast settling synchronizers was verified using circuit simulations.

1.4 Thesis Outline

Having introduced the topic of the thesis, making a case for this research and listing the thesis contributions, the rest of this report is organized as follows:

Chapter 2 presents a literature survey on design of long interconnects. Reported techniques of equalization at the transmitter and receiver have been described. Reported techniques of clock synchronization have also been presented. The chapter also describes reported works pertaining to the testability of mixed signal circuits.

Chapter 3 presents our work on equalization at the transmitter and at the receiver. A two tap capacitive feed-forward equalizer and a design optimization method for designing it is presented. Receiver equalization using a low latency DFE circuit, with integrated offset compensation is discussed.

Chapter 4 discusses a circuit for clock synchronization at the receiver. Clock domain transfer to the receiver clock domain is also presented. Its circuit implementation in UMC 130 nm CMOS technology is discussed.

Chapter 5 discusses the testability of repeaterless low swing interconnects. It is shown how, with a little additional circuitry, repeaterless interconnects can be tested with rest of digital systems using standard techniques that are used to test digital circuits. Fault coverage has been evaluated.

Chapter 6 discusses the settling time of mesochronous clock retiming circuits in the presence of timing jitter. Using Markov chain models, various types of jitter have been analyzed. The results have been confirmed with behavioural simulations. Techniques for reducing the settling time have been presented, and verified with circuit simulations.

Chapter 7 concludes this thesis and proposes future directions of research that stem from the work in this thesis.

Appendix A presents analysis and design of clocked sense amplifier comparators. The calculation of mean time to absorption and the variance of the time to absorption, for an absorbing Markov chain, is presented in Appendix B.

Chapter 2

Literature review

This chapter presents a literature review of the work on interconnects. As discussed in the previous chapter, in modern technologies, interconnect performance determines total system performance. With technology scaling, the delay of local interconnects, which also scale in length, remains unchanged to the first order. However, the delay of long interconnects increases with the square of distance. This has made long interconnects the performance bottlenecks in these systems.

Consider an interconnect segment as shown in Figure 2.1. The delay of the wire is propor-

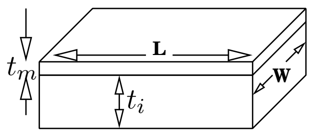


Figure 2.1: Illustration of a wire segment.

tional to its RC time constant. With a first order approximation, this RC product can be derived as

$$R = \rho \frac{L}{W t_m},$$
$$C = \epsilon \frac{LW}{t_i},$$
$$RC = \rho\epsilon \frac{L^2}{t_m t_i},$$

where ρ is the resistivity of the metal, t_i is the dielectric hieght, ϵ is the permittivity of the dielectric and L, W and t_m are the length, width and thickness of the interconnect. Hence, the delay of the wire can be written as AL^2, where A is a constant.

For short interconnects, the delay stays relatively unchanged with scaling, as t_m and t_i scale in the same way as L. For long interconnects, however, the delay increases with the square of the length. To alleviate this problem, repeater insertion is used [3]. This essentially breaks the long interconnect into a chain of short interconnects, whose delay scales with technology. Since we need the short interconnect's delay to shrink with scaling, it is easy to see that the that the number of repeaters needed scales up with technology.

If this long interconnect is buffered with periodic repeaters, its delay can be expressed as

$$\Delta = (n-1)\tau + nAL'^2,$$

where τ is the repeater's delay, n is the number of segments and $L' = \frac{L}{n}$ is the segment length. The optimum number of buffers can be found by differentiating this expression w.r.t n and finding its solution, as shown in the following.

$$\frac{d\Delta}{dn} = \tau - \frac{AL^2}{n^2} = 0,$$
$$\Rightarrow AL'^2 = \tau.$$

Thus, the segment length L' should be chosen such that its delay is equal to the repeater's delay. This makes the total buffered interconnect delay linear to the length. Since the delay of the repeater reduces as technology scales, the segment length reduces. This increases the number of repeaters required as technology scales.

While this first order analysis gives an idea of the trends, it is neither accurate nor sufficient for finding optimum repeater insertion. Finding the optimum number of repeaters for minimum delay, their placement and sizing is a difficult non-linear optimization problem [26]. This has led to the development of recursive algorithms to arrive at a solution [27] or use of heuristic algorithms [28]. However, the actual interconnect parasitics depend strongly on the actual implementation and can be found only by extracting the layout of the entire circuit. This increases the time taken for each design cycle iteration, increasing the design time significantly.

These difficulties have led to efforts in standardizing interconnects, and developing an 'interconnect aware' design methodology for VLSI chips. Prasad et al. in [29, 30] have attempted to create a standard library of interconnects. This is done by dividing the repeater insertion problem into three sub-problems, namely the pre-buffer stage, the middle stage and post-buffer stage. The pre-buffer and post-buffers are tapered buffers and the middle stage is an interconnect with periodic repeaters. This standardization reduces the number of iterations required for optimization, while ensuring minimum delay.

While buffer insertion reduces the delay of long interconnects, its design is non-trivial. Further, it also increases the power consumption of the interconnects. In modern designs the power dissipated by the interconnects, which includes the clock network, makes up the highest component of total system power consumption [5]. The problem of data communication over long interconnects has also led to exploration of alternative techniques that use low swing on the interconnect without repeaters. Use of low swing leaves headroom available for selectively boosting high frequency signal components, resulting in extended bandwidth. This allows high data throughput over a long repeaterless interconnect.

We shall now review reported works on repeaterless low swing interconnects. First, a brief description of early works on repeaterless interconnects is presented. This is followed by a more detailed review of low swing interconnects employing equalizers at the transmitter and at the receiver.

2.1 Current mode signaling

As the problems associated with the buffer insertion technique started to weigh down the benefits, a lot of research has been done on repeaterless links. Due to parasitic resistance and capacitance associated with the wires, interconnects act like low pass filters. In modern digital CMOS processes the interconnect resistance is typically more than 100 Ω/mm and the interconnect capacitance is about 200 fF/mm for the top metals which are typically used for such long interconnects [10, 19, 31]. Interconnect length of 10 mm is typically used as a benchmark for testing. If the interconnect is modeled as a first order RC low pass filter, this results in an RC time constant of 2 ns or a bandwidth of 80 MHz. Data transmission at giga bits per second over such interconnects, need to use signal processing techniques that can mitigate the effects of this low pass characteristic of the interconnect.

Figure 2.2 shows the frequency response of a 10 mm interconnect, which is modeled as a 20 section RC ladder (shown in Figure 2.3), with R=100 Ω/mm and C = 200 fF/mm. As seen from Figure 2.2, the 3 dB bandwidth of the repeaterless interconnect is about 195 MHz. The bandwidth of the repeaterless interconnect can be enhanced by terminating the line with a low resistance. Figure 2.4 shows the 3 dB bandwidth of the line as a function of terminating resistance. While terminating the interconnect in low impedance does enhance the bandwidth, it also increases the static power consumption in such links [32]. This means that the technique

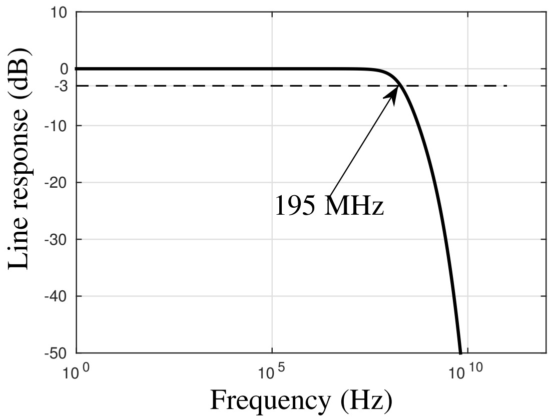


Figure 2.2: Frequency response of a 10 mm interconnect with R=100 Ω/mm and C = 200 fF/mm.

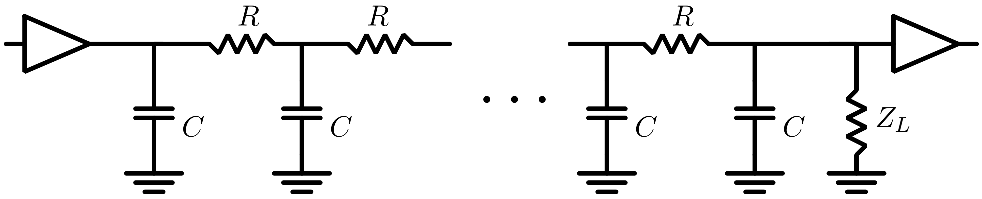


Figure 2.3: Multi section RC model of interconnect.

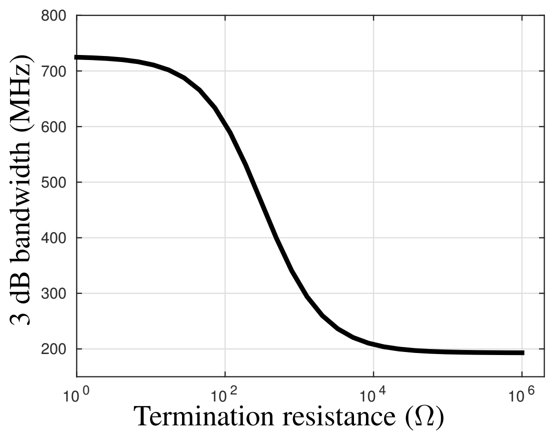


Figure 2.4: 3 dB cut off frequency of a 10 mm interconnect as a function of termination resistance.

is efficient only at high data rates with high data activity factors. Most modern systems use dynamic frequency scaling for optimizing power and performance. In such systems, the circuits are operated at low frequency when performance demands are low and at high frequencies whenever there is a demand for performance. Most systems have sporadic performance demands and hence, low power consumption at low frequencies of operation is important. Thus, enhancing the bandwidth of interconnects by terminating in low impedance at the expense of static power consumption is not a very attractive technique.

Early reported works on repeaterless interconnects used low impedance termination (dubbed as current mode signaling) for achieving high speed operation at low power [33–36]. However, due to high static power consumption, the performance advantage offered by them was not substantial enough to justify their choice over repeater insertion. Hence, further research was mostly focused on other implementations of repeaterless interconnects that used equalization circuits. Bandwidth of the interconnect can also be enhanced using equalizing filters. These techniques essentially use filters that compensate for the losses of the high frequency signal components by boosting them, thus extending the usable bandwidth. These filters can be implemented at the transmitter side or at the receiver side or some combination of both. When implemented at the transmitter side, these filters essentially pre-distort the signal before driving it over the interconnect. When these filters are implemented at the receiver, the data is already distorted by the interconnect and the filter must then recover the data signal by a using a filter that has a transfer function approximately equal to the inverse of the interconnect. The next section in this chapter discusses equalization at the transmitter and at the receiver.

2.2 Low swing interconnect: Equalization at the transmitter

Equalization at the transmitter boosts the strength of the high frequency components of the signal to be transmitted, before launching it onto the interconnect. This essentially pre-distorts the signal, which up on experiencing degradation in the channel, appears at the receiver input with desired noise margin. This can be implemented by compensating for previously transmitted bits (static) or based on the previous data transitions (dynamic). Both these techniques and their implementations are reviewed in this section.

2.2.1 Static equalization

Since the interconnect acts as a low pass filter, equalization essentially implements a high pass function. At the transmitter, this is implemented as a Finite Impulse Response (FIR) filter. Figure 2.5 shows a transmitter with a 1-tap feed-forward equalizer, implemented using charge steering current mode transmitter. It is easy to see that this effectively makes up a first order discrete time high pass filter. The transfer function of the transmitter can be written as

$$I_{line} = b_0 I_0 - b_{-1} I_\alpha,$$

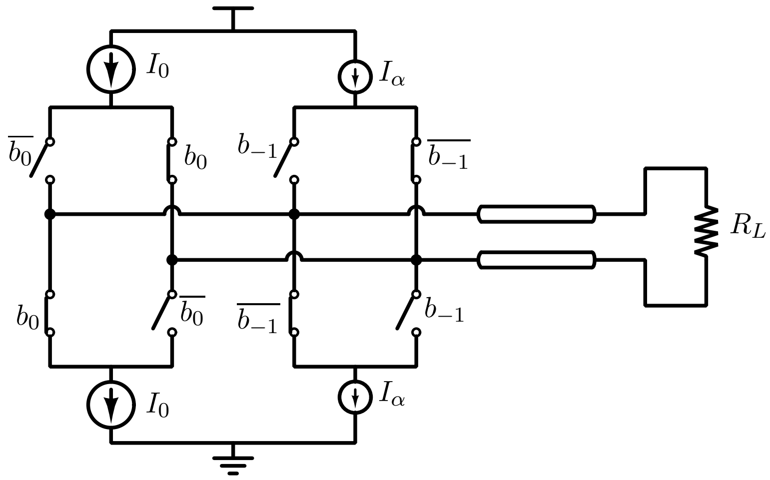


Figure 2.5: A 1 tap FFE transmitter implemented using charge steering current mode transmitter.

which is a first order difference equation. The $\mathcal{Z}$ transform of which can be written as

$$1 - \alpha z^{-1},$$

where the factor $\alpha = I_\alpha / I_0$ is chosen depending on the interconnect bandwidth.

One of the disadvantages of this implementation is that the subtraction is performed in the current domain. Hence, there are sequences which result in current drawn from the supply only to be drained to ground in the process of subtraction. Lee et al. in [37] use a current mode transmitter which sinks a constant current from a differential interconnect. In this architecture, the bias current for the receiver flows through the interconnect from the receiver to the transmitter. The sunk current is modulated according to the data, thus establishing the data link. This architecture is then extended to include a 1 tap Feed Forward Equalizer (FFE), which does not waste power during subtraction. Figure 2.6 shows the circuit diagram of this link. The total current sunk by the transmitter is constant and independent of data. However, while such implementation is very power efficient at high data rates, it has high constant static power consumption. The power consumption of this technique does not scale with frequency. This presents some limitations for using it in systems employing Dynamic Voltage and Frequency Scaling (DVFS) for servicing sporadic loads.

Kim et al. in [11, 38] report a 2 bit feed-forward equalizer using a current mode transmitter, which uses charge injection architecture. The charge injection architecture avoids analog current subtraction required for equalization. In this architecture, a decoder is used to compute the charge to be injected into the line for each combination of the present and previous bits. This decoded value then turns on a current source of corresponding value. This ensures that

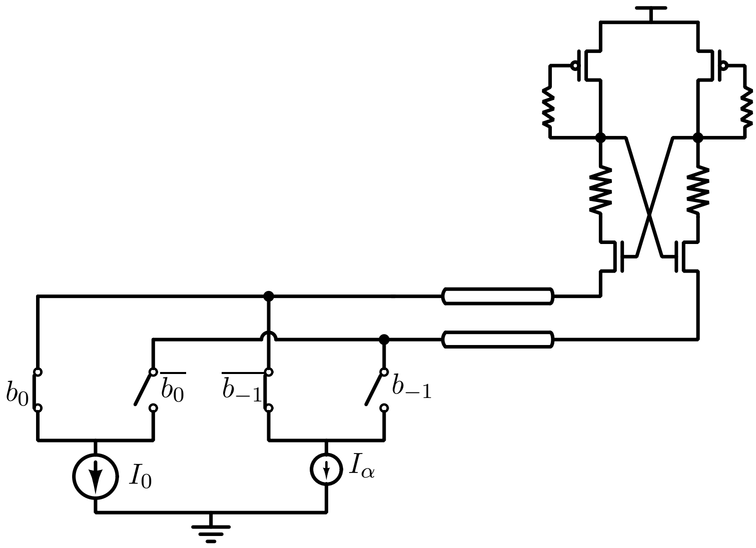


Figure 2.6: Current mode interconnect proposed in [37].

the current drawn from the supply is just as much as has to be injected into the line, resulting in low power consumption. This technique is however sensitive to the non-linearities in the current sources and the authors of [11, 38] use a pre-distortion DAC which compensates for the current source non-linearities. We shall later show in Chapter 3, that capacitive dynamic equalization, which also does not waste power during subtraction, achieves similar levels of equalization without needing decoders, resulting in a significantly simpler circuit that has better power efficiency.

2.2.2 Dynamic equalization: Current mode

Static equalization implemented the high pass FIR filter in discrete time (discretized to the clock period). High pass filters can also be implemented using transition detectors. Dynamic overdriving injects a large current into the line for short duration on the events of data transitions and a small current during the rest of the bit period. The large current injected on transitions enables the line capacitance to be charged quickly. The small current in the rest of the bit period is used to hold the logic level on the line. It may be noted that this allows the circuit to be used at arbitrarily low frequencies, contrary to the circuits commonly termed dynamic. The two driver circuits that implement the pulsed large current source (sink) and static weak current source (sink) are generally called the strong and the weak driver respectively. Since the data transitions contribute to the high frequency content on the line, this effectively boosts the high frequency content, thus performing equalization. The implementation of such transmitters can be done in feed-forward [39–41] or feed-back [42] fashion. The ratio of the strengths of the strong and weak drivers determines the amount of equalization performed, and is hence chosen depending on the interconnect bandwidth.

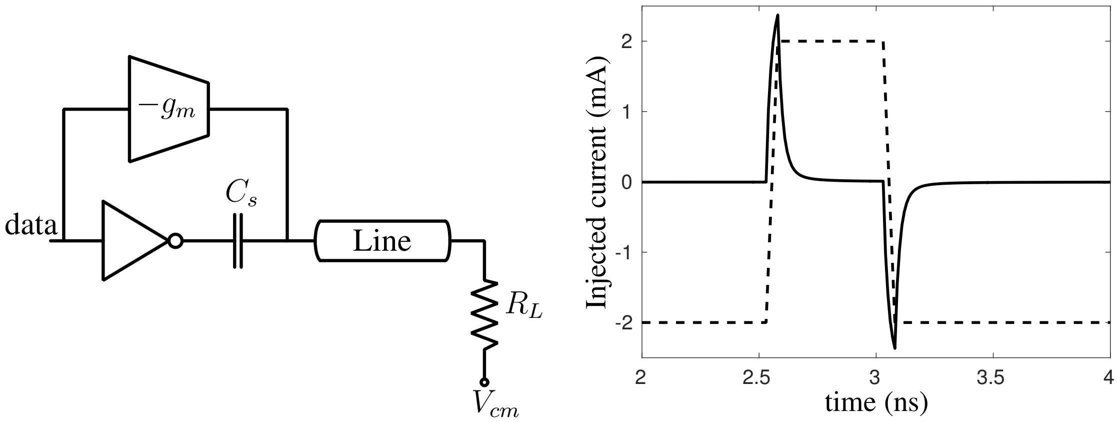
Kwon et al. in [39] report an implementation of a dynamic overdriving transmitter, which uses an inverter that is enabled for short duration during data transitions as the strong driver. The weak driver is implemented as a complementary source follower buffer. The swing on the line is hence reduced by the threshold voltage of the N type transistor for logic ‘0’ and by that of the P type transistor for logic ‘1’. The duration for which the strong driver is enabled is used as a design parameter for various lengths. The technique reports 40% improvement in power consumption. Katoch et al. in [42] report an implementation of the transmitter that uses a feedback to switch off the strong driver after the line voltage exceeds the trip point of an inverter. The delay of the feedback loop controls the amount of pre-emphasis as for that duration the strong driver pulls the line above the trip point of the inverter.

Dave et al. in [40] showed that the feedback technique has stability issues under certain conditions, which could occur due to process variations. Even the deviation in the relative strengths of the strong and weak drivers from designed values, due to process variations, can significantly affect the performance of this link [43]. Hence, a more robust implementation that is tolerant to process variations is required for the implementation of this concept. Dave et al. in [40] reported a variation tolerant current mode transmitter with dynamic overdriving, which used process invariant current references to generate the required bias currents for the transmitter.

Due to such stringent requirements on robustness against process variations, the reported implementations of current mode dynamic overdriving have been limited to equalizing for only the immediate previous data transition. Thus, it is very difficult to scale this technique for implementing higher order equalization, for higher data rates or for lines with large time constants.

2.2.3 Dynamic equalization: Capacitively coupled

A series capacitor can also serve as a transition detector and high pass filter, replacing the strong driver in the current mode dynamic equalization type transmitter. In the capacitively coupled transmitter, the circuit employs a tapered buffer, driving the line through a series capacitor. The weak driver in shunt with the capacitor, is used to maintain the logic level on the line, thus allowing arbitrarily low data activity factors. Figure 2.7(a) shows the block diagram of a capacitively coupled transmitter circuit. For brevity, single ended version is shown in Figure 2.7(a), simulations are done for a differential interconnect. The waveshape of the current injected into the line, when a data transition occurs, is depicted in Figure 2.7(b). Some works have avoided using



(a) Transmitter circuit

(b) Injected current

Figure 2.7: (a) Capacitively coupled line with weak direct coupled driver and (b) Waveshape of the current injected into the line.

the weak driver and instead use some type of encoding, like 8b10b [44], to ensure a minimum data activity factor [45, 46]. However, due to the latency and power consumption overhead of the encoders and decoders, the increased power consumption due to higher data activity factor and reduced effective data rate, this technique is inferior to using a weak driver.

The technique was first reported by Mensink et al. in [31, 47]. The authors used a first order RC model of the interconnect for the analysis. Using a first order model the optimum value of the series capacitor was evaluated. However, the first order approximation is too coarse and can result in a sub-optimal design which not only consumes higher power but also fails to meet the noise margin at the receiver for certain bit sequences [43]. As discussed in Section 2.1, repeaterless interconnects implemented with a very low termination impedance consume high static power. Hence, keeping the termination impedance in the order of a few kΩ's and using equalization for bandwidth enhancement results in better implementations. Designing and optimizing such interconnects can however be difficult. This is because for an interconnect driven with a sub-optimal transmitter, the time taken to reach steady state can easily run into 50 - 60 bit periods. This means that prohibitively long Pseudo Random Binary Sequence Generator (PRBS) sequences will have to be used to verify these links. Dave M. in [43] have proposed an alternate technique of optimizing such transmitters, by inspecting the interconnect's response to carefully identified worst case sequences that have very low activity factor. We shall review this design method in detail using the capacitively coupled transmitter as a design example. This technique can also be used to design current mode dynamic equalizers.

Worst case sequence based design method

When the value of the capacitor is less than the optimum, the bandwidth enhancement achieved is correspondingly less. On the other hand, higher than optimum values of the series capacitor result in overshoots on the line, which can degrade the noise margin for subsequent data bits. The design method of finding the optimum capacitor using worst case sequences, uses SPICE simulations with a more accurate model of the interconnect (20 section RLC). When the data contains a large number of 1's (0's), the voltage on the line settles to the low frequency logic level. A single 0 (1) at this time, will need maximum support from the series capacitor in order to be established at the receiver input with sufficient noise margin. Insufficient drive from the capacitor will result in this single 0 (1) to be missed. We designate this sequence, viz. "...1111'0'111..." as WC_1 and the single '0' as WB_1. If the drive from the series capacitor is too strong, then it results in overshoots at the receiver. A complementary bit appearing when the overshoot is maximum, ends up having the least noise margin. Hence, we identify this second sequence, viz. "...1111000'1'0000..." as WC_2, and the single '1' as WB_2 . By inspecting the noise margin at the receiver for these two sequences, and tuning the capacitor to equalize them, we arrive at the optimum value of the series capacitor.

Figure 2.8 shows the response of the interconnect to both the worst case sequences, emphasizing the bits WB_1 and WB_2, when the capacitor boost is more than optimum. It can be seen that the response to WC_1 is very good, but WC_2 has very poor noise margin. Figure 2.9 shows the response of the same transmitter on a longer time scale, in order to emphasize the long time taken to reach steady state in this condition.

On the other hand, when the series capacitor is less than optimum, the response to $WC1$ is poor, but $WC2$ shows a good response as shown in Figure 2.10. By tuning the capacitance to a value that makes the noise margin's of the receiver input to WC_1 and WC_2 equal, we can find the optimum as shown in Figure 2.11. For the optimum value of the series capacitor, the time taken to reach steady state is about 2 - 3 bit periods as seen in Figure 2.11.

Since the worst case sequence based design method uses an accurate model of the interconnect and uses simulations with low activity factor signals, it is a much easier and more

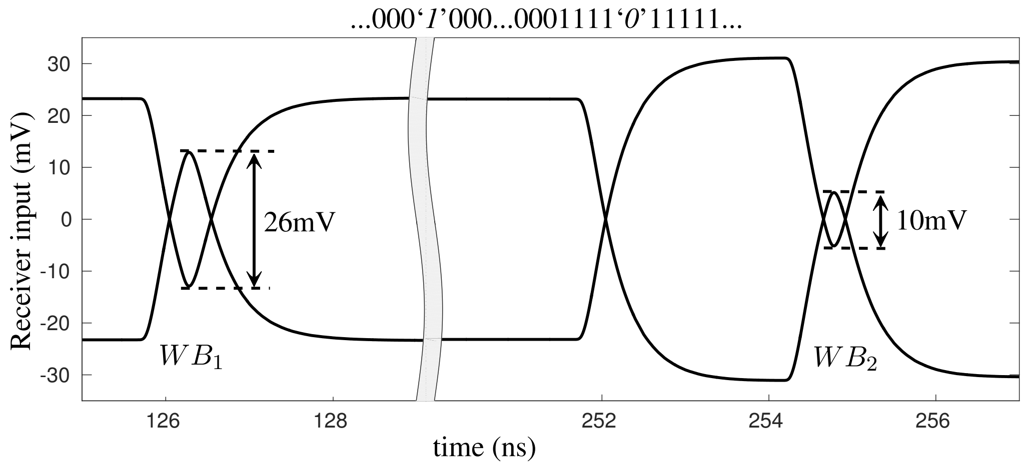


Figure 2.8: Interconnect response to WC_1 and WC_2 for more than optimum pre-emphasis: WB_2 has minimum eye opening at Receiver input.

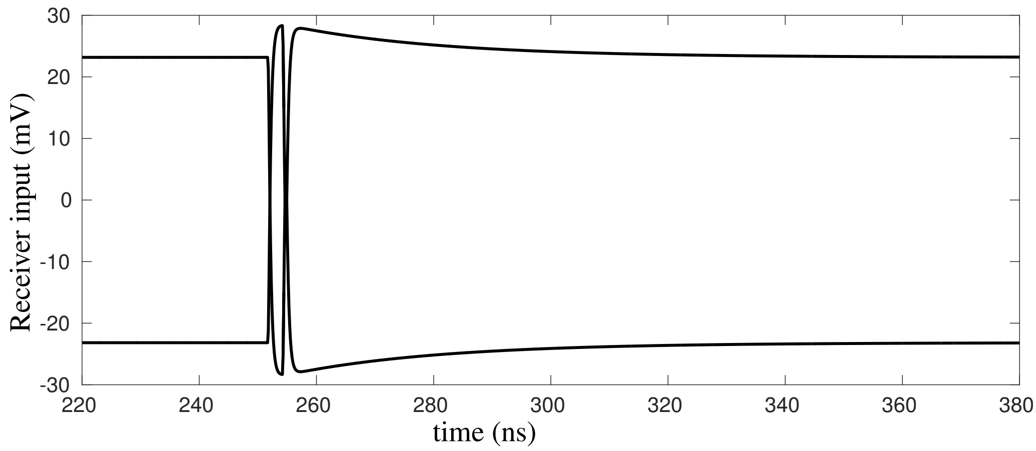


Figure 2.9: Interconnect response to WC_1 and WC_2 for more than optimum pre-emphasis, indicating the long time taken to reach steady state.

precise way to design such transmitters.

The capacitively coupled dynamic equalizer is quite robust against process variations [19, 48]. Due to its simplicity and robustness to variations, the capacitive dynamic equalization can be easily scaled to compensate for more of the previous data transitions, as has been reported for previous two data transitions in [19] and previous three transitions in [49].

2.3 Equalization at the receiver

The receiver circuit for low swing interconnects is typically a regenerative sense amplifier based comparator [10, 11, 31]. Equalization for extending interconnect bandwidth can also be done at

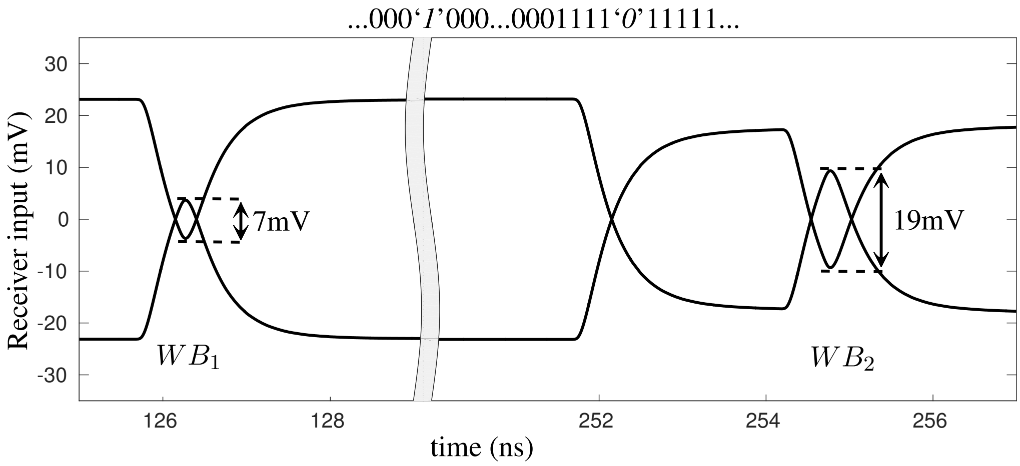


Figure 2.10: Interconnect response to WC_1 and WC_2 for less than optimum pre-emphasis: WB_1 has minimum eye opening at Receiver input.

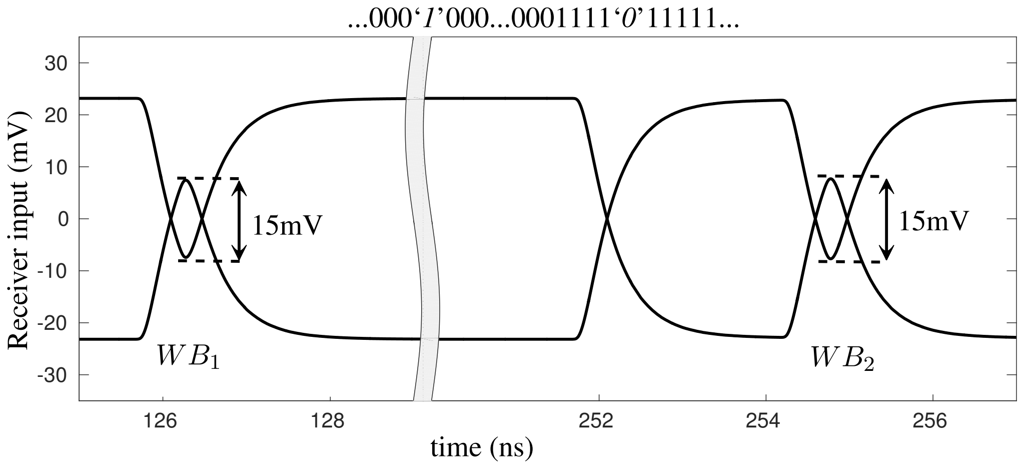


Figure 2.11: Interconnect response to WC_1 and WC_2 for optimum pre-emphasis: WB_1 and WB_2 have equal eye opening.

the receiver. This can be implemented in many ways which can be broadly classified as

- Inductive peaking type [10, 50].
- Feed-forward Equalization (FFE) based [51].
- Decision Feedback Equalization (DFE) based [11, 31],

Inductive peaking is a well known technique of extending the bandwidth of amplifiers [52]. The same principle can be used for extending the bandwidth of low swing interconnects, by terminating with an inductive input impedance. Typically, large values of inductance are required for extending the bandwidth, which necessitates the use of active inductors [50]. Lee et al. in [10] report an interconnect link with equalization at the receiver using active inductors. The bias current for the active inductor flows through the interconnect and is sunk at the transmitter as shown in Figure 2.6. However, despite the use of active inductors for achieving the large values of required inductance, the termination resistance needs to be kept very low for effective bandwidth enhancement [10, 50]. This increases the static power consumption of the link.

Feed-forward equalization can be done at the receiver using an FIR filter. Implementing FIR filters at the receiver needs delay circuits that faithfully generate delayed copies of the input signal, which are then weighted and summed, thus forming the filter. These delay circuits must have good linearity and wide bandwidth. Nazari et al. in [51] use sample and hold circuits as delay cells to implement a feed-forward equalizer. The authors use a capacitively coupled transmitter with the line common mode at V_{DD} and PMOS sampling switches at the receiver. The high common mode results in high linearity of the PMOS switches. However, since the line swings above and below V_{DD}, this technique can pose long term reliability problems.

DFE at the receiver is quite powerful as it allows non-linear transfer functions, which are not possible with the inductive peaking or feed-forward equalizers. In this technique, a hard decision is made on the input at every sampling instant. Scaled versions of the previous decisions are subtracted from the input before the next decision is made, thus nullifying the memory of the previous bits. Figure 2.12 shows a block diagram of a 1 bit DFE circuit.

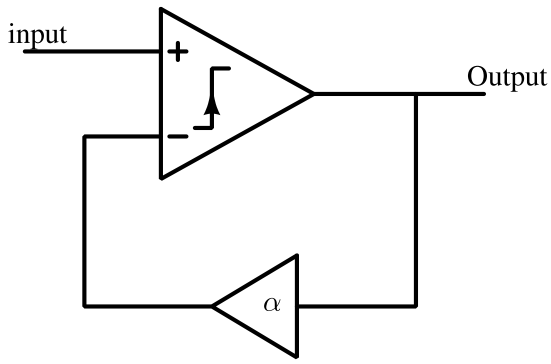


Figure 2.12: Block diagram of a 1-bit DFE circuit.

The order of the equalizer can be increased by subtracting scaled versions of more number of previous bits, at the cost of circuit complexity. Mensink et al. in [31] use an analog filter in the feedback path of the DFE, that emulates the transfer function of the line. This allows to correct for the memory of multiple previous bits without multiple feedback taps, with very low power and active area. The main limitation of this technique however, is that it increases the loop delay of the DFE circuit. This limits the scalability of the architecture.

The loop-delay of the feedback loop of DFE limits the maximum frequency of operation of the circuit. This can be tackled by 'unrolling the loop' [53]. This is also known as 'look ahead DFE'. In this technique, multiple comparators make decisions on the input data, each assuming a possible value of the previous decision. For example, for a 1-bit DFE, two comparators make decisions on the data, as shown in Figure 2.13. One of the two assumes the previous decision

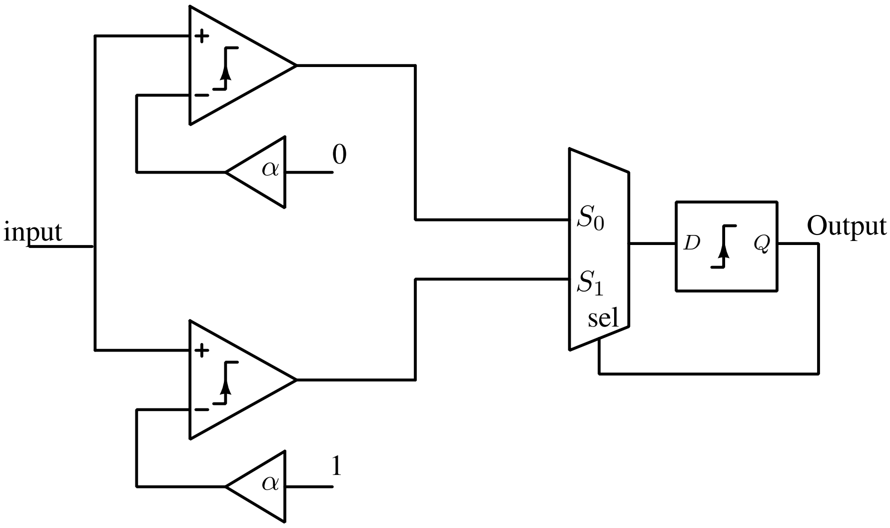


Figure 2.13: Block diagram of a 1-bit Loop unrolled DFE circuit.

was a zero, and the other assumes the previous decision was a one. Depending on the actual previous decision, one of the two outputs is selected. The decision device has to convert a low swing input to CMOS levels, while the selector device picks between two CMOS level data inputs. Hence, the latency of the selector is much lower than that of the decision device and the overall speed of the circuit can be enhanced.

The drawback of loop unrolled DFE is the fact that the number of comparators required increases. In scaled technologies, comparators need to have offset compensation mechanisms, to compensate for the random offset that arises from process induced mismatch in the transistors [10, 11]. This makes the implementation of loop unrolled DFE more complex with more number of comparators, each needing its own offset compensation circuits.

2.4 Synchronization

The receivers of low swing interconnects must convert the low swing input data to rail to rail levels. Clocked comparators are used for this because of they can achieve high speed operation at low power as opposed to static comparators. The latency of long interconnects can exceed multiple cycles. ISI and jitter result in a narrow region of optimum sampling instant for minimum Bit Error Rate (BER). Figure 2.14 shows the block diagram of a typical low swing interconnect system. Here, ϕ_{Tx} is the transmitter clock, ϕ_{Rx} is the receiver clock, and ϕ_d is

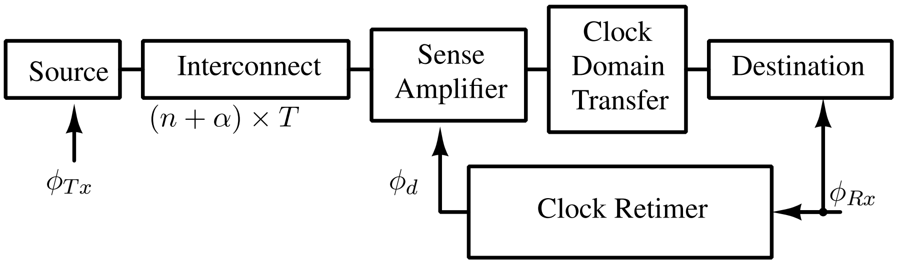


Figure 2.14: Block diagram of a repeaterless low swing interconnect system. ϕ_{Tx}: Transmitter clock, ϕ_{Rx}: Receiver clock, ϕ_d: retimed sampling clock, $((n+\alpha) \times T)$: Repeaterless interconnect delay, where $n \in \mathbb{Z}^{\geq 0}$, $\alpha \in \mathbb{R}$ & $\alpha < 1$, T: system clock period.

the desired sampling clock with its active edge at the center of the data eye. The delay of the repeaterless interconnect, that must be compensated for, is expressed as $(n + \alpha) \times T$, where T is the system clock period, n is a positive integer (including 0), and α is a positive real number which is less than 1.

In long interconnects with repeaters, repeaters are replaced with synchronizing flip-flops after section lengths of delay less than the critical path delay of the system [54]. This ensures that the long interconnect, while having multi-cycle latency, is still synchronous with the full system. However, insertion of synchronizing repeaters along a low swing interconnect will mean that the benefits of a low swing interconnect will not be leveraged to full potential. Mensink et al. in [31] estimate the delay using layout extracted simulations of the interconnect and add appropriate delay at design time itself. This needs sufficient over design so as to ensure proper operation at the desired frequency across process corners. Source synchronous schemes, akin to the one described in [55], are not compatible with low swing interconnects as converting the low swing clock to full swing clock will need buffers, whose delay again cannot be predicted accurately at design time. Hence, repeaterless interconnects need to use active clock retiming circuits at the receiver to ensure optimum sampling of the received data.

Clock and data recovery (CDR) circuits have been reported for off chip interconnects, and are well known [56]. These circuits are typically phase locked loops that lock a local oscillator's frequency to the incoming data frequency. However, for on chip interconnects, only the phase needs to be recovered and a clock running at the correct frequency is available at the receiver.

Lee et al. in [10] report a clock synchronizer for a low swing interconnect that uses a half rate forwarded clcok. The synchronizer uses foreground calibration before the interconnect is used. Figure 2.15 shows the block diagram of this clock synchronizer.

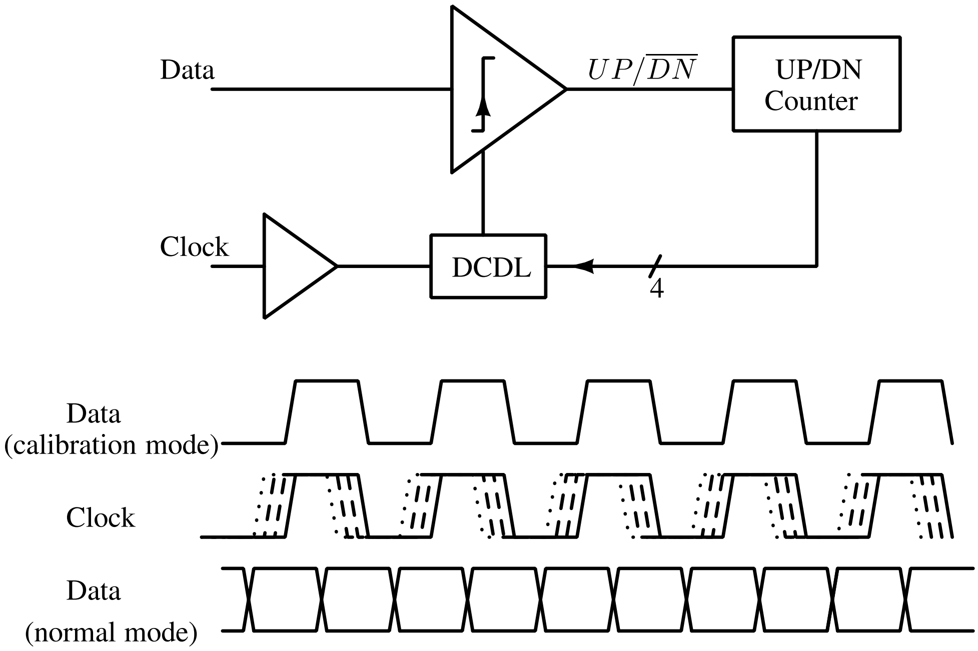


Figure 2.15: Foreground synchronizer reported in [10].

The circuit is first calibrated in training mode. For training, a half clock cycle delay is inserted in the data path and a toggling pattern of 1 and 0 is sent on the line. At the receiver, the comparator's output drives the $UP/\overline{DN}$ input of a 4 bit counter. The counter's output is used to select 1 of 16 delayed versions of the clock generated by a Digitally Controlled Delay Line (DCDL). The circuit converges when the clock and data are aligned. Once lock is achieved the half cycle delay at the transmitter is removed, thus bringing the sampling clock to the center of the eye. Since the technique uses a source synchronous clock, the low frequency jitter in the data and clock are correlated. In this system the recovered clock phase is quantized. Further, the system cannot track any changes in delay due to temperature or voltage fluctuations, without interrupting normal operation.

There are other types of synchronizers reported that use phase detectors to sense the phase difference between the data and the clock in normal operation, and use phase interpolation to derive the correct sampling clock [57–59]. Phase interpolation needs multiple phase sinusoidal clock signals or clock signals with large rise times. Such techniques inherently result in high power consumption and this high power can only be justified when small frequency differences have to be compensated as well. Kreienkamp et al. in [57] use a bang bang phase detector to sense the phase difference between the data and the sampling clock. The integrated error signal then controls an analog phase interpolator, that generates the correct sampling phase by interpolating between four phases of the receiver clock. This circuit is adaptive, and does not have phase quantization problem. It however occupies very large area and consumes large amounts of power. The technique predominantly uses analog circuits and may not be scalable easily. Takauchi et al. in [60] use a digital controller based phase interpolator for deriving the sampling clock. However, the required digital circuits are fairly complex and the limited phase resolution of digital circuits results in high phase noise and poor jitter performance.

2.4.1 Measuring latency and its variation with time

The synchronizers of on chip interconnects have a source clock of the correct frequency available (since it is routed from the same clock synthesizer). Measurement of clock skew between remote nodes is required for characterizing the delays to be compensated. Clock skew measurement can also be used for characterizing latency of data lines. Since the clock is branched from the same synthesizer, the low frequency jitter is correlated at all leaf nodes [61]. However, the high frequency jitter, which is introduced by thermal noise and supply induced jitter is not correlated [62]. Characterizing the skew and its variation with time, can provide useful information in designing clock synchronizers [63].

Many techniques have been reported for measuring time difference between two signals accurately. High accuracy measurements have relied on measurement using delay line of buffers [64] or using Vernier delay lines [65]. With a simple delay line of buffers, the accuracy is limited to the buffer delay. While high resolution can be achieved with a Vernier delay line, it is very sensitive to matching of devices. These techniques can achieve high accuracy, however, they are still accurate only when measuring skew between two local signals. Measuring skew between remote signals is more challenging.

One of the ways of measuring skew between periodic signals is to use asynchronous subsampling followed by histogram analysis of the XOR of the subsampled signals [66]. Asynchronous sampling ensures the samples cover the period completely and the number of times the subsampled outputs differ depends on the skew between them. The use of XOR operation however limits the minimum detectable skew in the presence of jitter. This can be understood by considering an example case of two clock signals with nominally zero skew. The random uncorrelated jitter will result in the samples being unequal occasionally. This will result in the XOR output being high, thus incrementing the histogram counter. The authors of [66] propose a work around for this, by measuring skew of the two signals of interest with respect to a third signal which has a large skew.

Amrutur et al. in [67] report a technique for measuring skew between remote nodes on a chip, using asynchronous subsampling. In this technique, the remote nodes are sampled with a sampling clock, which is offset in frequency with respect to the main clock. Figure 2.16 shows the block diagram of this method and Figure 2.17 illustrates the sampling of the clock with a clock that is slightly offset in frequency. This produces time amplified beat frequency signals.

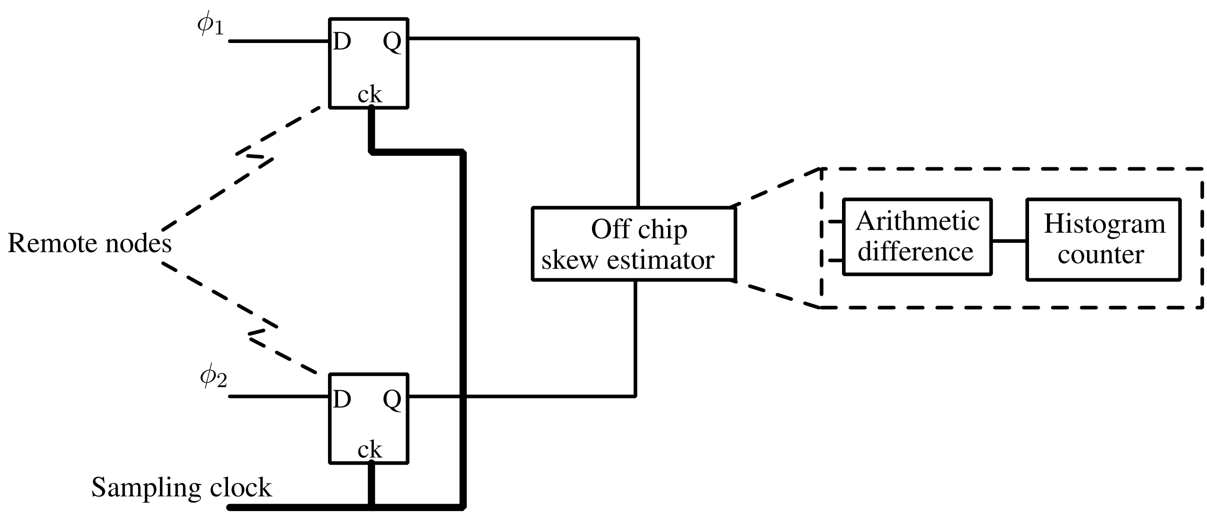


Figure 2.16: Subsampling technique reported in [67] for measuring skew between remote nodes.

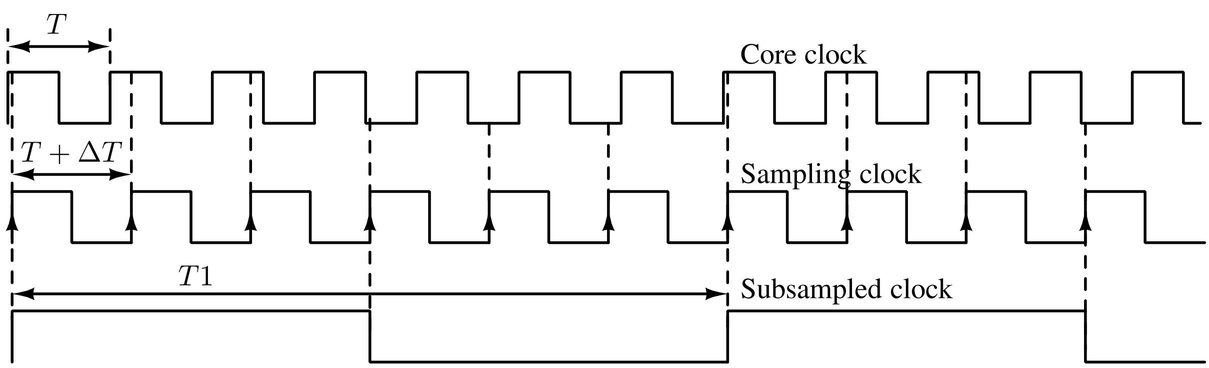


Figure 2.17: Illustration of aynchronous subsampling with a sampling clock that is offset in frequency from the core clock.

The skew between the signals is also amplified. The amplified skew between the subsampled signals can then be measured relatively easily. The authors perform histogram analysis of the arithmetic difference of the sampled outputs. Unlike histogram analysis over the XOR of the two sampled signals, this does not limit the minimum detectable skew. On the contrary, the jitter results in improved accuracy by the averaging inherently performed by the histogram analysis. Since the skew is amplified, the routing delays of the sampled signals introduce negligible error. However, the skew introduced by the routing of the sampling clock introduces significant error. Further, block averaging done at the backend, trades measurement time for accuracy. This limits the tracking bandwidth of the technique. The authors report a resolution of 0.84 ps with a measurement time of 140 ms. Vasudevamurthy et al. in [68] extend this method to reduce the measurement time by using multiple clock phases, at the expense of hardware complexity. Further, the measurement time is reduced appreciably only for small values of skew.

2.5 Testability of mixed signal circuits

Testability of pure digital circuits has been well studied. Most digital circuit tests use the stuck-at fault model for testing, which captures most catastrophic faults [13]. However, this cannot be applied to testing analog or mixed signal circuits as analog nodes in these circuit can assume any voltage value. Hence, most reported works on testing of mixed signal circuits use the structural fault model [14–16]. This fault model includes the following types of defects

- MOS faults
  - Gate-Source short (GSS)
  - Gate-Drain short (GDS)
  - Drain-Source short (DSS)
  - Gate open (GO)
  - Drain open (DO)
  - Source open (SO)
- Resistor faults
  - Resistor short (RS)
  - Resistor open (RO)
- Capacitor faults

– Capacitor short (CS)

All open faults are modeled as a $10M\Omega$ resistor in series with the open terminals. Shorts are modeled with a 1Ω resistor across the shorted terminals. Figure 2.18 illustrates all the faults in the transistors and their models. Figure 2.19 illustrates the faults in resistors and capacitors

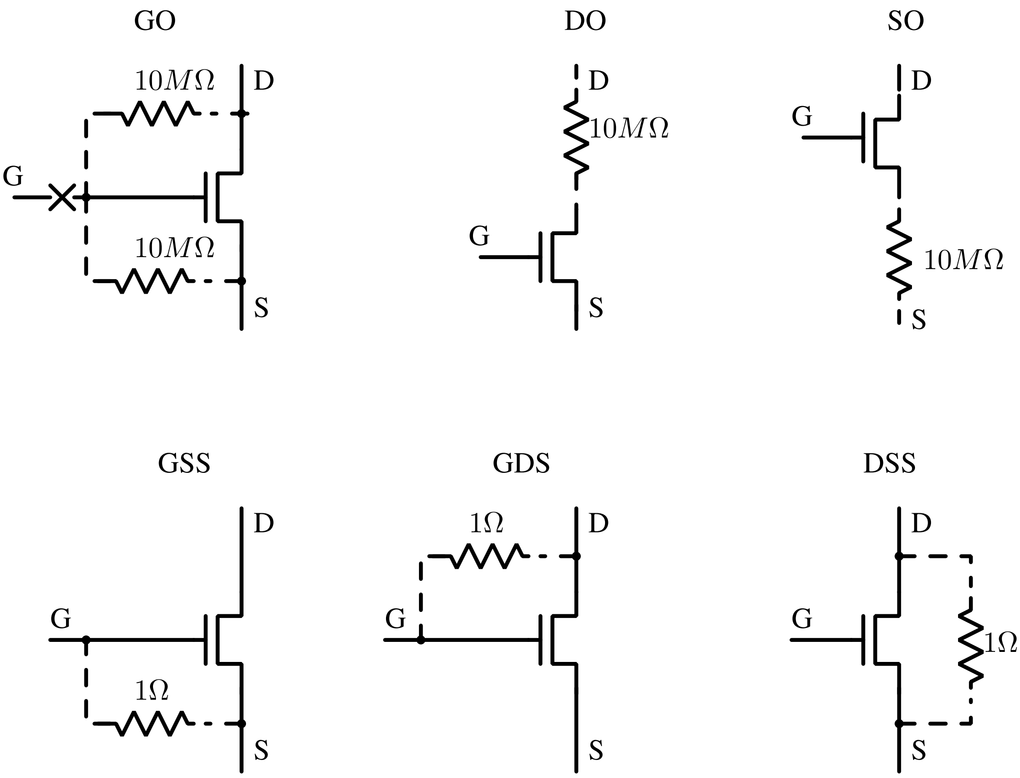


Figure 2.18: Structural faults in the transistors.

and their models.

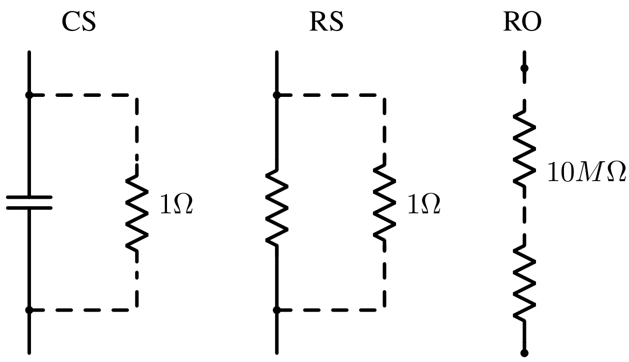


Figure 2.19: Structural faults in passive components.

Techniques of testing pure digital circuits have matured and have been standardized. CAD tools have been developed for automatically adding circuits needed for testability. These tools also perform automatic test vector generation. However, it is difficult to generalize the test of mixed signal circuits. This has resulted in researchers proposing different test methodologies for different classes of circuits.

For example, K. Arabi et al. in [69] show that many circuits which can be re-configured to break into oscillations, can be forced into oscillation for test. Observation of just the frequency of oscillation can reveal many types of catastrophic, as well as parametric defects. However, not all circuits can be reconfigured into an oscillatory unstable form, limiting this technique's application. For BIST of Analog to Digital Converter (ADC)'s, many works use some form of on chip test signal generation and pattern checking, like the ones reported in [18, 70] etc.

For testing PLL's and DLL's, many works have reported techniques of on chip measurement of various performance parameters like jitter, lock time etc. to qualify the performance [17, 71, 72]. These techniques however tend to have high complexity. Fault based testing of these circuits can achieve high fault coverage with marginal number of additional test structures. For example, by adding controlled delays to one of the clock inputs of a PLL, and capturing the divider's outputs as the PLL adapts itself to the new phase, most of the faults in the PLL can be captured [14, 15, 71]. Fault based testing of DLL's can be done with very little area overhead [16]. In this case a simple digital phase detector circuit is used to check of the DLL's output and input are within a predefined range of phase difference. This 1-bit output itself can reveal almost 95% of the faults in the circuit.

Testability of repeaterless interconnects has not been well explored in the literature. Kim et al. in [11] use on chip pattern generators and checkers for testing the performance of a repeaterless low swing interconnect. The on-chip test support circuits are configured in a scan chain for reading the test results. By comparing the error rates for different receiver threshold's the receiver input eye diagram is reconstructed. While the authors use such tests to validate the performance of their low swing interconnect, they do not investigate the fault coverage for different types of faults. The overhead of test is also significantly higher than the actual interconnect circuitry.

Techniques reported for testing other mixed signal circuits cannot be directly applied to test a repeaterless interconnect circuit.

2.6 Summary

Significant amount of research has been done to find a solution to the problem of data communication over long interconnects. Currently, all designs use repeater insertion for long interconnects, which is power hungry and significantly increases routing complexity. Further, the number of repeaters required increases as technology scales.

Low swing signaling with equalization has been shown to have the potential to replace repeater inserted links, with their low power, high throughput and simple layout. Among the equalization techniques, the capacitively coupled transmitter, with a weak driver, has been shown to be a simple and robust circuit [31]. Dave M. in [43] has reported a design method for this type of equalizer using the interconnect's response to worst case sequences. Kim et al. in [11] reported that higher order current mode equalization can achieve better equalization albeit with a more complex circuit. Higher order capacitive equalizers with weak drivers have not been explored in the literature. Design methods that allow quick optimization have also not been explored for higher order equalizers.

While low swing interconnects offer high throughput, the latency of these interconnects is still high. The optimum sampling region at the receiver's of these interconnects is narrow due to ISI and inherently present jitter. Sampling clock retiming circuits for such interconnects has not received sufficient attention in the literature. Lee et al. in [10] have reported a clock recovery circuit for low swing interconnects. This circuit needs foreground training, and hence, cannot track delay variation in normal operation. The recovered clock has phase quantization error. Hence, more investigation and work is needed in the area of clock retiming circuits for repeaterless interconnect receivers.

For high volume digital designs testability of designs is very important to screen out faulty parts early in the manufacturing cycle. Repeaterless interconnects with equalizers use mixed signal circuits for achieving high performance. The testability of these interconnects has however not received enough attention in the literature. Several techniques of testing mixed signal circuits have been reported in the literature. However, most of these are hand-crafted to individual application and cannot be generalized, or applied directly for testing low swing interconnects. Testability of low swing interconnects, hence needs more investigation.

Chapter 3

Design of low swing interconnect with equalization

In order to achieve high speed data transmission over repeaterless long interconnects, equalization at the transmitter or at the receiver or both, is required to compensate for the low pass characteristics of the interconnect. Equalization at the transmitter using capacitive boosting, along with a weak driver for allowing arbitrarily low data rates was shown to be an efficient and robust technique [31, 48]. Interconnects using higher order feed forward equalizers have demonstrated higher throughput than the first order capacitive equalizers [11]. In this chapter, we shall show that the capacitively coupled feed-forward equalizer can be extended to perform higher order equalization. A weak driver is used in shunt to allow arbitrarily low data activity factor.

A novel design method for quickly optimizing the first order dynamic transmitter end equalizers, using response to worst case sequences, was reported in [43]. However, this method was not generalized for higher order equalizers. In this chapter, we shall discuss a two tap capacitive feed-forward equalizer. Approximate small signal analysis of the transmitter is used to arrive at a simple implementation. The analysis is used to devise a design optimization method for designing the equalizer with worst case sequences.

Decision feedback equalization at the receiver can extend the data rate of interconnects with very small power overhead [11, 31]. Receiver side equalization is more power efficient than transmitter side equalization as the equalizer circuit does not have to drive the line. The feedback loop's delay determines the maximum frequency at which the equalization can be effectively done. At the cost of additional hardware, loop unrolling [53] can be used to increase the maximum frequency of operation [11]. In scaled technologies comparators have mismatch induced offset that has to be compensated [10, 11]. This makes loop unrolled DFE, with its increased number of comparators, less attractive.

In this chapter, we report a DFE circuit using the sense amplifier based comparator, that has reduced loop latency. In this circuit, the comparators offset is also corrected using the same feedback path. The circuit is designed, simulated and fabricated in the standard UMC 130 nm CMOS technology. Further, the circuit was also designed, laid out and the extracted netlist was simulated in TSMC 65 nm CMOS technology, to establish the scalability of the circuit.

3.1 A two tap capacitively coupled equalizer for transmitter end equalizaton

This section describes circuits for transmitter end equalization based on the capacitively coupled interconnect. Unless mentioned otherwise, the circuits are designed in the standard UMC 90 nm CMOS technology. The interconnect used for all simulations are all differential lines drawn in M8 layer with a width of 0.6 μm and a spacing of 0.4 μm. The channel is modeled as a 20 section RLC with R = 88.5 Ω/mm, C_{total} = 137 fF/mm, and L = 0.4 nH/mm and an adjacent channel capacitance of C_c = 52 fF/mm which were extracted from layout. Figure 3.1 shows the AC response of this interconnect. The interconnect was driven with a differential impedance of 200 Ω and terminated with a differential impedance of 16 kΩ for this simulation.

A first order dynamic transmitter equalizer, implemented using capacitive boosting, was discussed in Section 2.2 (refer Figure 2.7, which is reproduced here as Figure 3.2(a)). In order to implement a higher order equalizer, a copy of the driver circuit that is scaled in strength, is used to drive the line with the data inverted and delayed by one bit period. This essentially helps quickly erase the residue of the of the previous data bit on the interconnect, after 1 bit period. The scaling factor is chosen depending on the transfer function of the interconnect to be equalized. Thus, the memory of the line is reduced or in other words bandwidth of the interconnect increased. Figure 3.2(b) shows the block diagram of a capacitively coupled transmitter, with a second tap from the previously transmitted bit added.

We can perform an approximate analysis of this equalizer assuming the interconnect has zero impedance. With this assumption, the transfer function (transconductance) of the transmit-

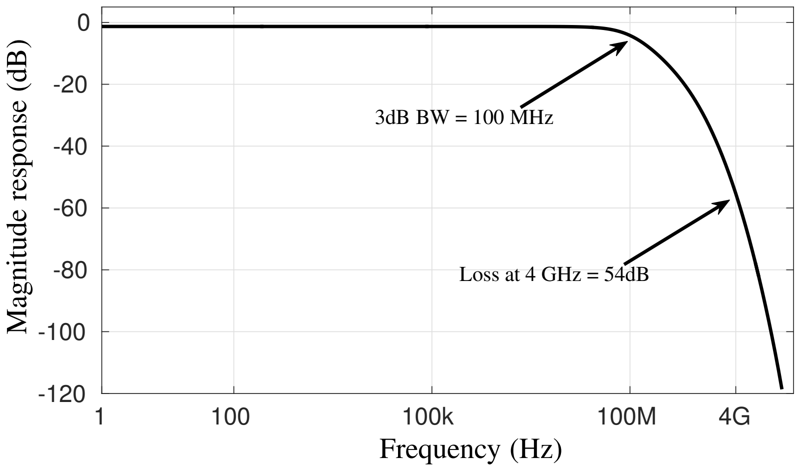
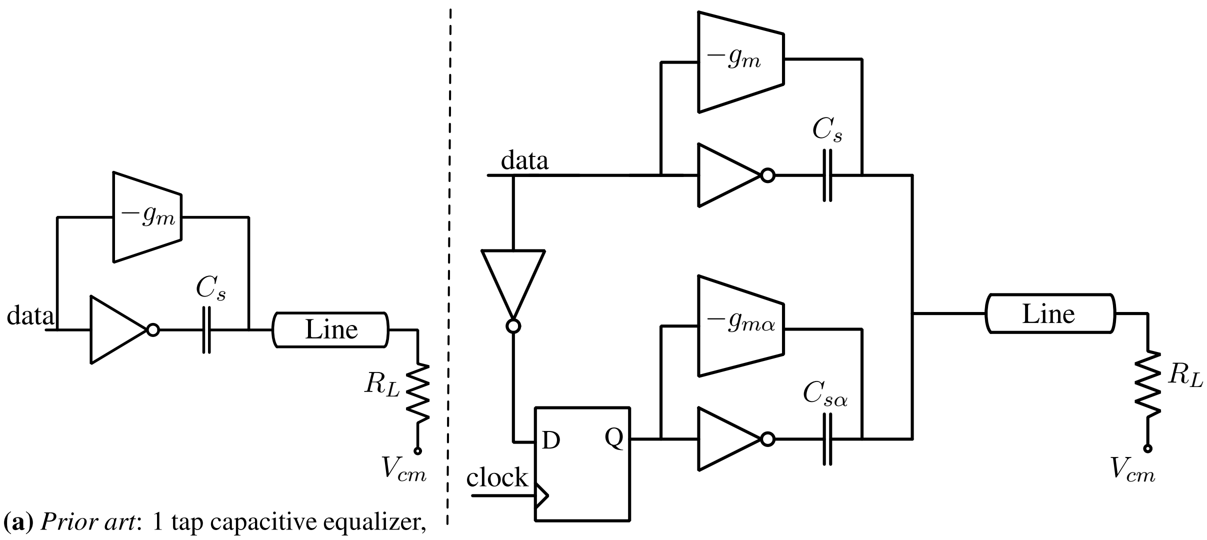


Figure 3.1: AC response of the repeaterless 10 mm interconnect when driven with a impedance of 200 Ω (differential) and terminated with 16 kΩ(differential).



(a) *Prior art*: 1 tap capacitive equalizer, reproduction of Figure 2.7(a).

(b) 2 tap capacitive equalizer.

Figure 3.2: Capacitively coupled interconnect with compensation for (a) previous transition, and (b) previous two transitions.

ter can be written as

$$\frac{I(s)}{V(s)} = g_m + sC_s - (g_{m\alpha} + sC_{s\alpha})e^{-sT_B}$$

Where g_m is the transconductance of the direct driver, C_s is the driver capacitor for the previous transition, $g_{m\alpha}$ is the transconductance of the previous bit driver, $C_{s\alpha}$ is its corresponding strong driver capacitor, and T_B is the bit period. Expanding the exponential term using Taylor series, we can collect the powers of s and write the transfer function as

$$
\begin{aligned}
\frac{I(s)}{V(s)} =& (g_m - g_{m\alpha}) + s(C_s - C_{s\alpha} + g_{m\alpha}T_B) \\
&+ s^2(C_{s\alpha}T_B - \frac{g_{m\alpha}T_B^2}{2!}) \\
&+ s^3(\frac{-C_{s\alpha}T_B^2}{2!} + \frac{g_{m\alpha}T_B^3}{3!}) + \cdots
\end{aligned} \tag{3.1}
$$

Defining G_m, λ_1 and λ_2 as

$$
\begin{aligned}
G_m &= (g_m - g_{m\alpha}), \\
\lambda_1 &= (C_s - C_{s\alpha} + g_{m\alpha}T_B), \\
\lambda_2 &= (C_{s\alpha}T_B - \frac{g_{m\alpha}T_B^2}{2!}),
\end{aligned}
$$

we can write the transfer function in equation (3.1) as

$$
\frac{I(s)}{V(s)} = G_m + \lambda_1 s + \lambda_2 s^2 + \cdots \tag{3.2}
$$

In equation (3.2) the transconductance component, viz. G_m along with the load resistance (R_L) determine the low frequency logic swing on the line. There is an optimum value for λ_1 and λ_2 that gives maximum bandwidth enhancement with minimum overshoot [73]. Deviation from these optimum values directly results in a throughput loss and sometimes higher power consumption. For the capacitively coupled link with equalization for previous two transitions, the effective transconductance is G_m. Repeaterless interconnects are designed with low swing (typically less than 100 mV) for high speed and low power. Termination resistances are typically less then 10 kΩ. In such a scenario the $g_{m\alpha}T_B$ term is much smaller than $(C_s - C_{s\alpha})$ and can be neglected, thus simplifying the design. With this approximation the terms of higher powers of s depend only on $C_{s\alpha}$.

3.1.1 Worst case sequence based design of the link

The analysis in the previous section assumed a very crude model with zero interconnect resistance. For designing the link, we use a macro model with VerilogA models for the transmitter and a 20 section RLC interconnect. Figure 3.3 shows the macro-model used to emulate the transmitter. The flip-flop and inverters in the model use VerilogA descriptions. The interconnect is modeled with a 20 section RLC ladder network. I_s is the weak driver current source, C_s is the equalizer's first tap capacitor, and $C_{s\alpha}$ is the second tap capacitor. The resistor R_s models the finite output impedance of the driving inverter, and the resistor R_o models the finite output impedance of the current sources. A value of 100 Ω is used for R_s and 1M Ω, is used for R_o, which are typical values for the standard UMC 90 nm CMOS technology, in which the circuit will eventually be designed. NO and NC are switches that are normally open and normally closed respectively. Note, that the previous bit driver does not have the scaled weak driver since its contribution is negligible, as explained above.

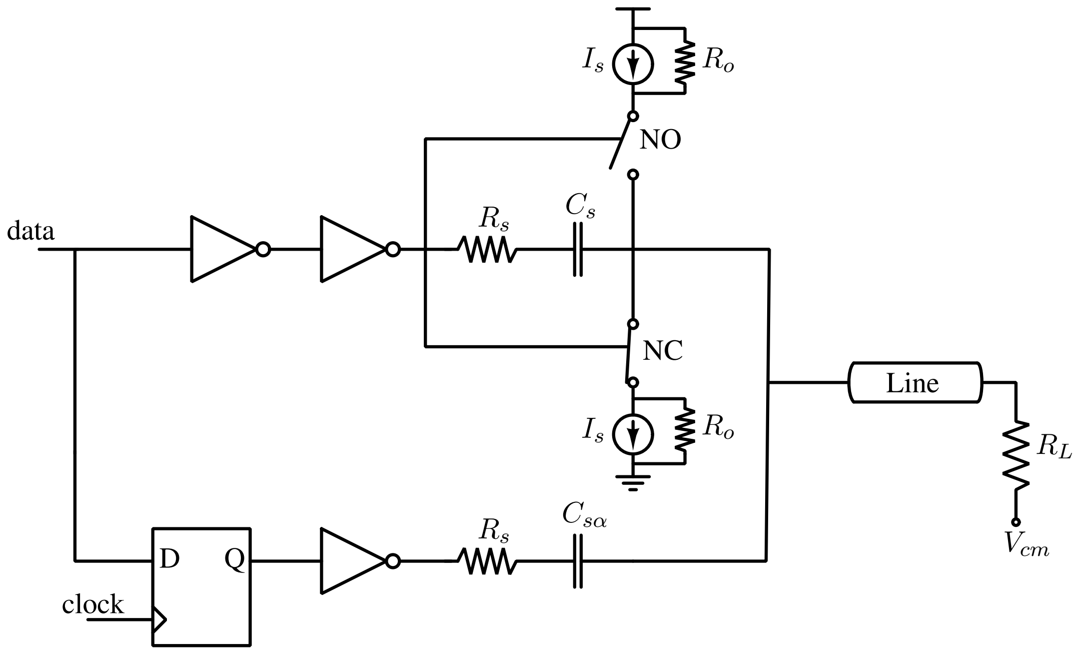


Figure 3.3: Macro-model of the Capacitively coupled transmitter.

For brevity, a single ended version is shown in Figure 3.3. All simulations, as well as analysis is however done for differential interconnects. The logic swing on the line for low frequency data can be written in terms of the weak driver current source I_s and the terminating resistance R_L as

$$\Delta V_{LF} = \pm I_s \times 2 \times R_L, \tag{3.3}$$

where the factor of 2 results from a differential implementation. We use R_L of 8 kΩ and an I_{static} of 1 μA. This sets $\pm$ 16mV as the logic levels for low frequency data. For effective equalization, the resistance R_s should be chosen to be much less than the impedance of the series capacitor at the operating frequency. Choosing too small a value will however require large sized buffers, which results in increased power consumption.

Worst case sequence based design method was shown to be a quick and efficient method for finding optimum design parameters for first order dynamic transmitter side equalizers [43], which was discussed in Section 2.2.3. In order to use this design method for a two tap equalizer, we need to identify worst case sequences that produce pronounced effects of the transmitter's design parameters at the receiver's input. We identify three such sequences that pronounce the effect of the two design parameters λ_1 and λ_2 of equation (3.2). The third design parameter, i.e. G_m only decides the low frequency logic swing on the line, which is set by I_s. The first pattern (called WC_1 henceforth) is a large number of 0's, which lowers the line to its low frequency logic level, followed by a 1 and a 0 and then followed by a large number of 1's (...0000'1'01111...). Figure 3.4 shows this sequence and its 1-bit delayed version. The bit

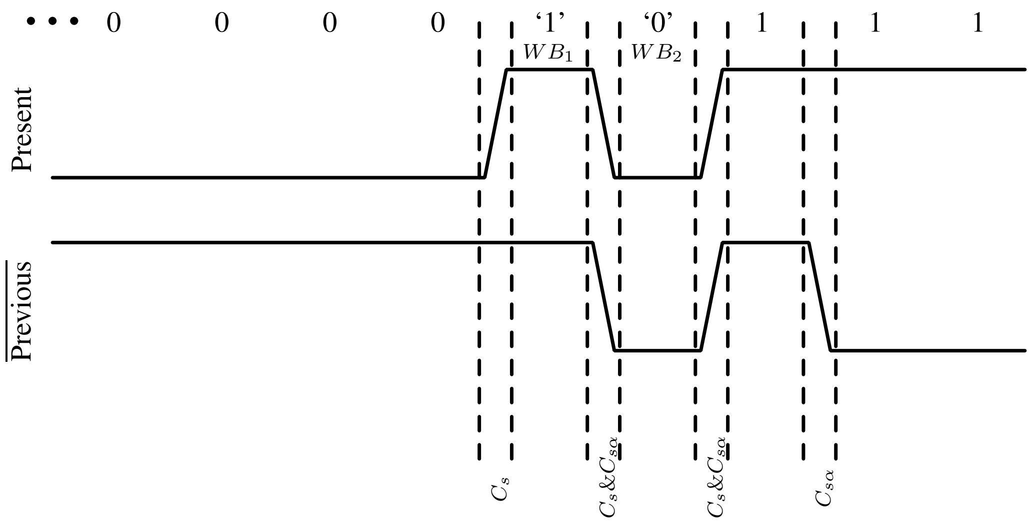


Figure 3.4: The worst case sequence WC_1 (which is same as WC_2 except for the bit under examination), used to design the two tap equalizer. The upper signal is the data signal and the lower signal is a 1-bit delayed and inverted version of the same.

under examination here is the single 1, shown in single quotes and labeled as WB_1, occuring after the large number of 0's. For this bit, the first series capacitor C_s, dumps charge onto the line at the transition corresponding to its beginning. At its leaving transition, both the capacitors C_s and $C_{s\alpha}$ discharge the interconnect in attempt to erase the memory of the bit just transmitted. This discharge reduces the effective received voltage for this sequence. Hence, the response to this sequence depends on the difference between C_s and $C_{s\alpha}$. For WC_1, a large difference between C_s and $C_{s\alpha}$ will improve the noise margin at the receiver for WB_1 in this sequence.

The second bit pattern (called WC_2 henceforth) is the same as WC_1, but the bit under examination is the single 0 after the single 1 (...00001'0'1111...). This bit is labeled as WB_2 in Figure 3.4. For this bit both C_s and $C_{s\alpha}$ dump charge together in phase, thus making the effective pre-emphasis proportional to (C_s + $C_{s\alpha}$). If $C_{s\alpha}$ is less then optimum this bit will result in the least eye opening at the receiver.

The third bit pattern (called WC_3 hence forth) is a large number of 1's followed by a few 0's a single 1 and many 0's (...1110000'1'0000...). This sequence and its delayed and inverted version is illustrated in Figure 3.5. Here, the single 1 after a few 0's is the bit under study, which

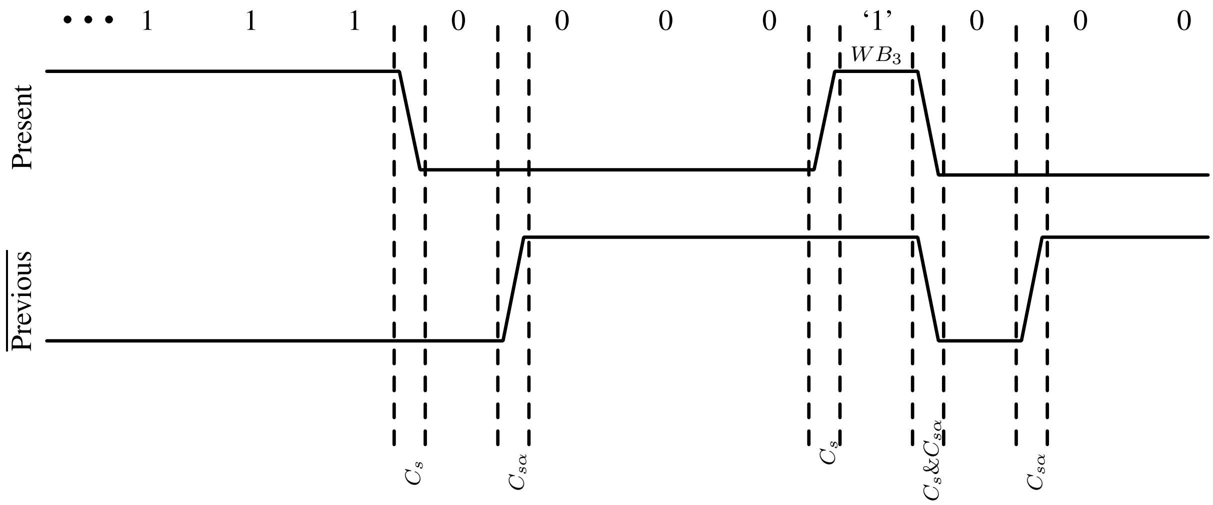


Figure 3.5: The third worst case sequence WC_3 that is used to design the two tap equalizer. WB_3 is the bit of interest. The upper signal is the data signal and the lower signal is a 1-bit delayed and inverted version of the same.

is labelled as WB_3. A large number of 1's take the line to the low frequency logic level, which when followed by a few 0's can overshoot the low frequency logic level. The overshoot will peak when the number of 0's is roughly equal to the memory of the line. After that the line will settle to its low frequency logic level. A single 1 imposed on the line when this overshoot is at its maximum can result in it being missed, thus making it a worst case pattern. In this case C_s dumps charge to change the state of the line quickly from a good 1 to 0, while $C_{s\alpha}$ sinks charge in the next bit to cut the overshoot. The residual overshoot is proportional to (C_s - $C_{s\alpha}$). Reducing the residual overshoot results in a better noise margin for this sequence. Hence, for WC_3, a large difference between C_s and $C_{s\alpha}$ degrades the noise margin at the receiver for the bit WB_3 in this sequence.

The transitions for the three worst case sequences shown in Figure 3.4 and Figure 3.5 are labeled with the capacitors that inject current when they occur. Together the response to these three sequences highlight the transmitter's design parameters. WC_1 and WC_3, depend strongly on the difference between the two capacitors. On the other hand, the transitions for WC_2 are driven by the sum of the two capacitors. The initial value for WC_2, however, depends on C_s, due to the charge it dumps for WB_1. WC_2 is therefore inversely proportional to C_s and directly proportional to $C_s + C_{s\alpha}$. The response to WC_2 is proportional to

$$\frac{C_s + C_{s\alpha}}{C_s} = 1 + \frac{C_s}{C_{s\alpha}}.$$

Hence, the response to WC_2 maps better to the ratio of the two capacitors. By symmetry of the circuit, the complements of the above three worst cases are also worst case sequences. Figure 3.6 and Figure 3.7 show the received waveforms when λ_1 is more and less than optimum, respectively. It is seen that WC_1 has poor noise margin when λ_1 is less than optimum and WC_3 has poor noise margin when λ_1 is more than optimum. Figure 3.8 and Figure 3.9 show the

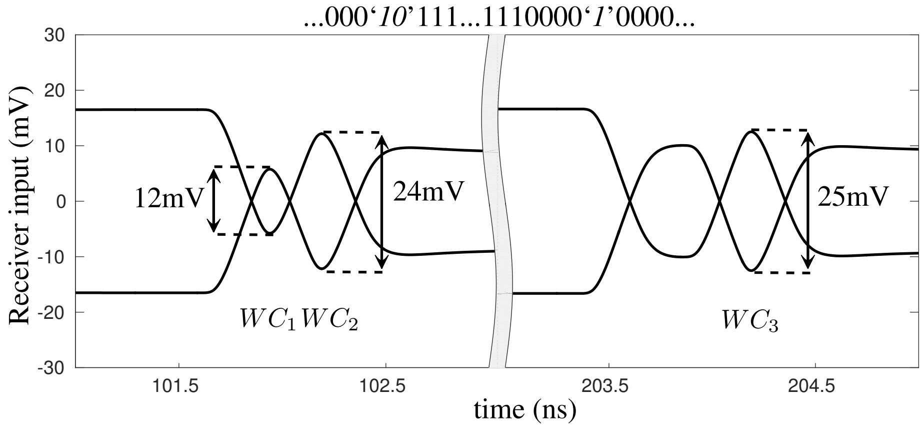


Figure 3.6: Receiver input voltage : λ_1 is less than optimum.

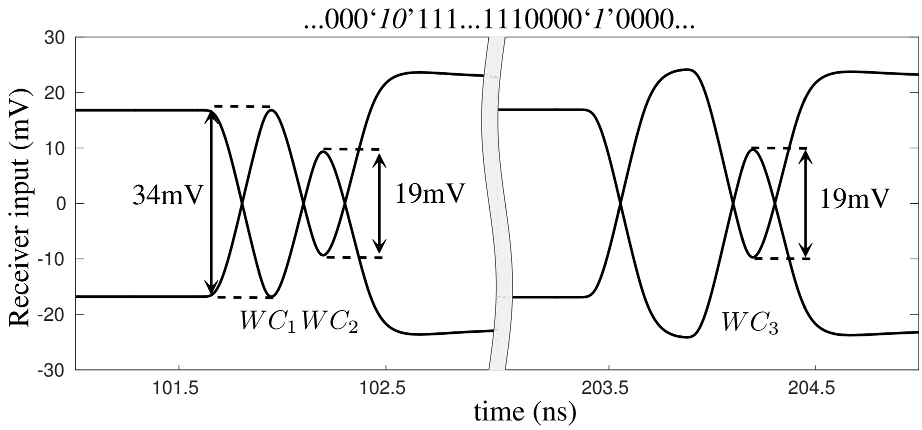


Figure 3.7: Receiver input voltage : λ_1 is more than optimum.

eye diagrams when the previous bit driver of the transmitter is less and more than optimum, respectively. It is seen that WC_2 results in good noise margin when λ_2 is less than optimum, but WC_1 and WC_3 have poor noise margin. When λ_2 is more than optimum, WC_1 and WC_3 have very good noise margin but WC_2 gets degraded.

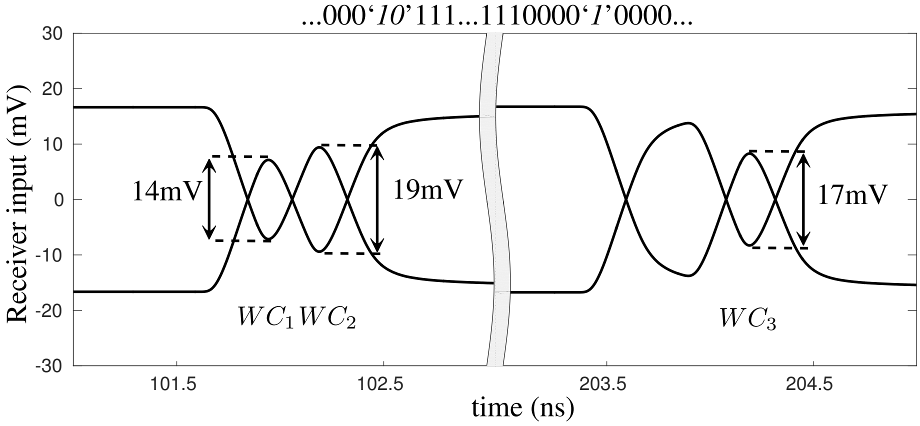


Figure 3.8: Receiver input voltage : λ_2 is less than optimum.

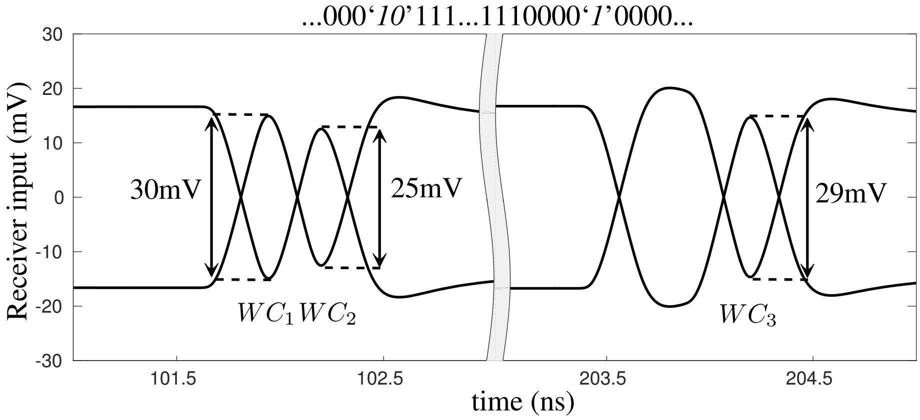


Figure 3.9: Receiver input voltage : λ_2 is more than optimum.

Figure 3.10 shows the received waveform when both λ_1 and λ_2 are optimum. Figure 3.11 shows the eye diagram at the receiver input with random bits with equal probability of 1's and 0's which shows the same vertical eye opening as the optimized value.

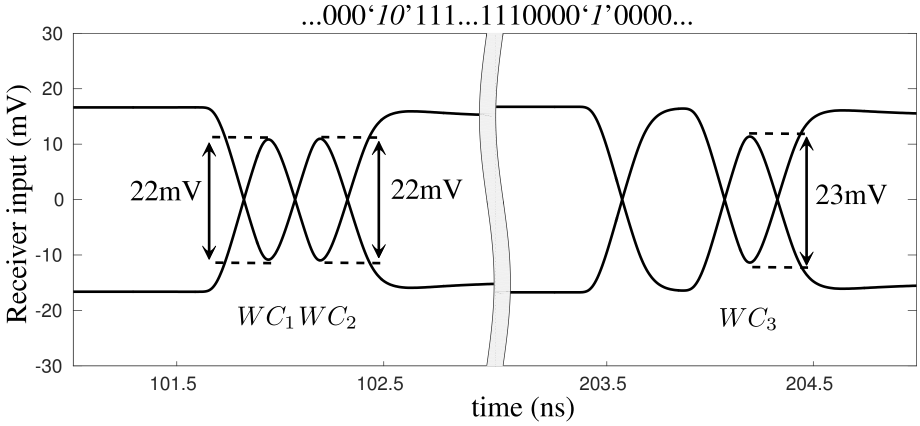


Figure 3.10: Receiver input voltage for the three worst cases, when both λ_1 and λ_2 are optimum.

The pre-emphasis capacitors C_s and $C_{s\alpha}$ are optimum when all the three worst case se-

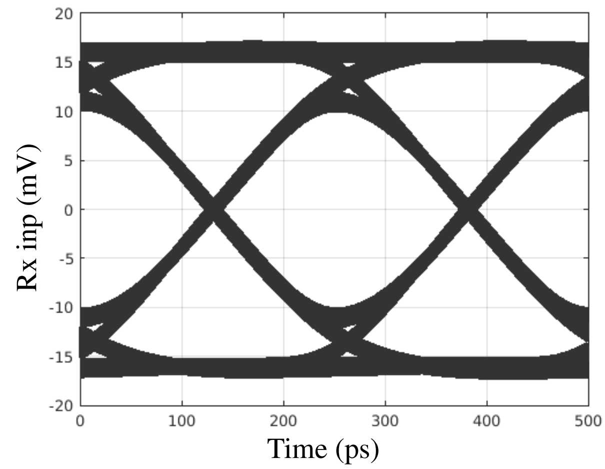


Figure 3.11: Receiver input eye diagram with random data with equal probability of 1 and 0.

quences result in equal eye height. Tuning C_s and $C_{s\alpha}$ independently to find the optimum can be difficult as they affect the three worst cases differently. While WC_1 and WC_3 depend strongly on λ_1 and weakly on λ_2, WC_2 depends strongly on λ_2. It is convenient to map the design variables as (3.4)

$$k = C_s - C_{s\alpha}, \qquad \alpha = \frac{C_{s\alpha}}{C_s} \tag{3.4}$$

k and α can be tuned to quickly find the optimum. The optimum values of k and α were found to be 51 fF and 0.46 respectively. From these values the two capacitor's values can be back calculated easily.

Figure 3.12 shows the pulse response of the interconenct with different types of drivers. The line is terminated in differential 16 kΩ in all cases. The voltage mode driver used a driver with a differential impedance of 200 Ω. The 1-Tap equalizer is an optimally designed 1 tap capacitively coupled transmitter and the 2-Tap equalizer is an optimally designed 2 tap capacitive transmitter discussed in this chapter. The last case in Figure 3.12 is when the interconnect is driven only with the weak driver which is used in the capacitively coupled transmitter.

3.1.2 Comparison of interconnect's response to worst case sequence and random data

Conventionally, random bit sequences are used to test the performance of interconnect links. PRBS generators are used to generate random data for these tests. To emphasize the importance of testing with worst case sequences, a suboptimally designed link was tested with random

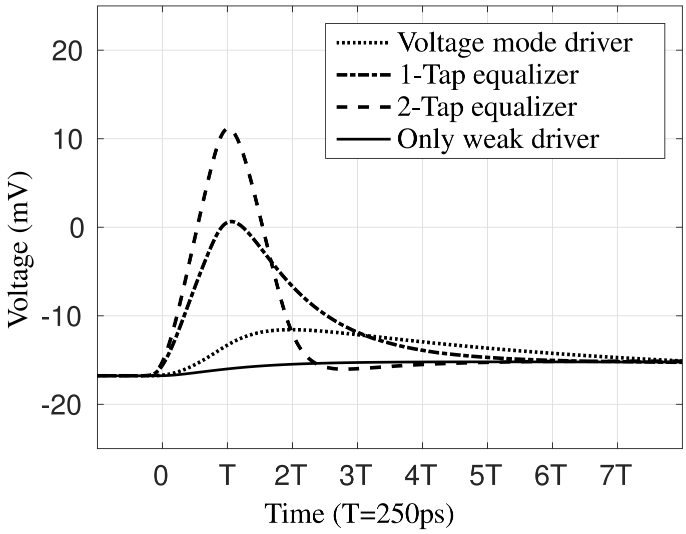


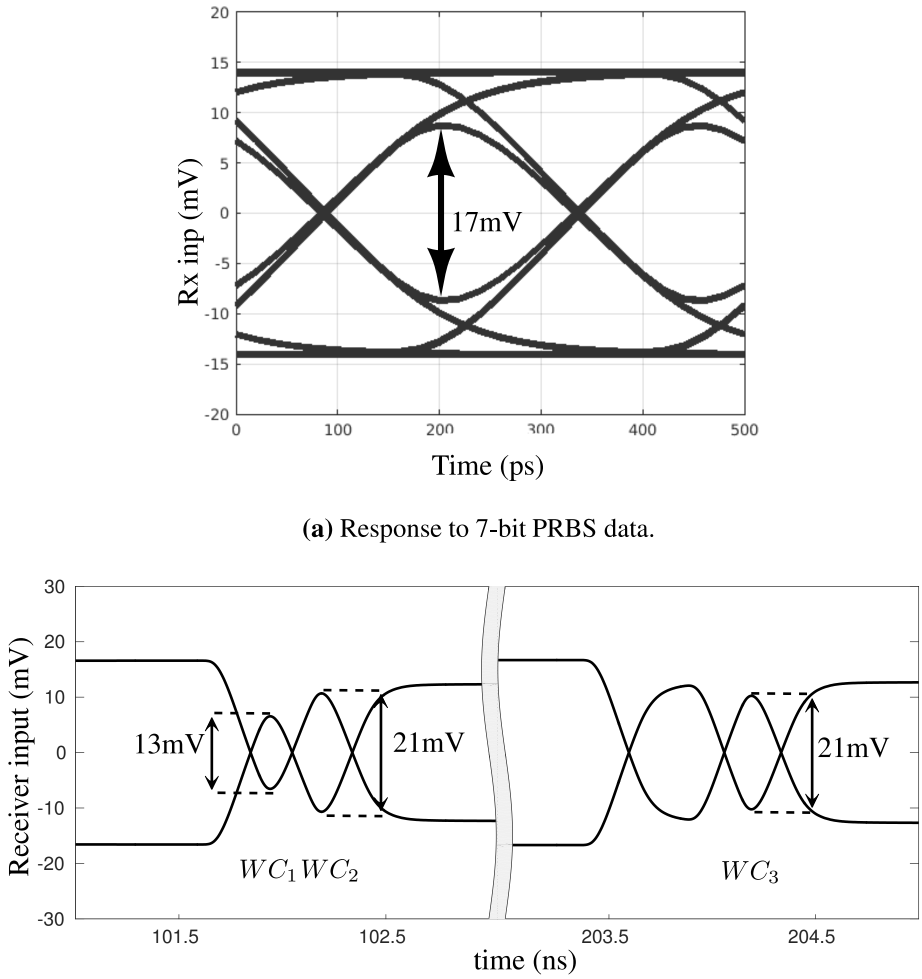
Figure 3.12: Pulse response of the interconnect with different types of drivers. The line is terminated in differential 16 kΩ in all cases.

sequence and its response compared to that of worst case sequence. Figure 3.13(a) shows the eye diagram at the receiver input with 7-bit PRBS sequence as the input data, for a suboptimally designed transmitter. As seen the receiver input eye diagram is very good.

When the same link was tested with the worst case sequence, the shortcomings of the link were clearly emphasized. Figure 3.13(b) shows the response of the interconnect to the worst case sequences. Longer PRBS sequence like 15-bit or 31-bit PRBS sequences can be used which are likely to capture the worst case response of the interconnect. However, the time needed to run a long simulation that ensures complete coverage of all possible sequences, is prohibitively long. The worst case sequences have very low data-activity factors and therefore, very short simulation time. Hence, the worst case sequence based design method is effective in quickly finding the optimum design parameters of transmitters with equalization. In the next section, we will discuss the implementation of the transmitter with actual circuits.

3.1.3 Circuit realization

Having found the optimum design parameters using the macro-model of the interconnect link, we now design the interconnect link at circuit level. Using the macro-model, one can modularly design each circuit component and incrementally construct the full circuit. The design of these components is discussed as follows.



(a) Response to 7-bit PRBS data.

(b) Response to Worst case sequence.

Figure 3.13: Response of a suboptimal link to (a) 7-bit PRBS data and (b) worst case sequence.

Series capacitor implementation: The capacitors are implemented with PMOS transistors with source and drain shorted. Two PMOS transistors are connected back to back so as to make the capacitance constant with voltage to first order.

Buffer Design: The line is driven with an inverter through a series capacitor. The output impedance of the inverter should be much smaller than the impedance of the capacitor at the operating frequency. To save power, the main driver is buffered to an output impedance of 100 Ω, while the previous bit driver is buffered to an output resistance of 250 Ω. For differential implementation, complementary bits are required[1] which are generated using a fork of inverters. The profile of the charge injected into the line is a function

[1] In some cases, complementary bits might be available in the system and a fork might not be required, for e.g. when flip-flops like sense amplifier flip-flop or the PowerPC flip-flop is used.

of the rise time of the buffer outputs. Hence, while designing the fork, the complementary outputs should not only be delay matched but also be rise time matched. Unequal rise times result in differential offsets at the receiver.

Weak driver: The bias for the current sources is generated by using an inverter with its input and output shorted. The weak driver current source is shared between the two differential lines by current steering. Current steering ensures that the current sources are always on, eliminating the effect of turn on transients of the current sources and minimizes charge injection. Figure 3.14 shows the complete circuit diagram of proposed transmitter.

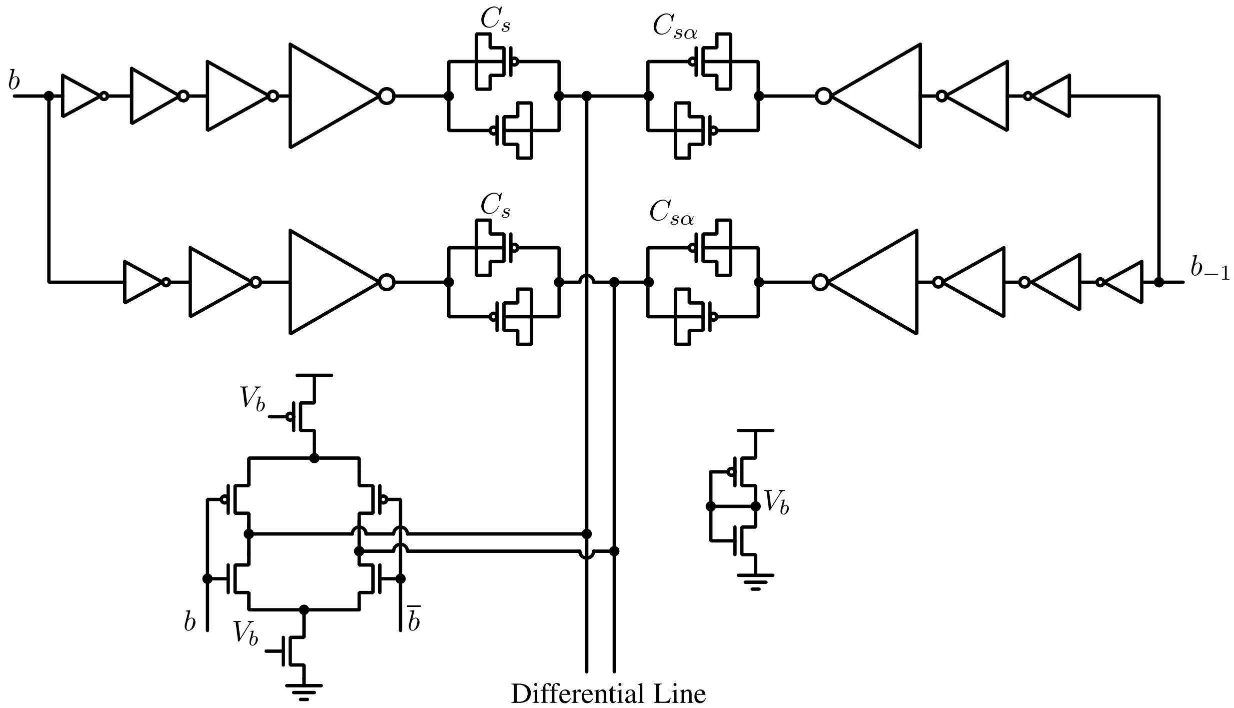
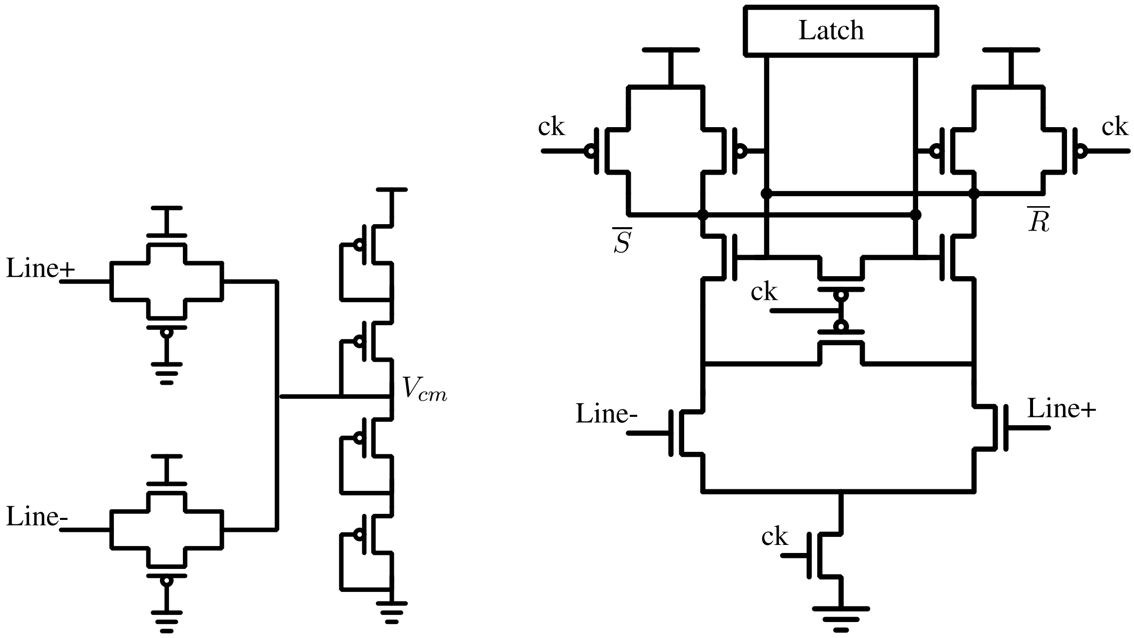


Figure 3.14: Transistor level circuit of the proposed transmitter.

Line termination: The line is terminated with a resistor which is implemented using transmission gates as resistors. Figure 3.15(a) shows the circuit diagram of the termination circuit.

Receiver comparator: The receiver is a clocked sense amplifier based comparator (Figure 3.15(b)) with the modified slave latch reported in [74]. The design and analysis of the sense amplifier comparator and slave latches is presented in Appendix A. Half rate architecture is used at the receiver. In clocked comparators, when the input of the comparator is not driven with very low impedance, kickback noise becomes very significant. The kickback



(a) Termination circuit.

(b) Sense amplifier comparator.

Figure 3.15: Reciever circuit for the interconnect.

can be differential in presence of a differential input, due to the voltage dependence of the MOS parasitic capacitance. Interconnect line terminated in too low a resistance would require stronger drive at the transmitter increasing the power consumption. The kickback noise can be reduced substantially by using half rate architecture. In half rate architecture, the two parallel comparators driven by complementary clocks, introduce kickback noise which is out of phase at the input. Figure 3.16 shows the waveform of the voltage at the drains of the input transistors of the comparator. The precharge phase is labeled as ϕ_1 and evaluation phase is labeled as ϕ_2. In half rate architecture, the precharge phase of one of the comparators overlaps the evaluation phase of the other, as they are driven by complementary clocks. Hence, the kickback at the input nodes is out of phase at the input, effectively canceling itself.

3.1.4 Results and comparisons

The link was designed in the UMC 90 nm CMOS technology and its performance was analyzed using post layout simulations. As mentioned earlier the interconnect used is a differential line drawn in M8 layer with a width of 0.6 μm and a spacing of 0.4 μm. The channel is modeled as a 20 section RLC with R = 88.5 Ω/mm, C_{total} = 137 fF/mm, and L = 0.4 nH/mm and an adjacent

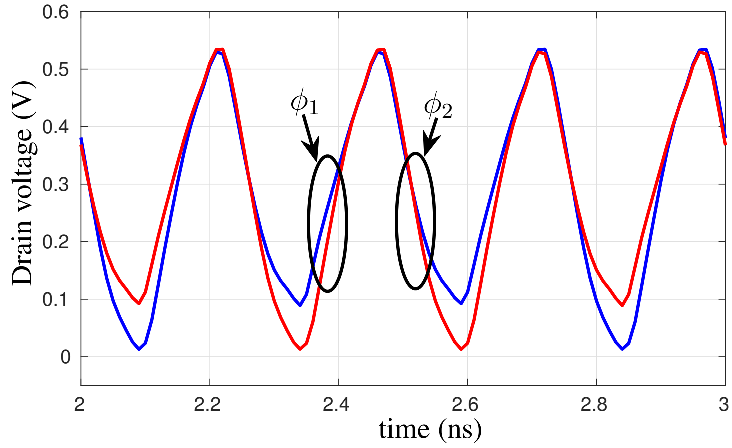


Figure 3.16: Voltage at drains of input transistors of the comparator.

channel capacitance of C_C = 52 fF/mm which were extracted from layout. Figure 3.17 shows the simulated eye diagram at the receiver input of the complete link with nominal transistor parameters, for random input sequence with a transition probability of 0.5. This simulation has

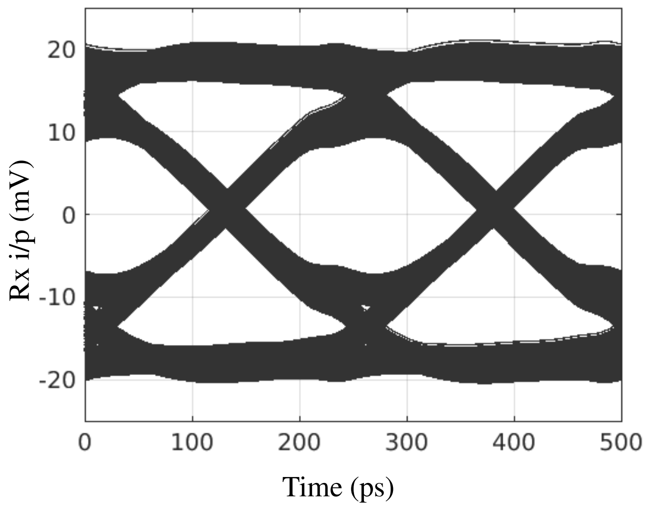


Figure 3.17: Eye diagram at the input of the receiver for random input bit stream.

been performed with a jitter free clock and the jitter in the received data is solely due to the residual ISI. From Figure 3.17, we can conclude that the macro-model is accurate and quick for finding optimum design parameters. The total energy consumption of the link is 313 fJ/bit in typical typical corner. The transmitter power is 223 fJ/bit and the receiver power is 90 fJ/bit. The latency of the link is 475 ps. Table 3.1 shows the comparison of the proposed design with other reported works.

Kim et al. reported a two tap feed-forward equalizer, using current mode implementation,

Table 3.1: Comparison with other works

Source	[46]	[11]	[31]	This Work
Technology	180nm	90nm	90nm	90nm
Length(mm)	10mm	10mm	10mm	10mm
Data Rate (Gbps)	1	4	2	4
Energy/bit (fJ/bit)	1000	371	280	313
Density[1] (Gbps/μm)	1.67	2	1.16	2
FoM = $(\frac{Gbps/\mu m}{pJ/bit})$	1.67	5.39	4.15	6.39

[1] Density is data rate per unit cross section of the interconnect. It quantifies the spatial efficiency of the interconnect.

in [11]. The authors use a decoder to calculate the current to be injected for avoiding current subtraction. This saves power. For the capacitive driver discussed in this Chapter, no such decoding logic is required, as the capacitors never inject charge in opposite directions. Thus, the difference in power consumption is approximately equal the extra power invested in the decoder.

The circuit was also tested for performance of the equalizer across corners of process variation. Table 3.2 shows the vertical and horizontal eye openings at the input of the receiver in all process corners. In SS corner the eye is closed, hence we have shown it at 2.8 Gbps. In SS corner however, the logic speed also falls to 2.7GHz. The logic speed is taken as the maximum clock frequency of a sequential circuit with a chain of 6 NAND gates between two flip-flops.

Table 3.2: Eye opening in different process corners

Process Corner	Vertical Eye opening	Horizontal Eye opening
TT	18.5 mV	224 ps (89.6%)
FF	19.0 mV	213 ps (85.2%)
SNFP	17.0 mV	206 ps (82.4%)
FNSP	14.5 mV	206 ps (82.4%)
SS (@2.8 Gbps)	23.0 mV	264 ps (75.4%)

The two tap capacitively coupled equalizer and its design using worst case sequences was designed in other technologies like UMC 130 nm CMOS technology and TSMC 65 nm CMOS technology, for different interconnect lengths. This helped in verifying the robustness of the architecture as well as its design method in different technologies.

3.2 Decision feedback equalization at the receiver

Equalization can also be done at the receiver, to compensate for ISI in the data. Equalization at the receiver requires less power than equalization at the transmitter, as the equalizer circuit does not have to drive the interconnect. One of the ways of doing equalization at the receiver is the DFE. In DFE, a hard decision is made on the input in every clock cycle. This decision is scaled and subtracted from the input before the next sampling event. The scaling factor is chosen based on the amount of previous bit ISI in the channel. If the initial decision is correct, this effectively erases the memory of the previous bit. This circuit can be extended, by adding more taps, to correct for interference from multiple past bits as well. The main implementation challenge in DFE circuits is the stringent loop delay requirement of the first feedback tap. The loop delay of the first feedback tap must be less than the clock period. In every cycle, this loop has to perform 1 decision, 1 scaling and 1 subtraction. Reducing this loop time can permit the circuit to be used at higher frequencies. It is worth noting that the ISI from the immediate previous bit is the highest, hence making the first tap significant.

In the decision device the comparator has to resolve the low swing input to rail to rail CMOS levels. The resolved signal is then scaled. This presents an unnecessary delay of scaling up the signal to CMOS levels and then scaling it down again, and bypassing this unwanted step can significantly reduce the loop delay of the circuit. This section describes a circuit that uses this idea to reduce the loop delay of a DFE circuit.

Further, comparators in scaled technologies suffer from offsets due to device mismatches, making offset calibration necessary. Offset calibration often reduces the frequency of operation of the circuit, as it adds capacitances at critical nodes in the comparators. In the implementation described in this section, the DFE circuit also allows to compensate for offset using the same transistors that perform equalization, without degrading the speed or power performance.

3.2.1 Concept of low latency DFE with offset compensation

In time domain, the output $y[n]$ of the DFE circuit can be expressed in terms of the comparator input $x[n]$ as

$$y[n] = Q\{x[n]\} = Q\{d_{in}[n] - \alpha y[n-1]\}$$
$$= \begin{cases} -1, & \text{if } x[n] < 0. \\ +1, & \text{otherwise.} \end{cases}$$

Here, $y[n-1]$ is the hard decision made by the comparator in the previous cycle and α is a constant that is less than 1. α is chosen depending on the amount of ISI present in the input data. The difference equation

$$x[n] = d_{in}[n] - \alpha y[n-1],$$

is a high pass function, which compensates for the ISI produced by the low pass nature of the interconnect. Since $y[n]$ is a hard decision, the term $\alpha y[n-1]$ can take only one of two values, i.e.

$$\alpha y[n-1] = \begin{cases} -\alpha, & \text{if } y[n-1] = -1. \\ +\alpha, & \text{if } y[n-1] = +1. \end{cases}$$

The analysis till now assumes an ideal comparator. Practically, comparators also suffer from offset (V_{offset}), which needs to be corrected. For a comparator with offset compensation included, we can write an equation for its input as

$$x[n] = d_{in}[n] - \alpha y[n-1] - V_{offset}.$$

Let $V_{fb} = \alpha y[n-1] - V_{offset}$. We can then write

$$x[n] = d_{in}[n] - V_{fb}[n],$$

where,

$$V_{fb}[n] = \begin{cases} +\alpha - V_{offset}, & \text{if } y[n-1] = -1. \\ -\alpha - V_{offset}, & \text{if } y[n-1] = +1. \end{cases}$$

We implement the DFE circuit using a switched capacitor circuit that uses the comparator output to select from pre-computed voltages that correspond to $-\alpha - V_{offset}$ and $+\alpha - V_{offset}$ for the feedback. Since most of the applications use differential input architecture, we will need a comparator with a double differential input, i.e. with a differential main input (d_{in}) and differential feedback input (V_{fb}).

When $y[n-1] = +1$, the differential feedback voltage $V_{fb} = (V_{fb}^{+} - V_{fb}^{-})$ can be written as

$$V_{fb}^{+} = V_{H}^{[+1]} = V_{cm} + \frac{\alpha}{2} - \frac{V_{offset}}{2}, \tag{3.5}$$

$$V_{fb}^{-} = V_{L}^{[+1]} = V_{cm} - \frac{\alpha}{2} + \frac{V_{offset}}{2}. \tag{3.6}$$

Hence,

$$V_{fb} = V_{fb}^{+} - V_{fb}^{-} = \alpha - V_{offset}.$$

Similarly, when $y[n-1] = -1$,

$$V_{fb}^{+} = V_{H}^{[-1]} = V_{cm} - \frac{\alpha}{2} - \frac{V_{offset}}{2}, \tag{3.7}$$

$$V_{fb}^{-} = V_{L}^{[-1]} = V_{cm} + \frac{\alpha}{2} + \frac{V_{offset}}{2}. \tag{3.8}$$

Hence,

$$V_{fb} = V_{fb}^{+} - V_{fb}^{-} = -\alpha - V_{offset}.$$

Here, V_{cm} is the common mode of the feedback input. From equations (3.5) through (3.8), it can be seen that the differential voltage V_{fb} can be generated by selecting two of four DC bias voltages $V_{H}^{[+1]}$, $V_{L}^{[+1]}$, $V_{H}^{[-1]}$ and $V_{L}^{[-1]}$. This selection can be done with switches that are controlled by the comparator's output as illustrated in Figure 3.18.

The loop indicated with the red trace in Figure 3.18 is the time critical loop. The comparator used here is the clocked sense amplifier comparator. In order to reduce the latency of this feedback loop, the switched capacitor network is driven by the intermediate signals of the first stage of the comparator, the details of which will be discussed soon. A brief description of the working of the sense amplifier comparator, a necessary background for a discussion on the DFE circuit's implementation, will now be given.

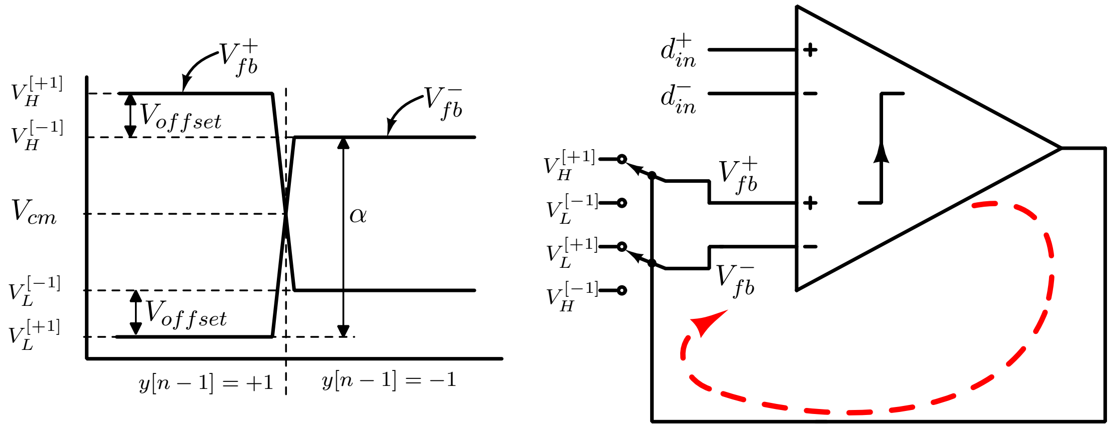


Figure 3.18: Conceptual block diagram of the DFE circuit with offset correction.

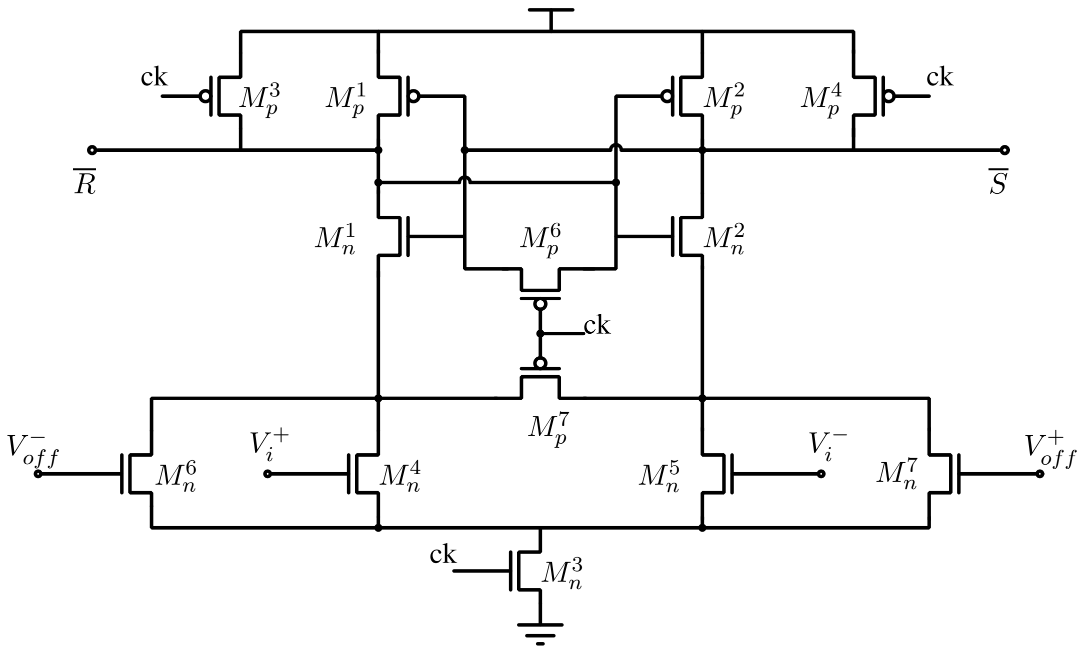


Figure 3.19: First stage of the sense amplifier comparator.

The sense amplifier comparator

Figure 3.15(b) shows the circuit diagram of the sense amplifier comparator. Transistors M_p^1, M_p^2 and M_n^1, M_n^2 form cross coupled inverters. Transistors M_n^4 and M_n^5 are the input transistors, while M_n^6 and M_n^7 are an additional input pair that can be used for offset correction or decision feedback. Transistors M_p^3 through M_p^6 are reset transistors. The operation of the circuit begins with a reset phase when the clock is low. This precharges nodes $\overline{S}$ and $\overline{R}$ to V_{DD}. The transistor M_n^3 is off during this cycle. When the clock goes from zero to one, the reset is released M_n^3 is turned on. The nodes $\overline{S}$ and $\overline{R}$ discharge through the input transistors and depending on the input, one of the nodes discharges faster than the other. When $\overline{S}$ and $\overline{R}$ are discharged below the trip point of the inverters formed by M_p^1, M_p^2 and M_n^1, M_n^2, the inverter positive feedback pulls $\overline{S}$ and $\overline{R}$ apart in the direction established by the input pair. Typical waveforms of $\overline{S}$ and $\overline{R}$ are shown in Figure 3.20.

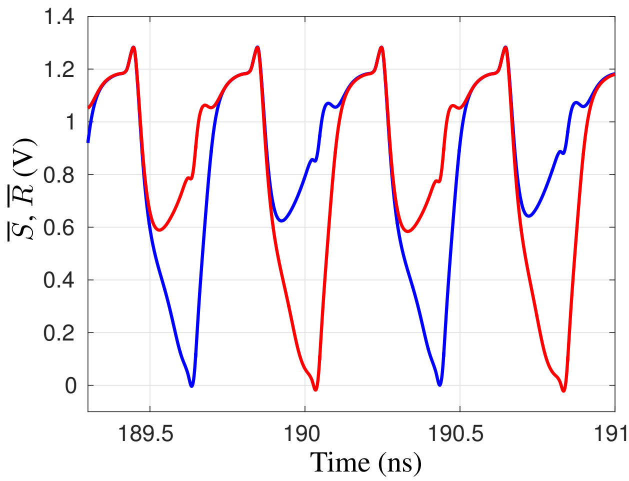


Figure 3.20: $\overline{S}$ and $\overline{R}$ signals of the first stage of the sense amplifier comparator.

These signals, which are dynamic due to the precharge in every cycle, are then forwarded to an SR latch to convert them to static digital data. A more detailed discussion on the analysis and design of this type of comparators is presented in Appendix A. In order to implement a DFE circuit with minimum latency, a switched capacitor circuit that uses the $\overline{S}$ and $\overline{R}$ signals of the comparator to drive a switched capacitor feedback network can be used. The next Subsection will discuss the implementation of such a low latency DFE.

3.2.2 Circuit implementation of switched capacitor DFE

In order to reduce the loop latency of the DFE circuit, the feedback is closed from the signals $\overline{S}$, $\overline{R}$ signals itself. $\overline{S}$ and $\overline{R}$ are used as select signals to select the required logic levels for performing equalization. Figure 3.21 shows the analog multiplexer circuit that selects the scaled logic value to be fed back to the auxiliary inputs of the comparator. Since the select transistors spend little time in the ON state before the precharge phase of the next clock cycle begins, the time available for the output to change states is limited. Low V_t transistors are used as the switches in order to improve the selector performance. When the comparator is reset, both $\overline{S}$ and $\overline{R}$ are at V_{DD}, which puts the multiplexer in a high impedance state. During input sample evaluation, one of $\overline{S}$ or $\overline{R}$ fall lower than the other, and the output of the analog multiplexer generates the scaled version of the resolved bit. This output is held dynamically on the parasitic

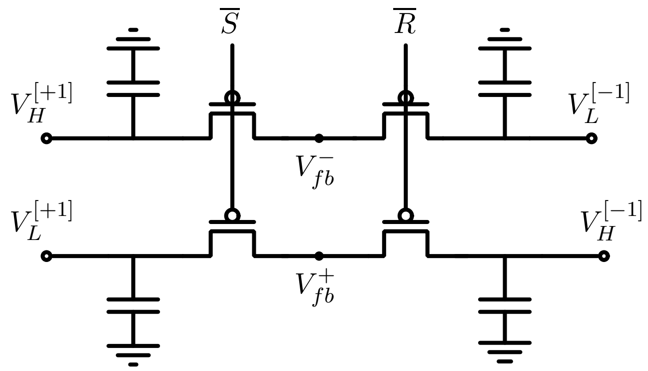


Figure 3.21: Feedback network using $\overline{S}$ and $\overline{R}$ signals as the select lines for an analog multiplexer. Low V_t transistors are used for the select switches.

capacitance of the node, as the comparator precharges for the next cycle. Hence, the next cycle evaluation subtracts the scaled previous bit value.

Since the common mode of $\overline{S}$ and $\overline{R}$ swings by large amounts, the Common Mode Rejection Ratio (CMRR) of the feedback network should be high. To ensure this, the first stage is modified to bias the auxiliary input pair with a tail current source. The bias to the tail current source is set to be same as the common mode of the data input to the comparator, so as to track their relative strengths. Figure 3.22 shows the circuit of the modified first stage of the comparator.

The circuit was designed and simulated in the standard UMC 130 nm CMOS technology. Since comparator circuits are sensitive to layout parasitics, post-layout simulations were used for the evaluation. A low swing transmitter with an interconnect interconnect that results in a completely closed eye at the receiver is used as the test bench. The feedback network is designed to attennuate by a factor of $\sim$2. Hence, the offset correction inputs and the equalizer feedback inputs are twice the desired values. Figure 3.23 shows the time domain waveforms of the data, the feedback signal and an effective input signal. The effective input is not a real signal, but is calculated numerically from the data and feedback signals by subtracting half the feedback signal from the data signal. This gives us an idea of the total input voltage seen by the comparator. Figure 3.24 shows the actual input signal eye diagram and the effective input signal eye diagram. As seen, the effective input eye is opened up, even when the actual input eye is closed. This eye diagram appears skewed after the samplng instant, which happens at the center of the eye. This is because from this instant the feedback input starts getting updated to the new value for the next cycle. As shown in Figure 3.25, the data is resolved correctly. From

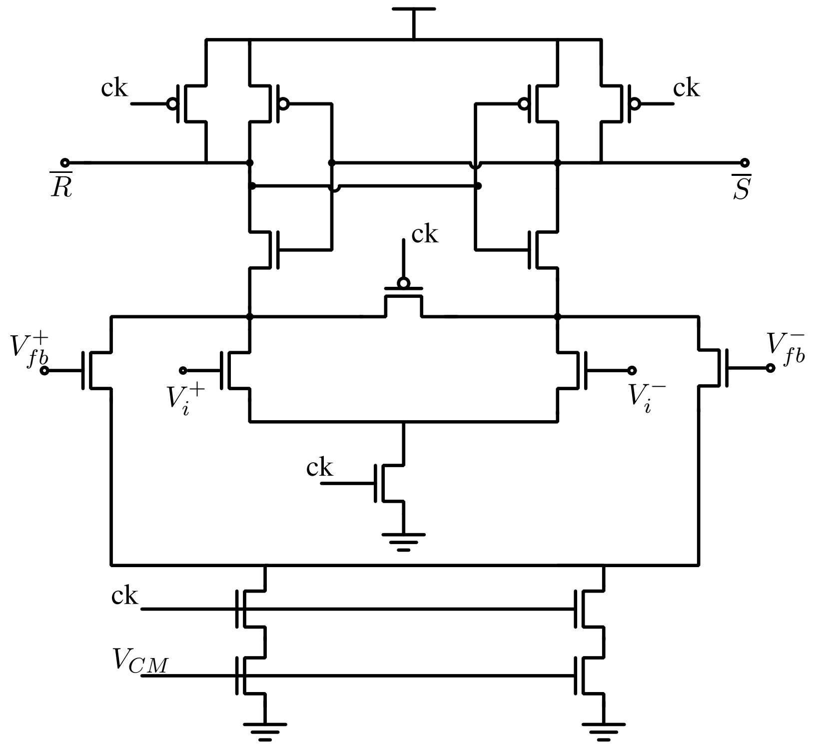


Figure 3.22: First stage of the sense amplifier comparator, with additional input pair for implementing DFE. The additional input pair is biased with a tail current source to improve its CMRR.

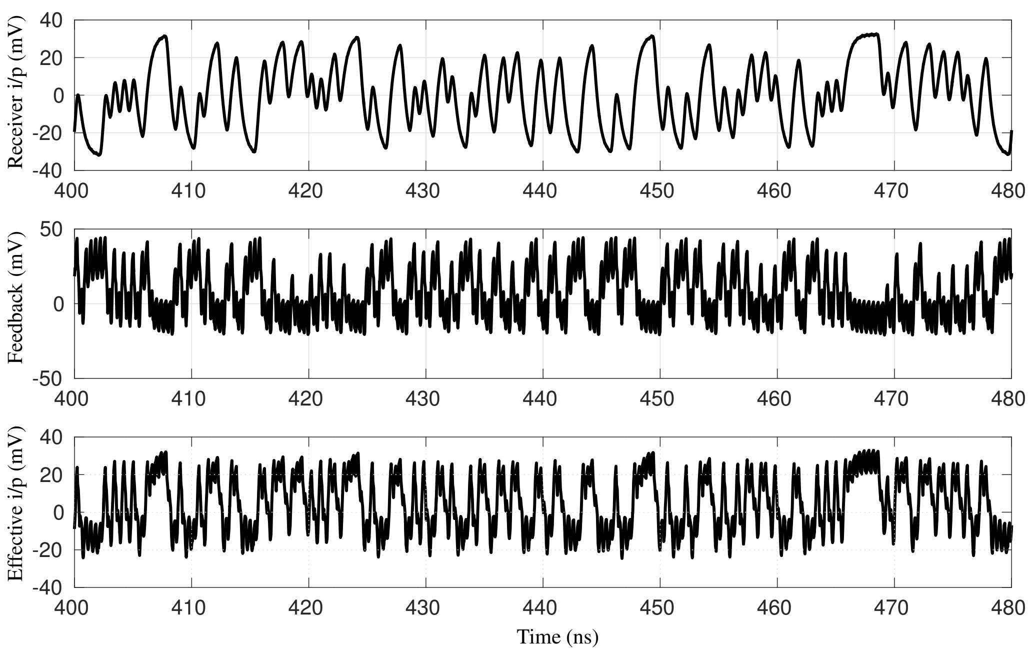


Figure 3.23: Simulated waveforms of the DFE circuit in the standard UMC 130 nm CMOS technology. From top to bottom, the signals are, the receiver input signal, the DFE circuits feedback signal and the numerically computed effective comparator input. Data rate = 2.5 Gbps.

simulations the loop delay was found to be $\sim$350 ps.

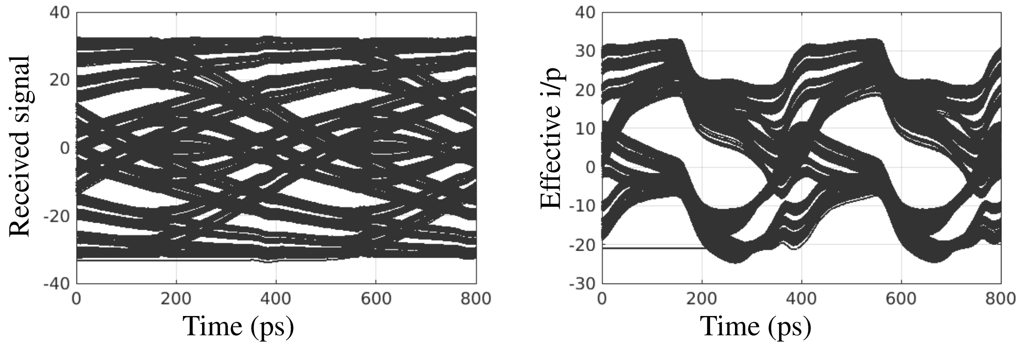


Figure 3.24: Eye diagrams of the receiver input voltage and the effective input signal seen by the comparator. Simulated in UMC 130 nm CMOS technology at 2.5 GHz.

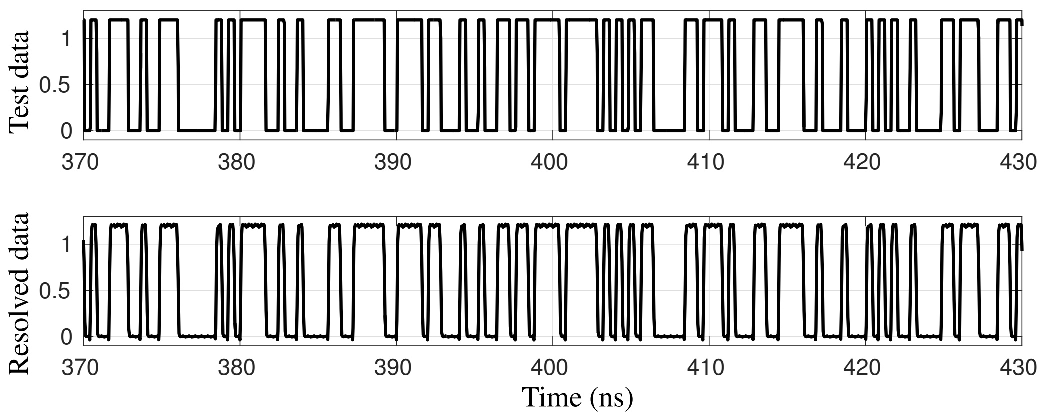


Figure 3.25: Top: Test input data for the interconnect with DFE at the receiver. Bottom: Recovered data after decision feedback equalizaton. Simulated in UMC 130 nm CMOS technology at 2.5 GHz.

3.2.3 Circuits for generating feedback voltages

The DFE circuit needs four feedback voltages, which are $V_H^{[+1]}$, $V_L^{[+1]}$, $V_H^{[-1]}$ and $V_L^{[-1]}$. These bias voltages can be generated using completely analog circuits or by using four Digital to Analog Converter (DAC)'s with digital inputs representing each of these bias voltages. Both these implementations are discussed in this Subsection.

Feedback bias generation using analog circuits

Figure 3.26 shows the circuit used to generate the feedback logic levels. Here, the current I_{bias} along with the resistance R, determines the differential output voltage between $V_H^{[+1]}$ and $V_L^{[-1]}$

(similarly between $V_H^{[-1]}$ and $V_L^{[+1]}$). This determines the magnitude of the feedback factor α (refer to Figure 3.18).

The two inputs V_{offset}^+ and V_{offset}^- set the common mode voltages of each of the two differential pairs ($V_H^{[+1]}$,$V_L^{[-1]}$) and ($V_H^{[-1]}$,$V_L^{[+1]}$) respectively. Together, this differential input (V_{offset}^+, V_{offset}^-), serves as the desired offset correction input. The capacitor C is a compensation capacitor. The opamp is a simple single stage differential input opamp.

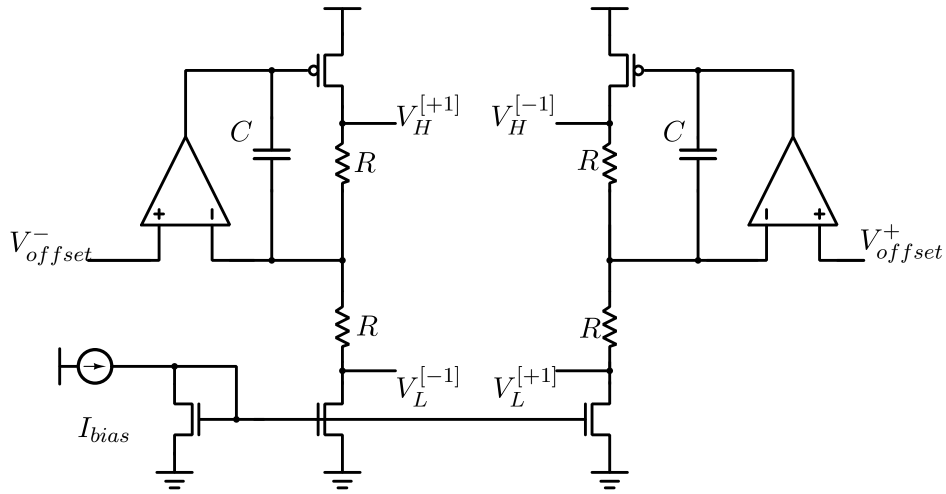


Figure 3.26: Circuit for generating the feedback logic levels using only analog circuits.

Feedback bias generation using DAC and digital logic

The bias voltages can also be generated using DAC's and some digital logic. A 5 bit resistive string DAC is used to generate the four bias voltages. The resistor string is driven by a current source is used to generate 32 levels of voltages, one of which is selected using a switch matrix. Figure 3.27 shows the circuit diagram of the resistor string used to generate multiple bias voltages. The generated bias voltages are centered around the common mode of the input of the comparator. Figure 3.28 shows the binary tree switch matrix constructed using transmission gate switches, which is used in the DAC. Four such switch matrices are used to generate 4 analog outputs using the same resistor string. The four switch matrix outputs are buffered with a single stage opamp. Four digital words, each 5 bit wide, are used to select appropriate bias voltages for the output. Two digital inputs, one for offset and another for the feedback tap weight are used to generate the required 4 digital words to drive the switch matrices. Of these, the offset control input is a 5 bit control word and the feedback factor is a 4 bit control word.

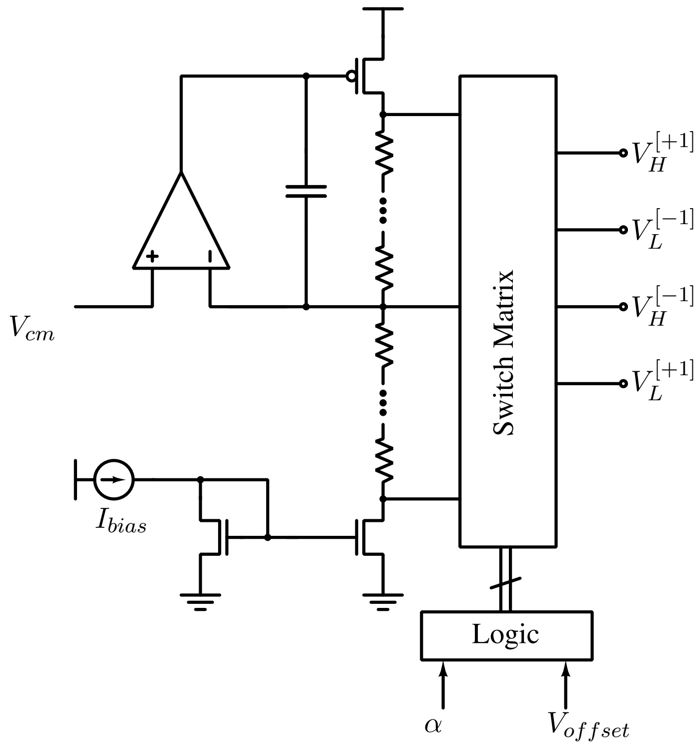


Figure 3.27: This circuit generates multiple voltages and the desired four bias voltages are chosen by the digital logic using the switch matrix.

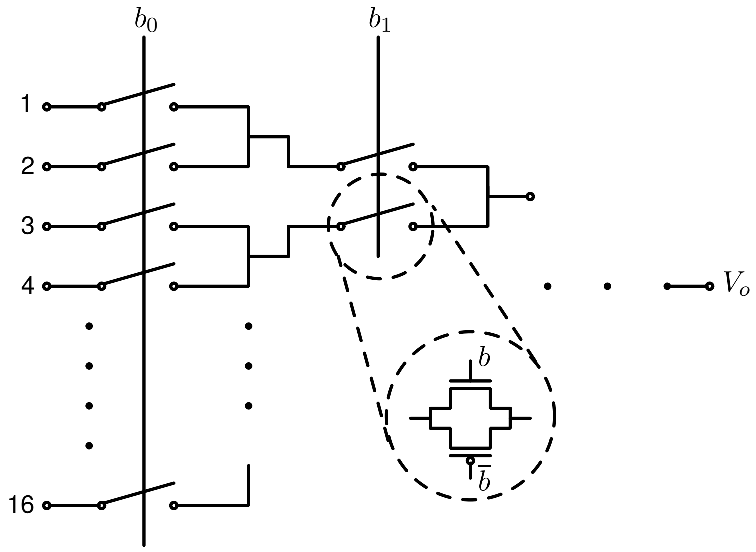


Figure 3.28: The circuit diagram of the switch matrix used in the DAC in Figure 3.27. Four such switch matrices are used to tap the resistor ladder to generate the required four bias voltages.

The required 4 digital words are calculated as

$$REG_{V_H^{[+1]}} = REG_{of} + \alpha, \qquad REG_{V_H^{[-1]}} = \text{“11111”} - REG_{of} + \alpha,$$

$$REG_{V_L^{[-1]}} = REG_{of} - \alpha, \qquad REG_{V_L^{[+1]}} = \text{“11111”} - REG_{of} - \alpha.$$

Here, REG_{of} is a 5 bit word that can take values from “01000” to “10111”, to select the offset input. α is a 4 bit word that can take values from “0000” to “1000” to chose the tap weight. This arrangement allows equal dynamic range for the offset and the feedback tap weights. Depending on expected offsets and desired tap weights, unequal splits can also be considered.

3.2.4 Implementation and results

The DFE circuit was implemented in the standard UMC 130 nm CMOS technology and fabricated. The DFE circuit was implemented using a resistive DAC to generate the bias voltages. A 5 bit DAC with a dynamic range of $\pm$25 mV was used. This allowed correction of offsets of up $\pm$12.5 mV and a feedback input voltage of $\pm$12.5 mV . A 10 mm differential interconnect with a 1 tap capacitive equalizer was used as the transmitter. The interconnect is in Metal 6 with a width of 0.6 μm and spacing of 0.35 μm. The parasitic resistance is 120 Ω/mm and its capacitance is 135 fF/mm. The swing on the interconnect was $\pm$30 mV. This implied a feedback factor of up to $\sim$ 0.2 is possible with this implementation. Figure 3.29 shows the circuit diagram of the main comparator with the transistor dimensions. Figure 3.30(a) shows a photograph of a

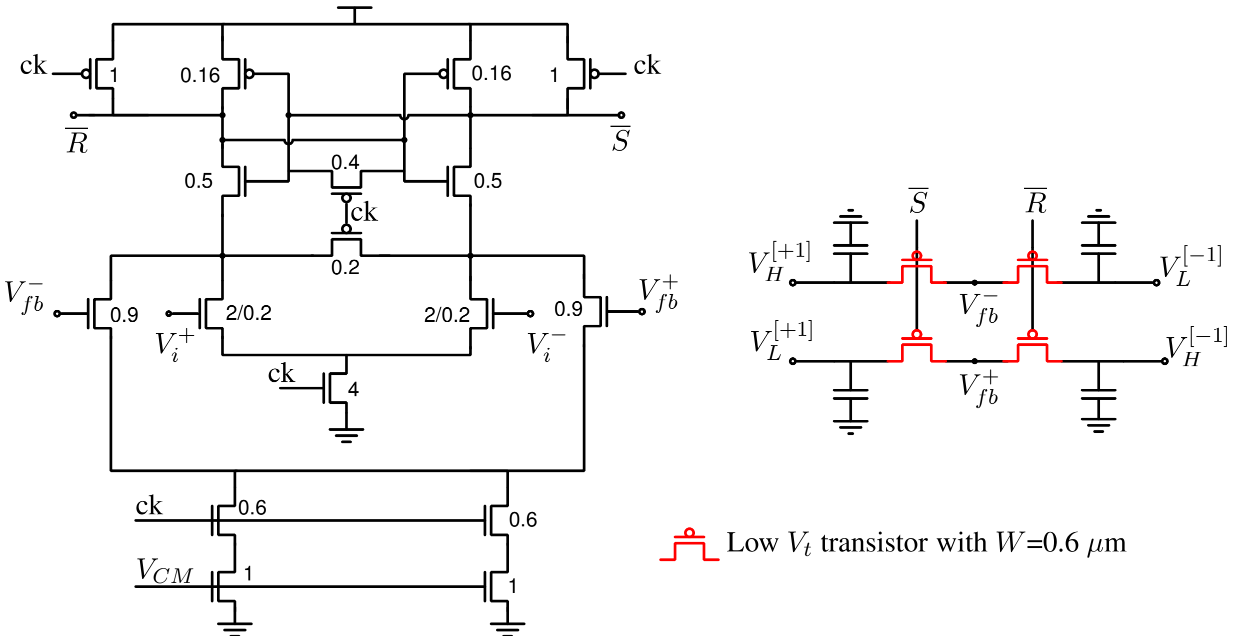
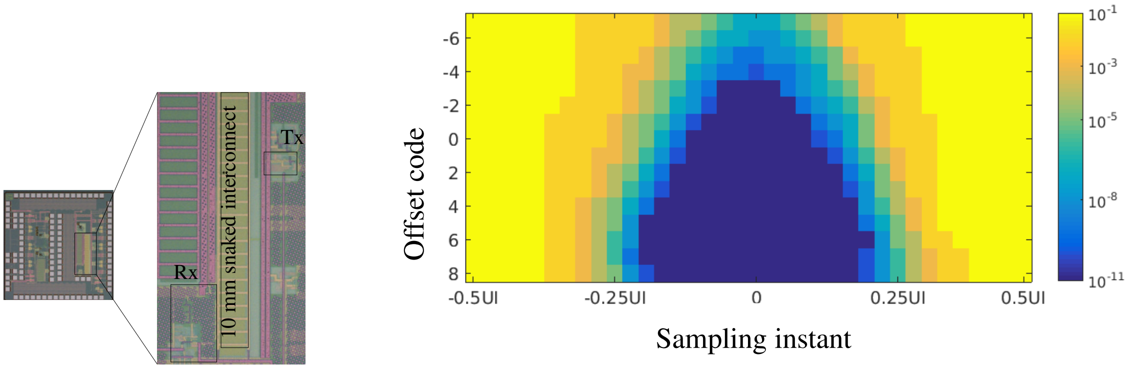


Figure 3.29: DFE circuit implemented in UMC 130 nm technology, with transistor dimensions. All dimensions are in μm. Unless explicitly mentioned, length of all transistors is minimum which is 120 nm.

bare die of the fabricated circuit.

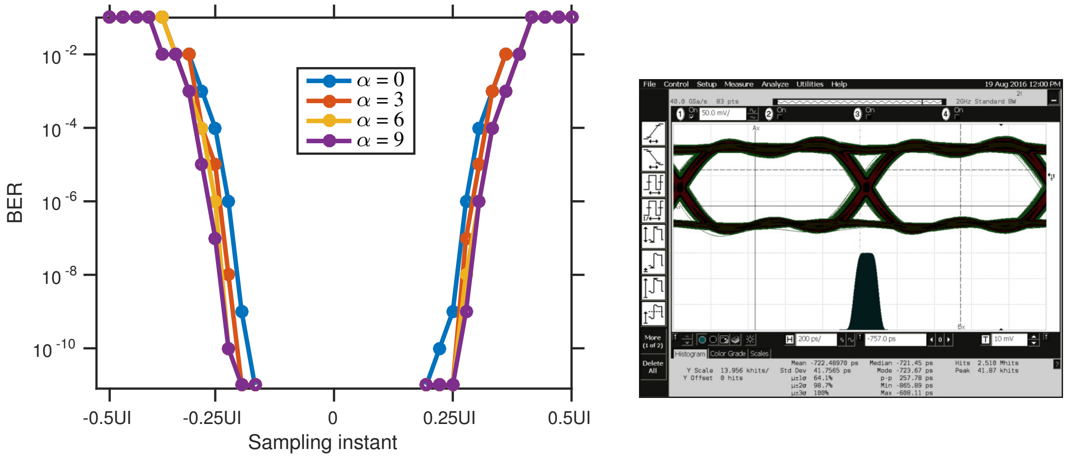
The circuit was tested at a frequency of 1 GHz with a supply of 1.2 V. First, the DFE feed-



(a) Die photograph. **(b)** BER as a function of sampling instant and offset code.

Figure 3.30: (a) Die photograph of the circuit fabricated in UMC 130 nm CMOS technology and (b) BER as a function of sampling instant and offset code. DFE is disabled for this measurement. Measured at a data rate of 1 Gbps.

back factor was set as zero and the offset was swept to find the code which showed the widest bath tub. Figure 3.30(b) shows the BER as a function of sampling instant and the offset code. After the offset code was found the DFE tap was increased and the bath tubs were measured again. Figure 3.31(a) shows the bath tub plots obtained for various values of the DFE feedback factor. The wider tubs in Figure 3.31(a) correspond to higher values of the DFE tap weight. The



(a) BER bath tub curves. **(b)** Eye diagram of recovered data.

Figure 3.31: Measured bath tub curves for a few values of the tap weight, and measured eye diagram of the recovered data. Technology: UMC 130 nm CMOS technology, Data rate: 1 Gbps.

eye opening increases by $\sim$15% for the maximum DFE tap weight. Figure3.31(b) shows the measured eye diagram of the recovered data.

3.2.5 Scalability

In order to verify the scalabilty of this circuit, the circuit was also designed and simulated in TSMC 65 nm CMOS technology. The circuit was laid out and simulation of layout extracted netlist was used for evaluation. A 10 mm differential interconnect in Metal 6 with a width of 0.43 μm and spacing of 0.5 μm is used which has a parasitic resistance of 88.5 Ω/mm and a capacitance of 189 fF/mm. Figure 3.32 shows the eye diagram of the data input and the effective input signal to the comparator with DFE. The input eye diagram is closed due to ISI produced by the interconnect. In addition a large offset of $\sim$50 mV is applied to the input. As seen in

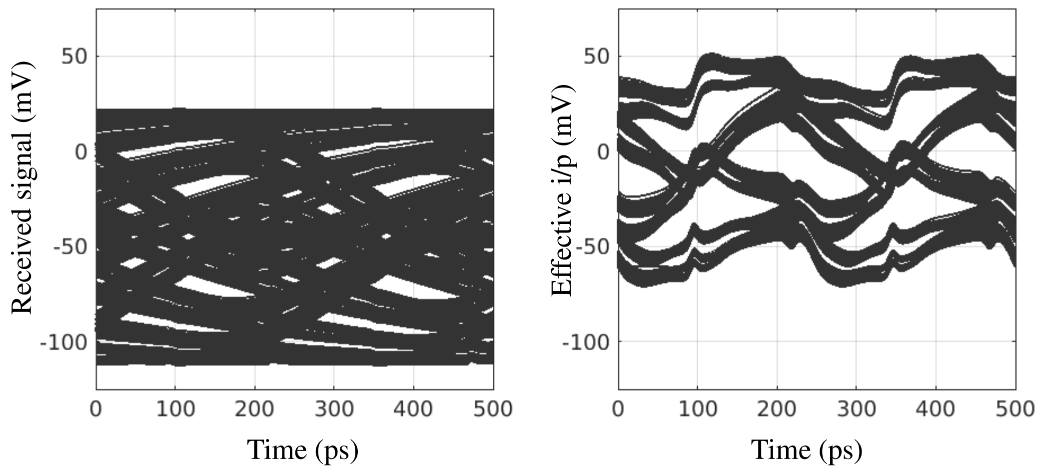


Figure 3.32: Eye diagrams of the receiver input voltage and the effective input signal seen by the comparator. Simulated in TSMC 65 nm CMOS technology at 4 Gbps.

Figure 3.32, the effective input of the circuit is open as the circuit compensates for the offset in addition to performing equalization. Correspondingly, the resolved data is also correct, as shown in Figure 3.33. From the layout extracted simulations, the loop delay in this technology was found to be $\sim$235 ps.

3.3 Summary

A low swing interconnect with capacitively coupled feed-forward equalization and a weak driver is presented in this chapter. The architecture is analyzed using small signal approximation. Insights from this analysis lead to design simplifications. A method of optimizing the design using worst case sequences has been reported, which allows quick and easy design of

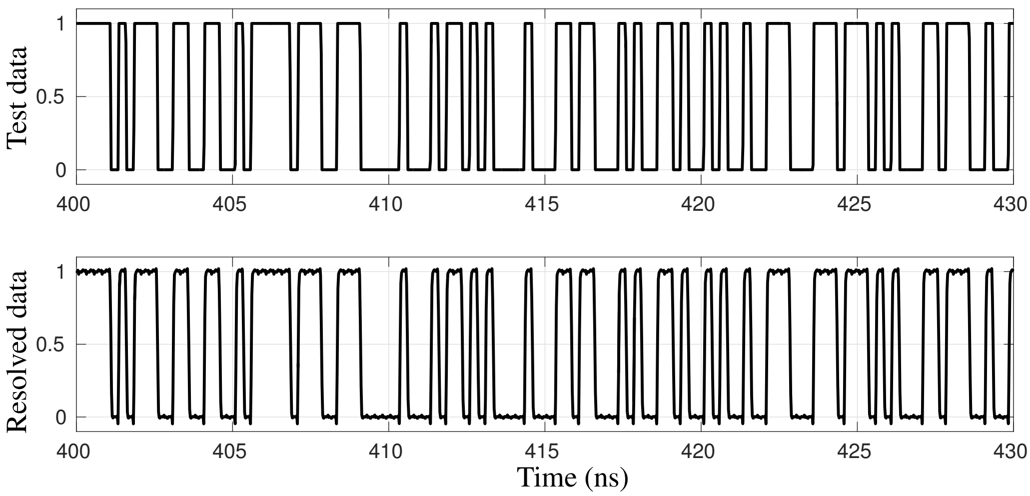


Figure 3.33: Top: Test input data for the interconnect with DFE at the receiver. Bottom: Recovered data after decision feedback equalizaton. Simulated in TSMC 65 nm CMOS technology at 4 Gbps.

the circuit parameters for a given interconnect. The circuit is designed and simulated with layout extracted netlists in UMC 90 nm CMOS technology. A comparator with reduced kickback noise is used as the receiver comparator. This circuit architecture is robust to process variations and is simple and power efficient when compared to current mode equalizers of the same order. The versatility of the reported design method and the scalability of the architecture have been verified by designing the circuit in different technologies, like UMC 130 nm CMOS technology, UMC 90 nm CMOS technology and TSMC 65 nm CMOS technology.

Subsequently, the chapter reports a low latency DFE circuit. The reported circuit is built around a sense amplifier with a switched capacitor feedback network. The switched capacitor circuit uses signals from the first stage of the sense amplifier comparator, thus resulting in low latency. In addition, offset correction can be done using this same feedback network. This avoids the need for extra offset correction transistors in the core of the comparator. A 5 bit DAC, along with a little logic circuitry, is used to generate the required four bias voltages. The circuit is designed, fabricated and tested in UMC 130 nm CMOS technology. To confirm the scalabilty of the circuit, it was also designed and simulated with layout extracted parasitics in TSMC 65 nm CMOS technology.

Chapter 4

Sampling clock retiming and clock domain transfer

Repeaterless interconnects have been extensively researched and many circuit architectures have been proposed that enable high speed data transmission over a repeaterless interconnect. These interconnect architectures were reviewed in Chapter 2. However, most of these works have not addressed the problem of synchronization of these interconnects. Due to the large latencies associated with repeaterless interconnects, the data arriving at the receiver is not synchronous with the receiver's clock. A suitable clock synchronizing circuit, that retimes the sampling clock to sample the data at the center of the data eye, is thus required.

Lee et al. in [10] report a circuit that uses a calibration routine to tune the phase of the receiver's sampling clock in training mode. As described in Section 2.4, the recovered clock has a residual phase quantization error and the technique cannot track environmental changes without interrupting normal operation. The latencies of the transmitter and receiver circuits change with temperature and supply voltage. The interconnect's latency is also not constant, as its resistance depends on temperature. Hence, environmental changes can result in loss of synchronization. Further, once the data is sampled correctly using a synchronized sampling clock, it should also be transferred to the receiver's clock domain using an appropriate synchronizer. Serial synchronizers are generally used for transferring data between mesochronous islands [12], which trade latency for the MTBF, and need additional handshaking signals.

In this chapter, we first establish the need for synchronization. We show that due to the process variations in modern CMOS technologies, the yield loss incurred without active clock synchronizing circuits is substantial. We then report a clock synchronizing circuit that provides a sampling clock positioned at the center of the data eye. The design includes a low latency clock domain transfer from the sampling clock domain to the receiver clock domain. The circuit was designed, fabricated in UMC 130 nm CMOS technology and tested. To confirm the scalability of the suggested scheme, the circuit was also designed and simulated along with layout extracted parasitics in TSMC 65 nm CMOS technology. The robustness of the circuit to high and low frequency jitter has been verified with circuit simulations as well as experimental measurements.

4.1 Need for clock retiming circuits

A low swing interconnect receiver should sample the incoming data at the center of the eye to ensure minimum BER. The necessary delay in the sampling clock to ensure optimum sampling can be inserted during the design phase. However, ensuring the sampling instant is optimum across process, temperature and voltage variations is difficult. In this section we estimate the yield loss incurred if active synchronizing circuits are not used for repeaterless interconnects. We use interconnects at 4 Gbps in UMC 90 nm CMOS technology and UMC 65 nm CMOS technology for this analysis. This choice was governed by the availability of interconnect physical parameters variability models.

Figure 4.1 shows the sketch of an eye diagram at the input of a receiver for a typical low swing interconnect. Typically, low swing interconnects have an available eye opening of about 40% of the bit period, which defines the noise margin [10, 11]. At a data rate of 4 Gbps this corresponds to a sampling point at $t_s \pm 50$ ps, where t_s is the optimum sampling point at the center of the eye. The standard deviation of the system clock in typical high performance microprocessors is about 6 ps [75]. The clock jitter is assumed to be random with a Gaussian distribution. If the sampling point is exactly at the center of the eye, the BER can be estimated by the number of occurrences of the clock outside the allowed window of $t_s \pm 50$ ps. The ratio (ρ) of peak deviation to standard deviation of the clock is $\frac{50}{6} = 8.33$. The BER can then be looked up from Table 4.1 [76]. From Table 4.1, the BER is $< 10^{-15}$. This is the BER when the sampling instant is nominally at the center of the data eye diagram, which is the case under typical process parameters or when an active synchronizer is used. This assumes that the clock jitter in the global grid and the local recovered sampling clock is of the same order.

In the absence of a clock synchronizer, the design of an interconnect system needs to

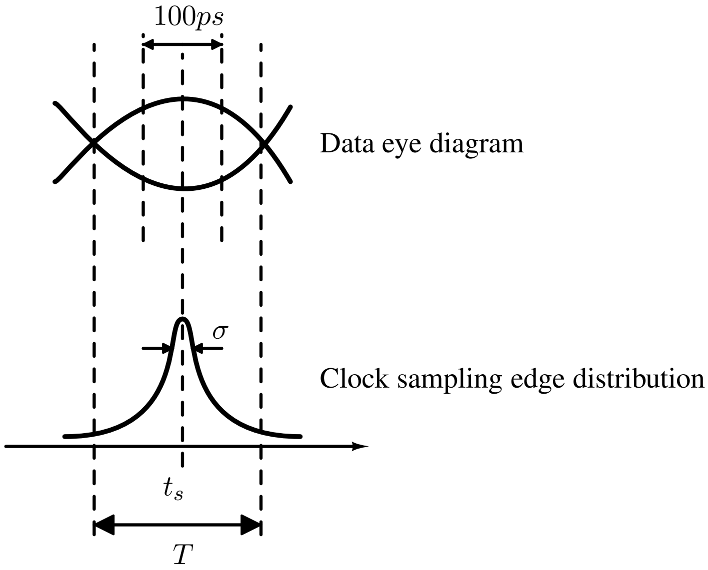


Figure 4.1: Illustration of typical eye diagram at receiver input for low swing interconnects. T is the clock period and σ is the standard deviation of the jitter in the clock.

Table 4.1: ρ to BER look-up table. *Source : K. Kundert [76]*

ρ	BER	ρ	BER
3.9	10^{-4}	6.4	10^{-10}
4.4	10^{-5}	6.8	10^{-11}
4.9	10^{-6}	7.1	10^{-12}
5.3	10^{-7}	7.4	10^{-13}
5.7	10^{-8}	7.7	10^{-14}
6.1	10^{-9}	8.0	10^{-15}

estimate the interconnect delay using parasitic extraction from layout. However manufacturing variabilities will affect the accuracy of the clock sampling instant. Long interconnects in scaled technologies often have multi-cycle delays. In order to estimate the variability in the latency of long interconnects, we laid out 10 mm interconnects in UMC 90 nm CMOS technology and UMC 65 nm CMOS technology, and extracted the parasitics across the interconnect process corners. Table 4.2 shows simulated delay for a few metal layers.

As seen from Table 4.2 the delay of the interconnect can vary by as much as 50 ps. The variation in the higher metal layers is less then that of the lower ones. Timing uncertainty due

Table 4.2: Delay of 10 mm interconnect for a few metal layers across interconnect process corners

Technology	UMC 90 nm	UMC 90 nm	UMC 90 nm	UMC 65 nm
Metal level	ME8	ME7	ME6	ME8
Thickness (μm)	0.5	0.5	0.25	0.36
T_{max} (ps)	278	293	425	362
T_{min} (ps)	271	278	373	312
Δ = ($T_{max} - T_{min}$)(ps)	7	15	52	50

to process variation of clock buffers, voltage variation and temperature variation will add to this uncertainty. Assuming that the worst case variation of the interconnect delay is the 6σ value, the standard deviation of the interconnect delay as 8.33 ps. Hence, 68% of the fabricated chips will have the interconnect delay variation of $< \pm$ 8.33 ps. For a delay variation of 8.33 ps, the BER is $< 10^{-11}$. Thus, for the chips with a delay variation $< 1\sigma$ the BER is $< 10^{-11}$. Similarly, for the chips with a delay variation between 1σ and 2σ the BER is $< 10^{-7}$. Table 4.3 shows the percentage of chips and their predicted BER.

Table 4.3: Delay variation and estimated BER

σ	ρ	Percentage of chips	Worst case BER
1σ	6.9	68%	$< 10^{-11}$
1σ to 2σ	5.5	27%	$< 10^{-7}$
2σ to 3σ	4.2	4.8%	$< 10^{-4}$

Hence as seen from Table 4.3, without active synchronizing circuits, chips designed with repeater-less long on chip interconnects will have a poor yield, despite the rather optimistic calculations shown, which ignores the effects of transistor process corners, voltage and temperature variation.

4.2 Sampling clock recovery

In order to ensure that the sampling clock samples the incoming data at the center of the eye diagram, a clock retiming circuit is required. Clock and data recovery circuits are well known for off-chip interconnects [56]. These circuits use phase detectors that sense the phase difference between the clock and the data and produce an error signal. The error signal is then integrated and used to control a local oscillator's clock frequency. The negative feedback loop converges when the recovered clock's frequency is equal to the incoming data frequency and it's active edge is at the center of the data eye. For on chip interconnects however, a clock of the correct frequency is available at the receiver and only the phase has to be recovered.

Figure 4.2 shows the functional block diagram of a phase synchronizer for a repeaterless link. Here, ϕ_{Tx} is the transmitter clock, ϕ_{Rx} is the receiver clock, and ϕ_d is the desired sampling clock with its active edge at the center of the data eye. The delay of the repeaterless interconnect, that must be compensated for, is expressed as $(n + \alpha) \times T$, where T is the system clock period, n is a positive integer (including 0), and α is a positive real number which is less than 1. At the

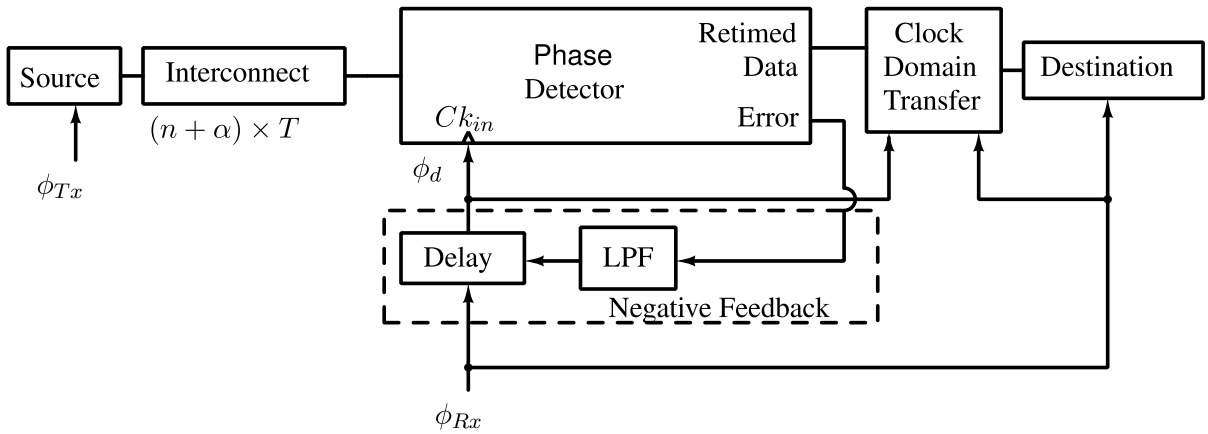


Figure 4.2: Conceptual block diagram of the clock recovery system at the receiver of a low swing interconnect system. ϕ_{Tx}: Transmitter clock, ϕ_{Rx}: Receiver clock, ϕ_d: retimed sampling clock, ($(n + \alpha) \times T$): Repeaterless interconnect delay, where $n \in \mathbb{Z}^{\geq 0}$, $\alpha \in \mathbb{R}$ & $\alpha < 1$, T: system clock period, LPF - Low pass filter.

receiver a phase detector senses the phase error between the data and the clock and generates an error signal. This could be one of the Alexander phase detector [77] or the Hogge phase detector [78] or the Meghelli phase detector [79] etc. The error signal is integrated using a Low Pass Filter (LPF) and used to control a delay circuit that delays the receiver clock ϕ_{Rx} to generate the sampling clock ϕ_d. The negative feedback minimizes the error, resulting in the sampling clock position at the center of the eye. Once the phase detector samples the data and resolves it, the resolved data must be transferred to the receiver's clock domain using an appropriate synchronizer, which is denoted by the 'Clock Domain Transfer' block. The implementation of the system shown in Figure 4.2 can be done in two ways in which the sampling clock is controlled in discrete steps or continuously. Both these implementations will be discussed in this section.

4.2.1 Clock synchronizer with discrete phase control

Figure 4.3 shows the block diagram of the clock synchronizer with discrete phase control. In

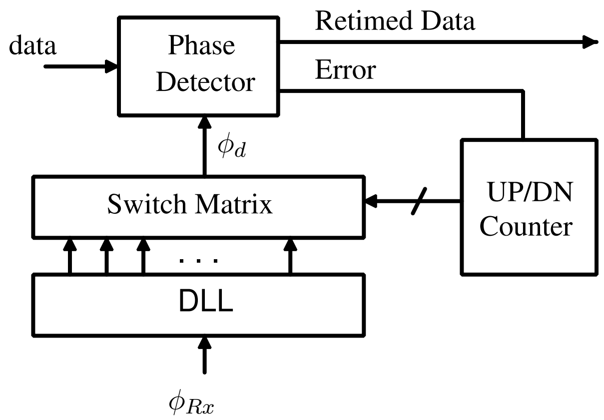


Figure 4.3: Block diagram of the discrete phase clock synchronizer.

this circuit, a DLL generates multiple phases of the clock, of which the one closest to the center of the eye will be chosen as the sampling clock. The recovered clock phase will therefore be quantized to the DLL phases. A bang-bang phase detector is used to generate an $UP/\overline{DN}$ signal, that drives the $UP/\overline{DN}$ input of a one-hot ring counter, which is shown in Figure 4.4. The ring counter size is equal to the number of phases of the DLL. This ring counter is driven

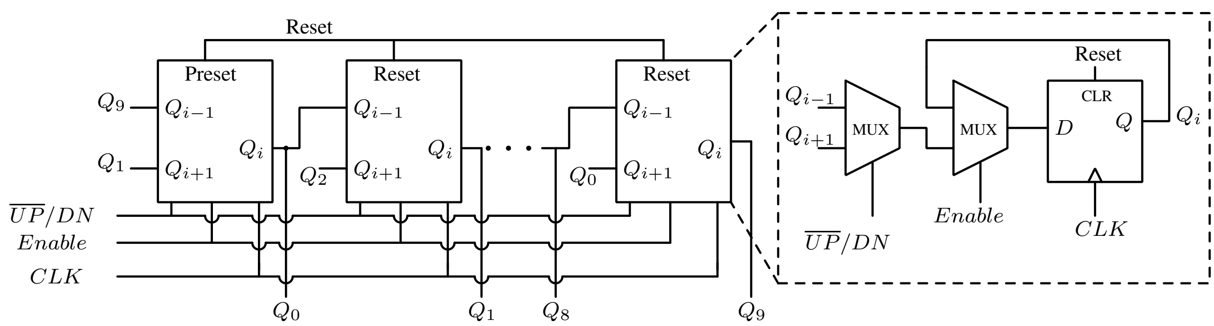
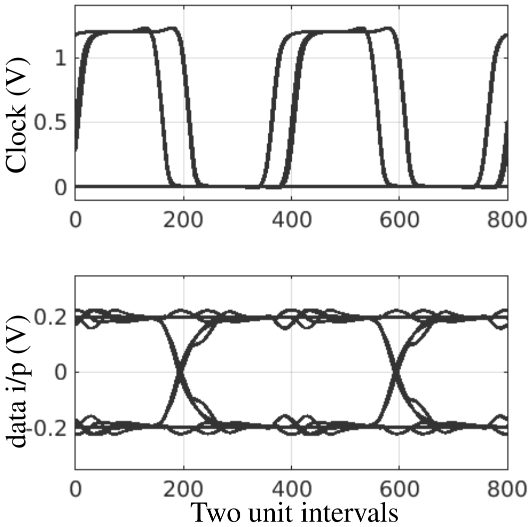


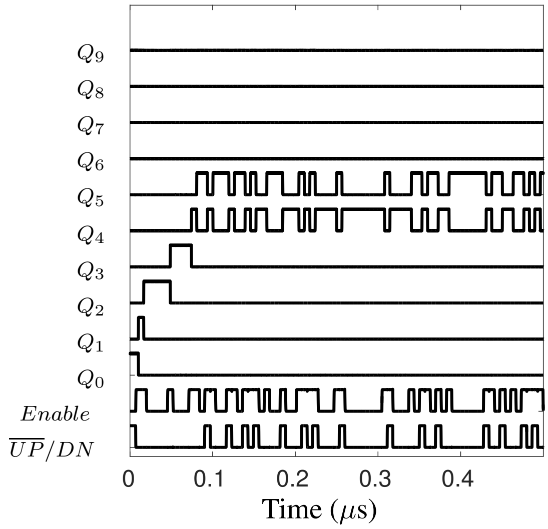
Figure 4.4: One hot ring counter.

by a divided clock to reduce the power consumption. A division factor of 16 was used for simulation. The output of the ring counter drives the switch matrix to chose the corresponding phase number of the DLL as the sampling clock.

Figure 4.5(a) shows the eye diagram of the data and the clock signal after lock has been achieved. As can be seen the clock phases are quantized. Figure 4.5(b) shows the ring counter status and its control inputs $UP/\overline{DN}$ and $Enable$.



(a) Eye diagram of data and clock.



(b) Ring counter state.

Figure 4.5: (a) Eye diagram of data and clock of the discrete phase clock synchronizer after lock has been achieved. (b) Ring counter outputs from start-up to lock of the discrete phase clock synchronizer.

To reduce the startup time of the link, the status of the ring counter can be saved and preloaded at every subsequent power up. This enables quick restart of the interconnect, after going into sleep mode. The design being predominantly digital, is simple, scalable and has low power consumption. The main limitation of this system is the phase quantization error and the phase noise in the recovered clock. The phase quantization can be reduced by using a DLL with more phases. However DLL's with large number of phases are difficult to construct and come with a power overhead. Even if a 10 phase DLL is used, which can be constructed with a reasonable power budget, the accuracy of the recovered phase is 10% of the bit period. From the approximate analysis discussed in Section 4.1, a 10% offset in the sampling instant can degrade the BER from 10^{-15} to 10^{-11}. The phase noise of the system can be improved by using a digital low pass filter to integrate the $\overline{UP}/DN$ signal before giving it to the ring counter. This integrator will have to operate at the data rate of the interconnect. Digital integrators designed for such high input frequencies and large time constants are very power hungry. Due to these reasons, this system puts increased demands on the sensitivity of the comparator or the minimum eye opening of the interconnect system, both leading to increased power consumption. The next subsection will discuss an alternate implementation which solves the phase quantization problem without offsetting the other advantages of the digital implementation.

4.2.2 Clock synchronizer with continuous phase control

The phase quantization error can be eliminated by using an analog Voltage Controlled Delay Line (VCDL) to generate the delayed sampling clock ϕ_d. However, in order to retain the advantages of low power consumption and quick startup time of the discrete phase synchronizer discussed, we use a synchronizing circuit that uses coarse digital correction and fine analog correction. Figure 4.6 shows the block diagram of the clock synchronizing circuit that employs a fine analog control using a VCDL, over the coarse correction quantized to the DLL phases. The system thus comprises of two loops, which are the coarse tuning loop and the fine tuning loop, as demarcated in Figure 4.6. The heart of the coarse tuning loop is the DLL that generates multiple phases of the clock. The phase closest to the center of the data eye is picked by this loop. The picked phase is delayed by the VCDL, which is the heart of the fine tuning loop. The fine tuning loop positions the sampling clock at the exact center of the data eye. Since the system is of the first order, the loop filter is a single capacitor, as shown between the fine and coarse tuning loops in Figure 4.6.

In this circuit, the phase detector's error output is integrated with a charge pump. The integrated voltage acts as the control voltage V_c for the VCDL. This forms the fine tuning loop. The circuit operation starts with this fine tuning loop trying to get the sampling clock ϕ_d to the center of the eye. If the VCDL range is not sufficient to achieve lock, its control voltage will exceed the preset bounds. This is detected by the coarse tuning loop and it responds by picking the next DLL phase as the source clock for the fine tuning loop. It also resets the control voltage to lie within the preset bounds. This process repeats until the phase closest to the center of the eye is selected and the VCDL range is sufficient to lock the clock to the exact center of the eye. Referring to Figure 4.6, the input low swing data first goes to a phase detector. The UP and DN pulses from the phase detector are averaged using a weak charge pump to produce the control voltage (V_c). The control voltage then modulates the delay of the VCDL. The VCDL source clock comes from the coarse tuning loop, and is one of the phases of the DLL. Convergence of the loop is assured when the VCDL is designed to have a range greater than 1 phase step of the DLL.

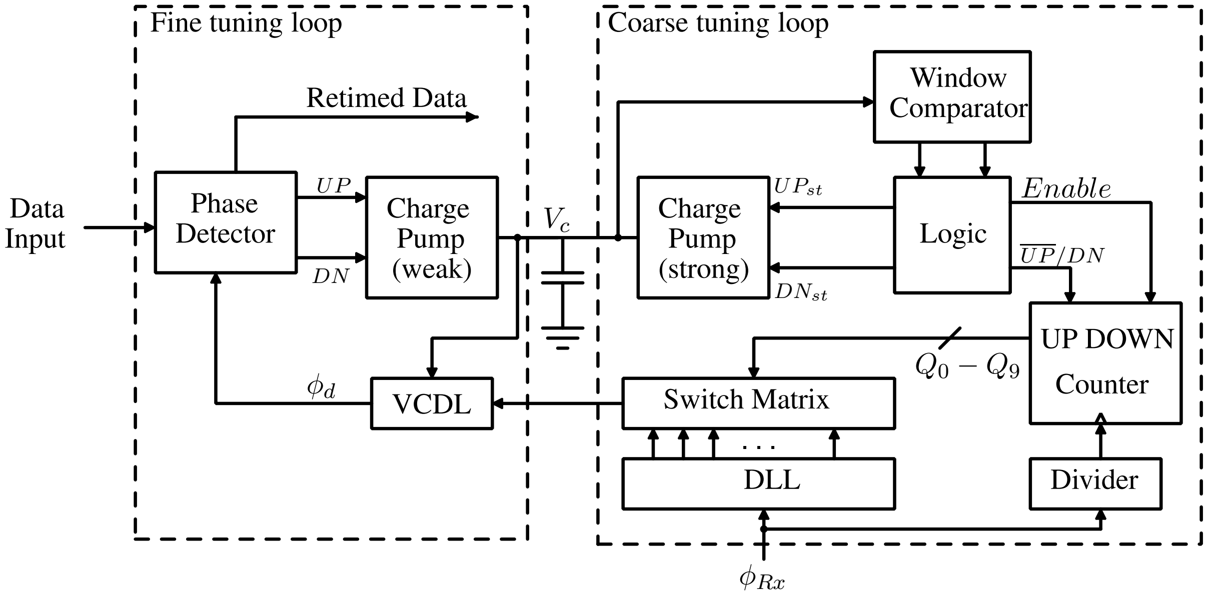
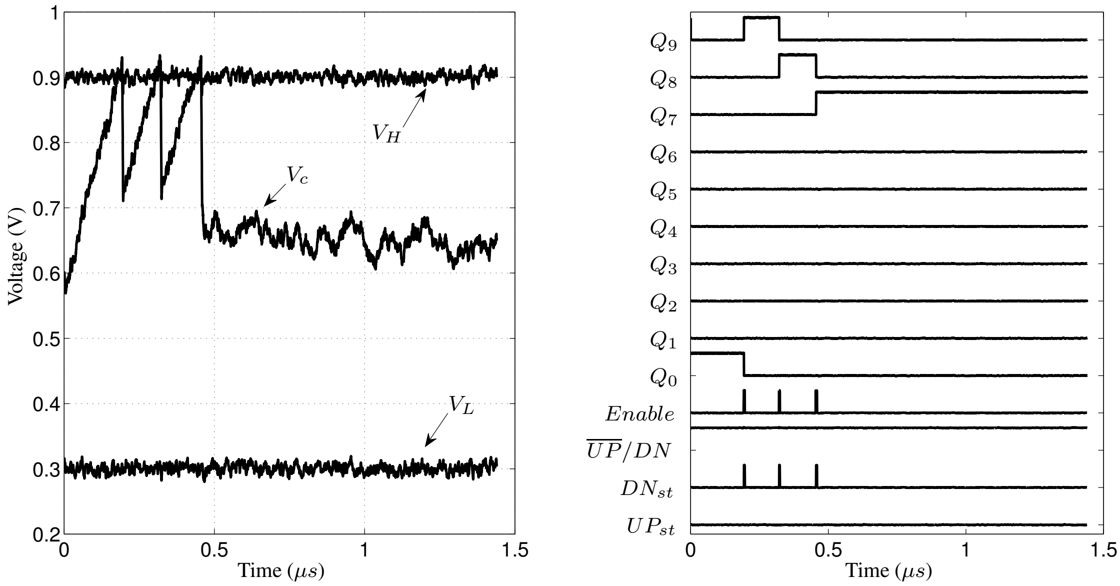


Figure 4.6: Block diagram of the continuous phase clock synchronizer system, divided into fine tuning and coarse tuning loops. VCDL - voltage controlled delay line, Vc - control voltage.

The coarse tuning loop uses a window comparator to monitor the control voltage. The comparator thresholds define the preset range of the control voltage for the VCDL. When the fine tuning loop is unable to achieve lock with the control voltage within the window comparator thresholds, the window comparator detects this and triggers a coarse phase correction through the Logic block. The logic block is a Finite State Machine (FSM) that generates the $Enable$ and $\overline{UP}/DN$ signals for the one hot ring counter. As in the case of the discrete phase clock synchronizer, the ring counter's state is used to select one of the phases of the clocks generated by the DLL. The direction in which the ring counter counts depends on whether the control voltage exceeds the upper threshold or is below the lower threshold of the window comparator. In addition to picking the next DLL phase, the Logic block also generates UP_{st} and DN_{st} signals for the secondary strong charge pump, which is used to reset the control voltage (V_c) to bring it within the window. When the control voltage is within the window thresholds, the digital circuits retain their state by de-asserting the $Enable$ signal. Figure 4.7(a) and (b) show the trajectory of the control voltage and the progression of the ring counter from startup to lock condition. Figure 4.8 shows the simulated eye diagram of the data and the clock after lock has been achieved.

Figure 4.9 shows the diagram of the window comparator and the logic which controls the strong charge pump. The comparators used in this circuit are traditional static comparators without a clock. The comparators trip when the control voltage crosses the preset thresholds.



(a) Control voltage. **(b)** Ring counter state.

Figure 4.7: Control signals of the continuous phase clock synchronizer obtained from layout extracted simulations. 5% noise was added to the supply for this simulation.

The clock for the flip-flops in this circuit is obtained by dividing the system clock by a factor K. The clock is divided to make the speed acceptable for the digital logic and the strong charge pump, in addition to saving power. The ring counter is enabled only if any one of the comparator's outputs is high. This happens only when the control voltage is not within the allowed range. The $\overline{UP}/DN$ signal for the ring counter, and UP_{st} & DN_{st} signals for the strong charge pump are derived from the comparator outputs. When V_c is more than the upper threshold (V_H), $\overline{UP}/DN$ is high. At this time the upper flip-flop is enabled. On the next clock after this event, the ring counter counts down and in order to bring V_c within the window, the DN_{st} signal for the strong charge pump is asserted. Once the loop filter capacitor discharges sufficiently so that V_c is within the window comparator thresholds, the comparator outputs go low. This resets the flip-flops and de-asserts $Enable$. Depending on the chosen clock division ratio K, the strong charge pump is designed such that this exercise is completed in one cycle, taking into consideration the time taken by the comparators to resolve. Similar process follows when V_c is less than the lower threshold (V_L), when the ring counter counts up.

The status of the coarse tuning loop which is the state of the ring counter can be captured in a snapshot register and used to initialize the counter just like the discrete phase clock synchronizer. Thus, this system eliminates the phase quantization error without compromising on

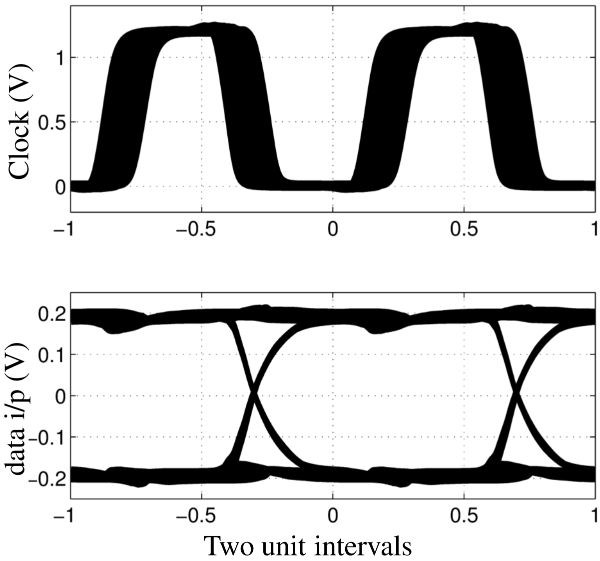


Figure 4.8: Eye diagram of the data and the recovered clock obtained from layout extracted simulations. The jitter in the recovered clock is mainly due to the 5% noise added to the supply for this simulation. Technology: CMOS 130 nm, Data rate: 2.5 Gbps, V_{DD} = 1.2 V.

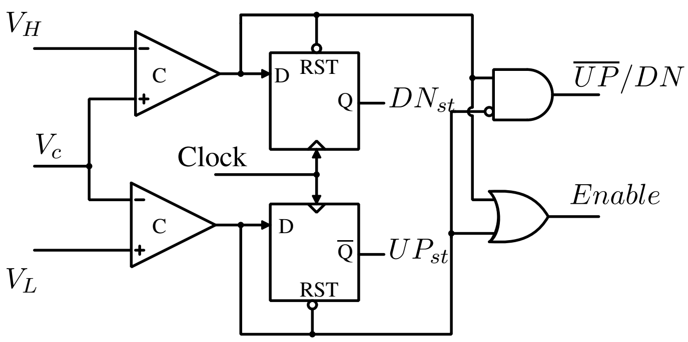


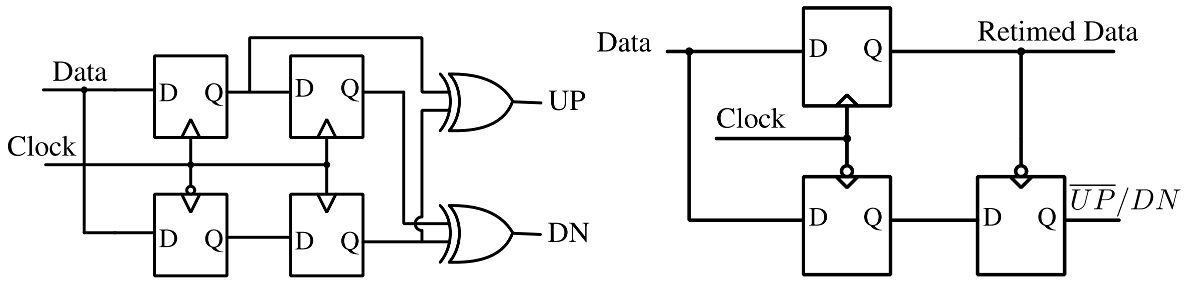
Figure 4.9: Control logic for generating $\overline{UP}/DN$ and $Enable$ signals for the ring counter and UP_{st} & DN_{st} signals for the strong charge pump. V_H, V_L are the upper and lower thresholds of window comparator respectively.

the other advantages of the discrete phase implementation.

4.2.3 Choice of phase detectors

Phase detectors that sense the phase difference between clock and data can be broadly classified into linear and bang-bang type phase detectors. In principle any phase detector can be used in the synchronizers described in this chapter. However, linear phase detectors like the Hogge phase detector [78] use the data signal as the clock input internally. This requires the data to be a large signal with good rise time. Amplifying the low swing data input to sufficient amplitudes consumes considerable power. Hence bang-bang phase detectors are more suitable for clock recovery circuits with low swing data inputs.

The Alexander and Meghelli phase detectors are shown in Figure 4.10 (a) and (b) respectively. In the Alexander phase detector when there is no transition in the data, the phase detector de-asserts both UP and DN signals. Thus, no charge is dumped on the loop filter capacitor. The Megheli phase detector uses fewer components, however it has only one output which is $UP/\overline{DN}$ and it does not have a high impedance output state. Hence, the phase detector always asserts either UP or DN signal even when there is no transition in the data. Hence, the noise on the control voltage is higher when the Megheli phase detector is used, which translates to a higher jitter.



(a) Alexander phase detector.

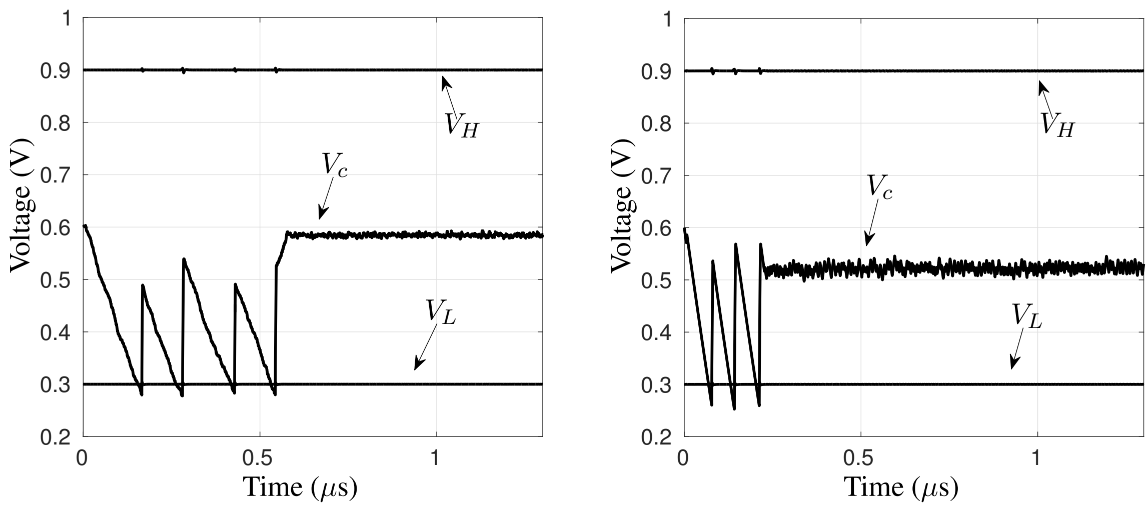
(b) Meghelli phase detector.

Figure 4.10: Circuit diagrams of the Alexander and Meghelli phase detectors.

Figures 4.11 (a) and (b) show the control voltage trajectories of the continuous phase clock synchronizer with the Alexander and Meghelli phase detectors respectively. The standard deviation of the control voltage after lock is achieved is 2.6 mV when the Alexander phase detector is used and 7.1 mV when the Meghelli phase detector is used. Figure 4.12 show the eye diagrams of the recovered clock for each of the two phase detectors. As expected, the jitter in the recovered clock is higher when the Megheli phase detector is used. The peak to peak jitter in the recovered clock is 4 ps when the Alexander phase detector is used, and 8 ps when the Megheli phase detector is used.

4.3 Clock domain transfer

Once the data has been sampled with a synchronized clock and converted to CMOS levels, it should be transferred to the receiver's clock domain with an appropriate synchronizer. Se-



(a) Control voltage with Alexander phase detector. **(b)** Control voltage with Meghelli phase detector.

Figure 4.11: Control voltage trajectory with two types of bang-bang phase detectors.

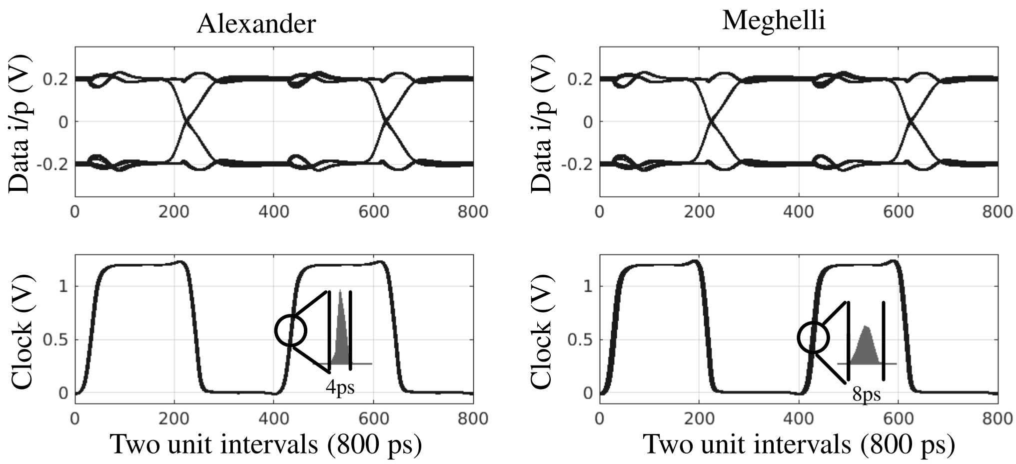


Figure 4.12: Eye diagrams of the recovered clock with two types of bang-bang phase detectors.

rial synchronizers with multiple flip-flops that are used for such clock domain crossings trade latency for MTBF. In the system discussed here the clock domain transfer can be performed with a single flip-flop using the multiple clock phases available. This flip-flop is clocked with a phase intermediate to the sampling and receiver clock phases. Figure 4.13 shows the serial synchronizer along with the clock domain transfer flip-flop. Here the DLL generates N phases of the clock. The clock recovery system selects the phase closest to the center of the data (ϕ_n) and delays it appropriately, to generate ϕ_d such that it is positioned at the center of the eye. The retimed data from the phase detector is available at this phase ϕ_d. The receiver samples the data at the receiver clock which is ϕ_{Rx} (which is the first phase of the DLL). The flip-flop

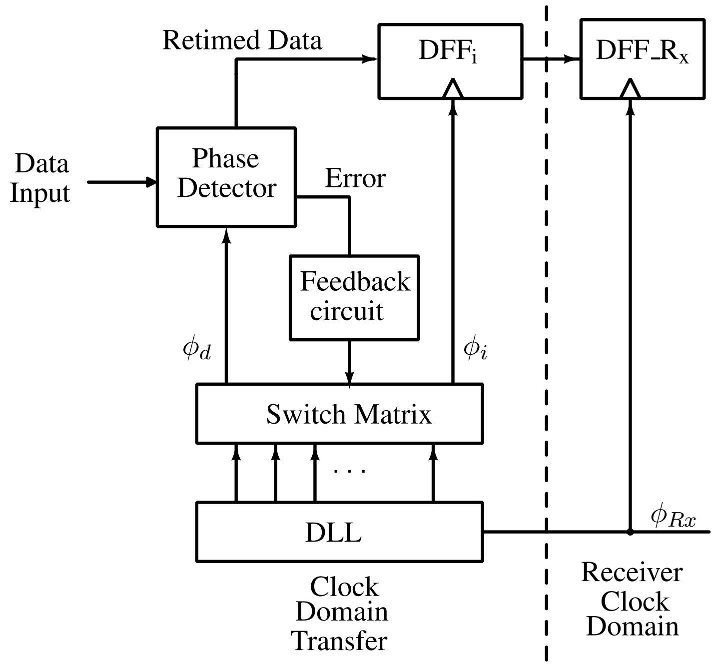


Figure 4.13: Clock domain transfer from sampling clock ϕ_d to receiver clock ϕ_{Rx}.ϕ_i - clock of intermediate phase between ϕ_d and ϕ_{Rx}.

DFF_i should be clocked with a phase that will guarantee that no flip-flop samples during a data transition. ϕ_i is selected as

$$\begin{aligned} \phi_i &= \phi_{(n+2-\frac{N}{2})} && \text{if } n+2 > \frac{N}{2} \\ &= \phi_{Rx} && \text{otherwise.} \end{aligned}$$

Figure 4.14 shows the sampling clocks for the worst case latency, for an example case that uses an 8 phase DLL ($N = 8$). One can see that $\phi_{(n+2-\frac{N}{2})} \sim \overline{\phi_{Rx}}$. Using $\overline{\phi_{Rx}}$ simplifies the implementation in two ways. First, it reduces the load on the DLL and switch matrix. In addition, it also compensates for the delay introduced by the clock buffers from switch matrix output to the phase detector's clock input, with the delay introduced in generating $\overline{\phi_{Rx}}$ from ϕ_{Rx}.

Here, an assumption that a flip-flop sampling a valid data input is able to resolve its output in time $\frac{T}{2} - T_{setup}$ is made. Since pipelined microprocessors generally have a logic depth of more than 5 NAND gates, this assumption is not very demanding. The reason for choosing $\phi_i = \phi_{(n+2-\frac{N}{2})}$ for $n+2 > \frac{N}{2}$ comes from the fact that the fine tuning loop of the clock recovery system could delay the sampling clock by at most 2 phase steps of the DLL. This is because the VCDL is over-designed for two phase steps so as to meet the range requirements across process

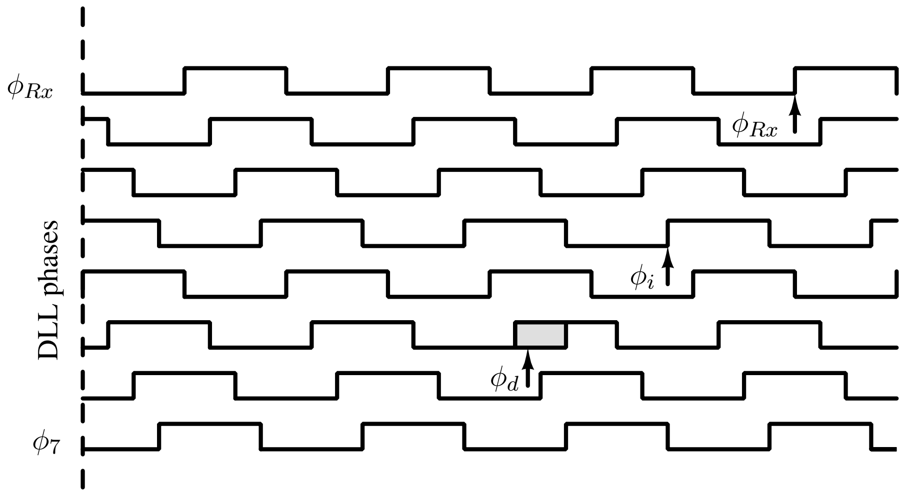


Figure 4.14: Illustration of the timing diagram of continuous phase control clock recovery system with clock domain transfer, under the worst case latency.

corners. This is explained later in section 4.5 when discussing the VCDL implementation.

The total latency of the above synchronizer is at most 3 clock cycles. The phase detector takes 2 clock cycles (as will be explained in section 4.5) and the following DFF's output is sampled half a clock cycle later. Under the worst conditions the third flip-flop in the chain will introduce another half a clock cycle delay. This makes the total latency $\leq 3T$.

Clock domain transfer for clock synchronizer using discrete phase control

The discrete phase clock synchronizer's sampling clock is quantized to one of the DLL's phases. When this synchronizer is used the clock (ϕ_i) for the synchronizing flip-flop $\mathrm{DFF_i}$ is chosen as

$$\begin{aligned} \phi_i &= \phi_{(n-\frac{N}{2})} && \text{if } n > \frac{N}{2} \\ &= \phi_{Rx} && \text{otherwise.} \end{aligned}$$

Figure 4.15 shows an illustration of the sampling clocks for worst case latency.

4.4 Jitter Analysis

Clock recovery circuits that use the data transitions to estimate the correct sampling phase are sensitive to jitter in the received data. The jitter in the received data has a deterministic and a random component. The random component comes from the phase noise in the transmitter clock and the deterministic component is due to ISI in the channel. Also, wander in the clock

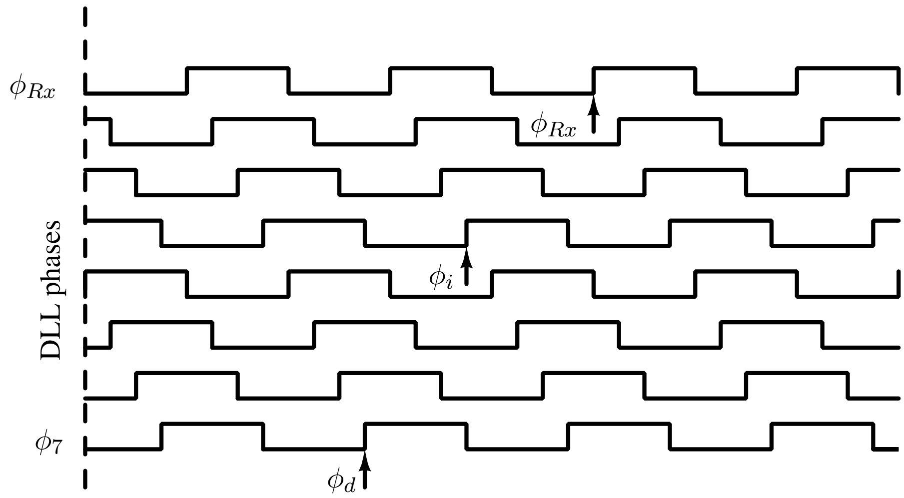


Figure 4.15: Illustration of the timing diagram of discrete phase control clock recovery system with clock domain transfer, under the worst case latency.

generating oscillators results in low frequency jitter. Typically, clock recovery circuits are required to tolerate small amplitude of high frequency jitter (typically under 0.1 Unit Interval (UI)). However, the tolerance to low frequency jitter, which is within the tracking bandwidth of the circuit, needs to be much higher (0.5 UI or more).

In repeaterless on-chip interconnect system the transmitter and the receiver share the same clock source albeit in arbitrary phase. Hence the low frequency jitter due to wander in the clock generating PLL is correlated between the transmitter and the receiver. Thus, the receiver synchronizer does not have to be designed to track the low frequency jitter. The loop bandwidth of the DLL in the receiver determines the frequency up to which the jitter in the clock source stays correlated between the transmitter and the receiver.

The synchronizer was tested with a low frequency jitter of 1 MHz frequency and 0.5 UI amplitude. In order to reduce the simulation time, first, extracted layout of the DLL alone was simulated with an input clock having low frequency jitter, which confirmed that the DLL clock phases tracked the low frequency jitter. Then the full loop of the synchronizer with the schematic level netlists and an ideal DLL model was simulated. Figure 4.16 shows the eye diagram of the received data with jitter, and the control voltage V_c for this simulation. As the low frequency jitter is correlated between the transmitter and the receiver, the control voltage does not have to track it.

High frequency jitter which is generated in the clock distribution networks due to thermal noise in transistors and power supply noise will however not be common to the transmitter and

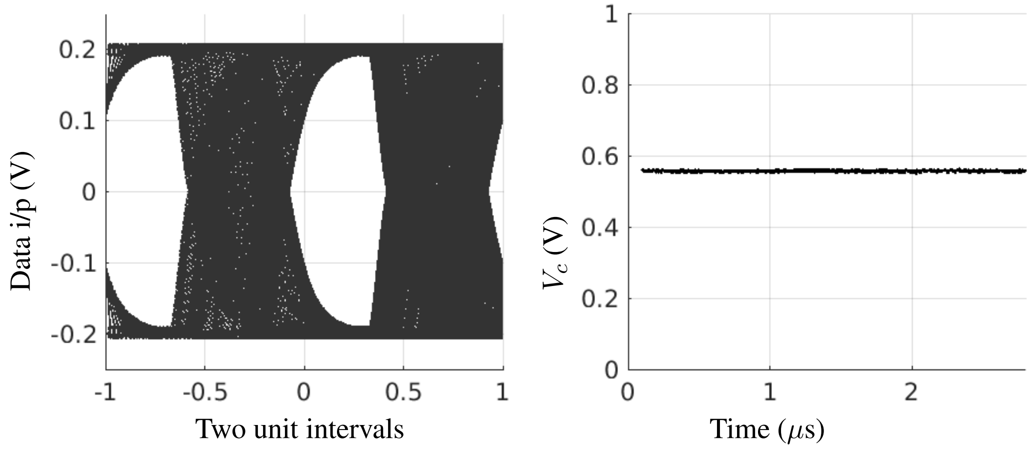


Figure 4.16: Simulated eye diagram and V_c with a sinusoidal jitter of frequency 1 MHz and amplitude of 0.5 UI. The transmitter and receiver clock jitter are correlated.

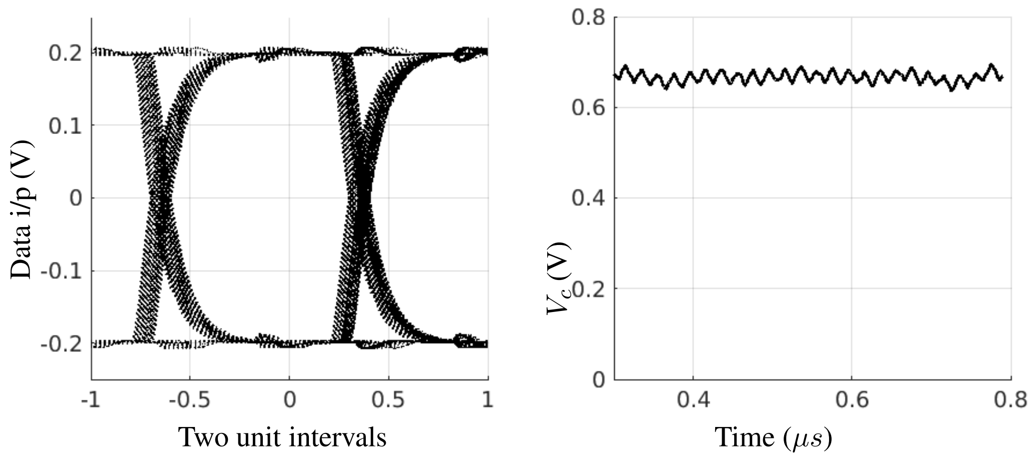


Figure 4.17: Simulated eye diagram and V_c with a sinusoidal jitter of frequency 50 MHz and amplitude of 0.1 UI. Jitter is added only to transmitter clock.

receiver. This high frequency jitter is filtered out by the filter capacitor in the synchronizer circuit. Figure 4.17 shows the eye diagram and control voltage for a sinusoidal jitter of an amplitude of 0.1 UI at a frequency of 50 MHz. The circuit was also simulated for high frequency uncorrelated jitter of 50 MHz and 200 MHz for 10 μs with various jitter amplitudes. The circuit has no errors even for jitter amplitudes as high as 0.4 UI.

In conclusion, for an acceptable level of low frequency jitter tolerance, the DLL in the synchronizer must be designed with a loop bandwidth that is more than the expected low frequency jitter in the clock generating PLL.

4.5 Implementation details and Results

4.5.1 Implementation Details

The clock synchronizer has been designed in UMC 130 nm CMOS technology, with a supply voltage of 1.2 V. A clock division ratio (K) of 16 was used for the coarse tuning loop. The window comparator thresholds in the coarse tuning loop were $V_{DD}/4$ and $3V_{DD}/4$.

Figure 4.18 shows the circuit diagram of the static comparator, which is used to construct the window comparator. The lower threshold (V_L) of the window comparator is $V_{DD}/4 =$

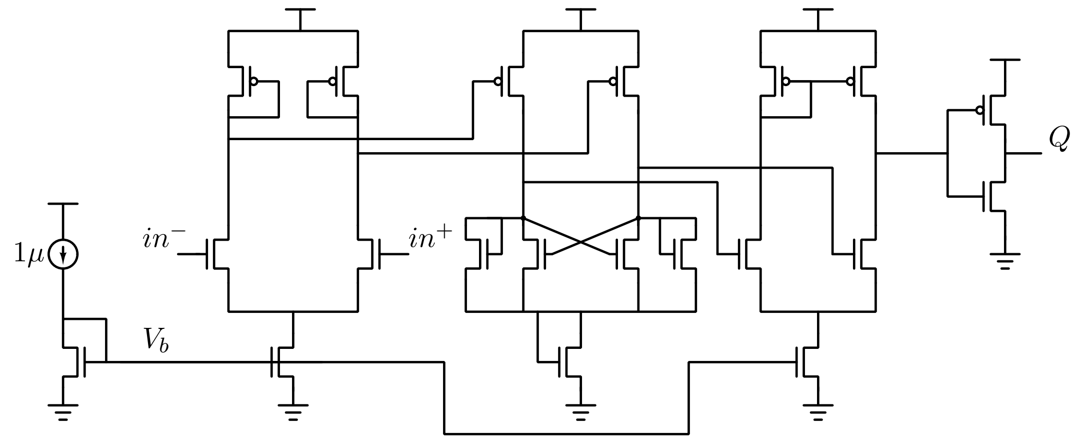


Figure 4.18: Circuit diagram of the static comparator.

$300mV$. The gain of the comparator for such low common mode is very low. (V_{th} in this technology is $370mV$). Hence a level shifter, shown in Figure 4.19, is used for the comparator with the lower threshold.

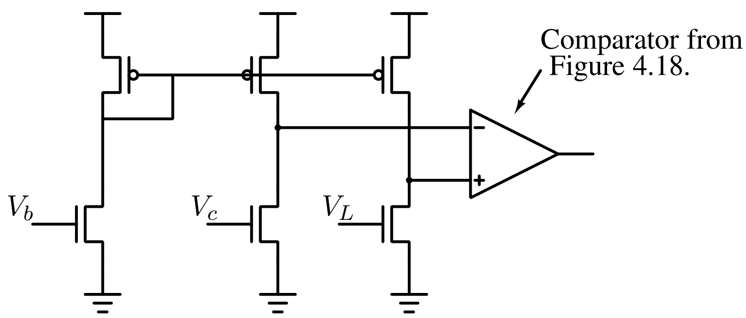


Figure 4.19: Circuit diagram of the level shifter for the static comparator with the lower threshold V_L.

The comparator is tested for an input slope of 1 μA/200 fF. Figure 4.20 shows the simulated response of the comparator under these test conditions. The comparators resolve in about 6 ns after the input crosses the threshold.

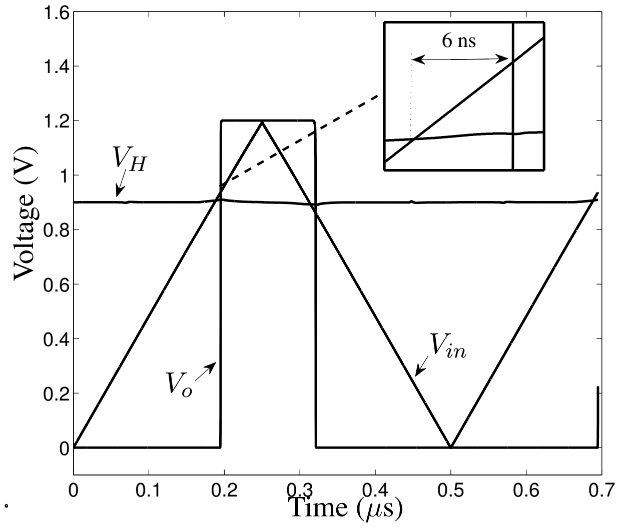


Figure 4.20: Simulation of the static comparators with an input slope generated by the weak charge pump driving the loop filter capacitor.

The Alexander bang bang phase detector was used as the phase detector. Figure 4.21 shows the diagram of the phase detector used in the design. Sense-amplifier based clocked comparators [74] were used and were followed with another flip-flop clocked by the same clock. This is done because the sense-amplifier comparator can take more than half a clock cycle to resolve, and for proper generation of the UP and DN pulses the comparator is required to resolve in less than half a clock cycle. It is interesting to note that sometimes the clock recovery loop can remain stuck with the wrong edge at the center of the eye diagram. This happens when the data is corrupt with jitter and the inertia of the system remaining stuck in this region depends on the amount and type of jitter. This effect is analyzed in detail in Chapter 6.

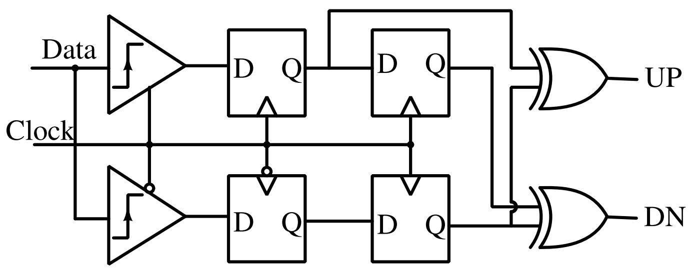


Figure 4.21: Alexander bang-bang phase detector for low swing data input.

Figure 4.22 shows the circuit diagram of the charge pump. This is the well known active amplifier charge pump circuit [80], modified to add the strong current source and sink, which injects current into the loop filter capacitor along with the weak current sources. DN, $\overline{DN}$,

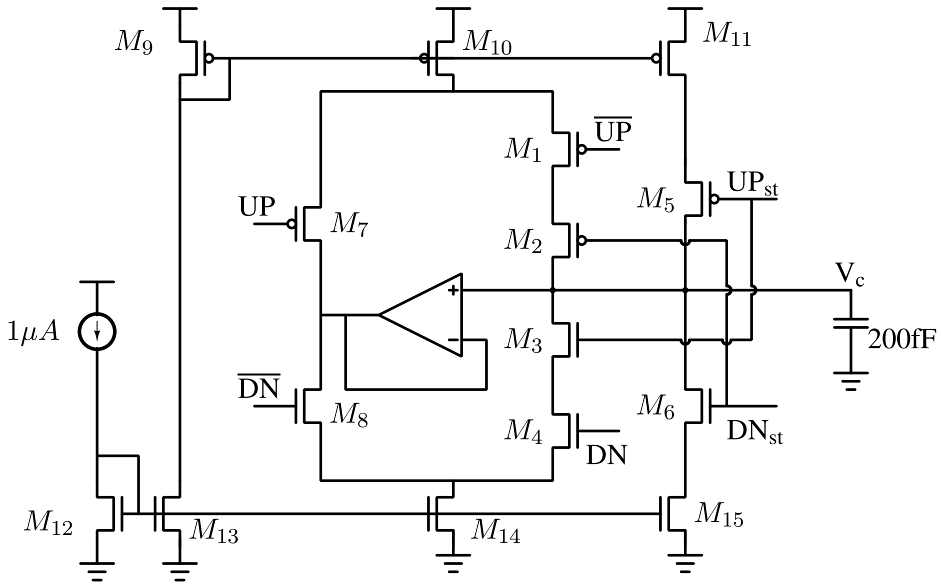


Figure 4.22: The weak and the strong charge pump. DN, $\overline{\mathrm{DN}}$, UP and $\overline{\mathrm{UP}}$ are from the fine tuning loop and $\mathrm{UP_{st}}$ and $\mathrm{DN_{st}}$ are from the coarse tuning loop.

UP and $\overline{UP}$ are driven by the fine tuning loop and UP_{st} and DN_{st} are driven by the coarse tuning loop. The strong charge pump is designed to be 16 times the strength of the weak charge pump. The weak charge pump is of 1μA. The loop filter capacitor is a MIMCAP of 200 fF. The geometries of the transistors in the charge pump are given in Table 4.4. The extra transistors M_2

Table 4.4: Transistor sizes of the charge pump circuit in Figure 4.22

Transistor	W	L
M_1, M_2	950 nm	120 nm
M_3, M_4	320 nm	120 nm
M_5	1 μm	120 nm
M_6	300 nm	120 nm
M_7	700 nm	120 nm
M_8	160 nm	120 nm
M_9, M_{10}	2 μm	500 nm
M_{11}	17 μm	500 nm
M_{12}, M_{13}, M_{14}	500 nm	500 nm
M_{15}	7 μm	500 nm

and M_3 are used to disable the weak charge pump when the strong charge pump is active. This is done so as to prevent the weak current source (sink) and strong current sink (source) from forming a path from V_{DD} to GND, which would push the transistors to their linear region. The opamp used is a traditional single stage differential amplifier.

Figure 4.23 shows the implementation of the VCDL of the fine tuning loop. The delay cells

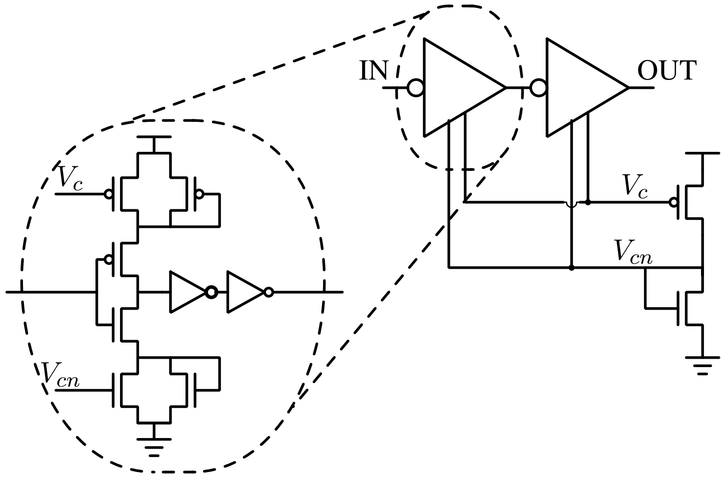


Figure 4.23: Circuit schematic of the VCDL.

are current starved inverters, with the current sources in parallel with diode connected transistors which act as bleeder resistors as shown. Two stages are used in cascade to get the required tuning range. Figure 4.24 shows the range of delays for the allowed range of control voltages across different process corners. The system does not impose any linearity requirement on the

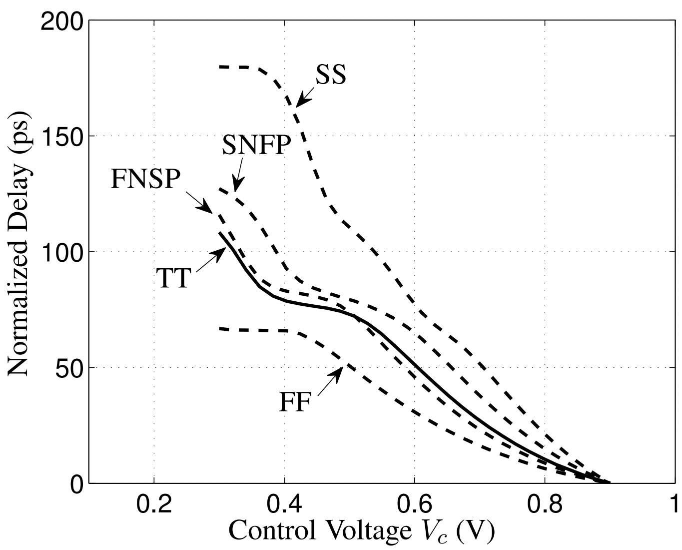


Figure 4.24: Transfer characteristics of the VCDL across process corners.

VCDL. However, the transfer characteristics must be monotonic. The VCDL must be designed to have a range of greater than 1 phase step of the DLL. In order to meet this requirement the VCDL must be over-designed. When designed for a range of 1 phase step in the fastest corner, the VCDL has a range of 2 phase steps under typical process parameters. A DLL of 10 phases was implemented. Linear delay cells reported in [81] are used, with a precharge phase detector [82]. The complete synchronizer has an area of 76μm $\times$ 80μm.

4.5.2 Results

Test Chip 1

In order to test the clock synchronizer the test chip used a low swing transmitter. Since the chip was primarily designed to test the clock retiming circuit, a short interconnect was used to keep the ISI minimal. The logic swing was set to 400 mV$_{\text{pp}}$ so that offset in the comparators can be ignored. In order to emulate the latency of the long interconnect, the transmitter clock is derived from a delay line with external tap select. Figure 4.25 shows the block diagram of the test circuit. The control voltage V_c of the synchronizer and the control voltage of the DLL that generates the multiple phases of the clock, were buffered with internal opamps and brought out on pins for testing. The status of the ring counter in the receiver is also encoded and brought out on pins. The data source for the transmitter was one of 7 and 15 bit PRBS, with a low swing transmitter. A short interconnect is used between the transmitter and receiver. For testing the receiver's phase tracking the transmitter's clock is deliberately shifted using an inverter based delay line, with a programmable tap.

Figure 4.26 shows a photograph of a bare die of the fabricated chip. The fabricated chip was tested at a frequency of 1.3 GHz, and the power consumption of the complete synchronizer was 1.4 mW off a 1.2 V supply. The clock was generated using a three stage inverter ring oscillator, whose frequency was controlled by modulating its supply. No duty cycle correction circuit was included. The tracking of the phase detector was confirmed by deliberately shifting the clock phase of the transmitter and observing the control voltage of the circuit. Figure 4.27(a) shows the trajectory of the control voltage (V_c) when the introduced phase shift in the transmitter does not need a DLL phase increment or decrement. Figure 4.27(b) shows the trajectory of the control voltage when the introduced phase shift in the transmitter needs DLL phase decrements for achieving lock. The results obtained with 7 bit and 15 bit PRBS are consistent. Once lock is

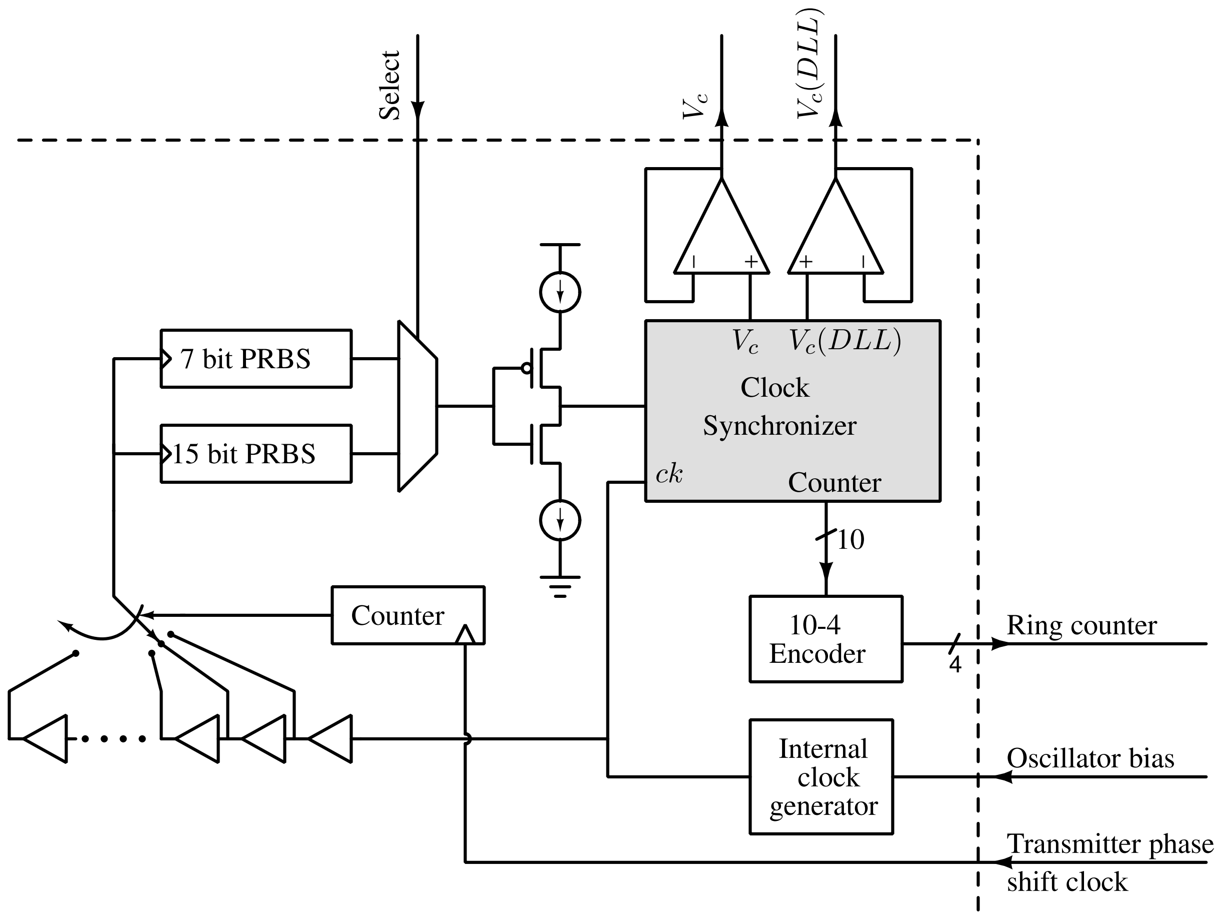


Figure 4.25: Test setup for characterizing the fabricated synchronizer circuit. The circuit under test is shaded gray.

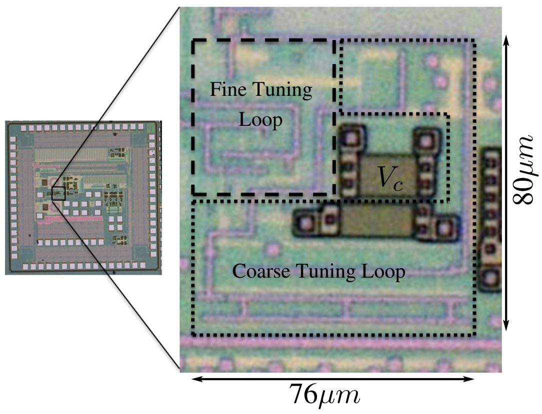
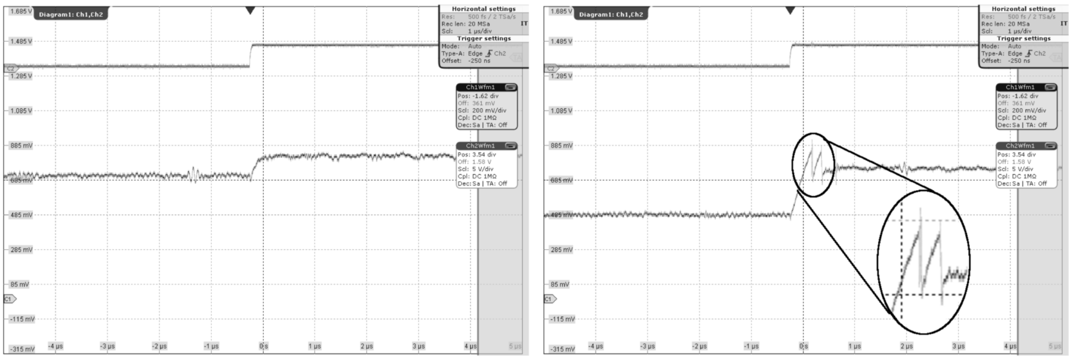


Figure 4.26: Photograph of a die of the fabricated circuit (Test Chip 1).

achieved, it was observed that the circuit remains locked over long periods of time, which was tested by monitoring the lock over a period of up to 30 minutes.

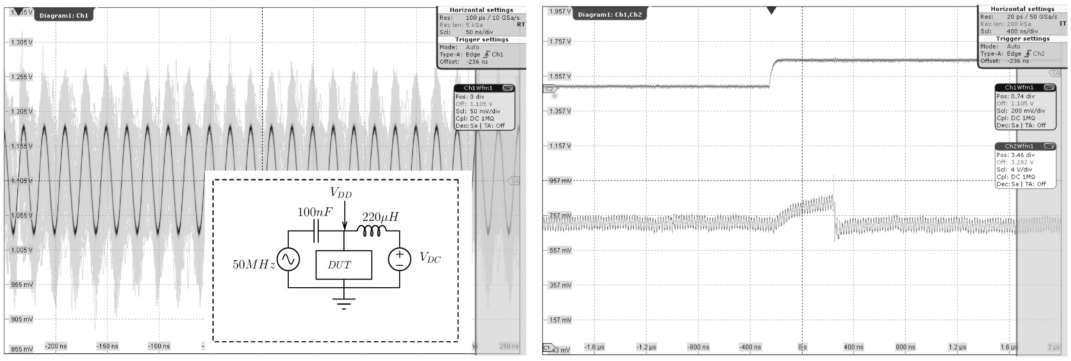
The circuit was also tested with supply voltage fluctuations by adding a 50 MHz ± 80 mV sine wave to the supply voltage. Figure 4.28(a) shows the supply voltage as observed on an



(a) Coarse correction not required. **(b)** With two coarse correction steps.

Figure 4.27: Measured trajectory of V_c with an introduced phase shift in the transmitter clock. The upper waveform is the transmitter phase shift clock. (a) When a DLL increment/decrement is not required and (b) When DLL phase decrements are required.

oscilloscope along with an inset of the circuit used to add a sine wave over the DC supply. Figure 4.28(b) shows the trajectory of the control voltage for the measurement under these conditions.



(a) Modulated supply. **(b)** Control voltage response.

Figure 4.28: (a) Waveform showing the supply voltage modulated with a 50 MHz, ±80 mV sine wave. Inset is the circuit used to generate the modulated supply. (b) Measured trajectory of V_c under these test conditions. The upper waveform is the transmitter phase shift clock.

Test Chip 2

The first test chip results served as a proof of concept. We subsequently designed a second test chip, which was designed to enable measurements like BER, jitter tolerance and settling time (settling time measurement will be discussed in Chapter 6). While the circuit under test was identical to the first test chip, the test data and the transmitter and receiver clocks were generated externally and input via pins. We used Centellax TG1B1-A BER tester and Centellax TG1C1-A clock synthesizer for the testing. The chip was tested at a frequency of 1.9 GHz and with a 31 bit PRBS as the test data sequence. Figure 4.29 shows a photograph of a bare die of the fabricated chip.

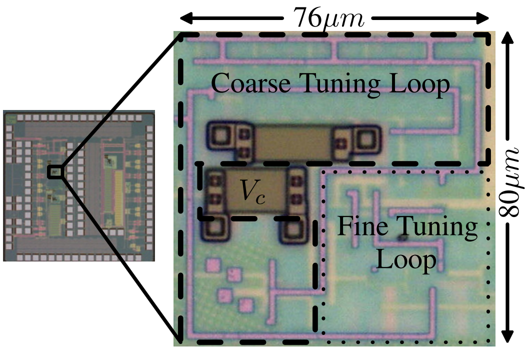


Figure 4.29: Photograph of a die of the fabricated circuit (Test Chip 2).

Bit error rate: The circuit shows good measured BER performance with BER of $< 10^{-12}$ at a frequency of 1.9 GHz. Figure 4.30 shows a photograph of the test setup.

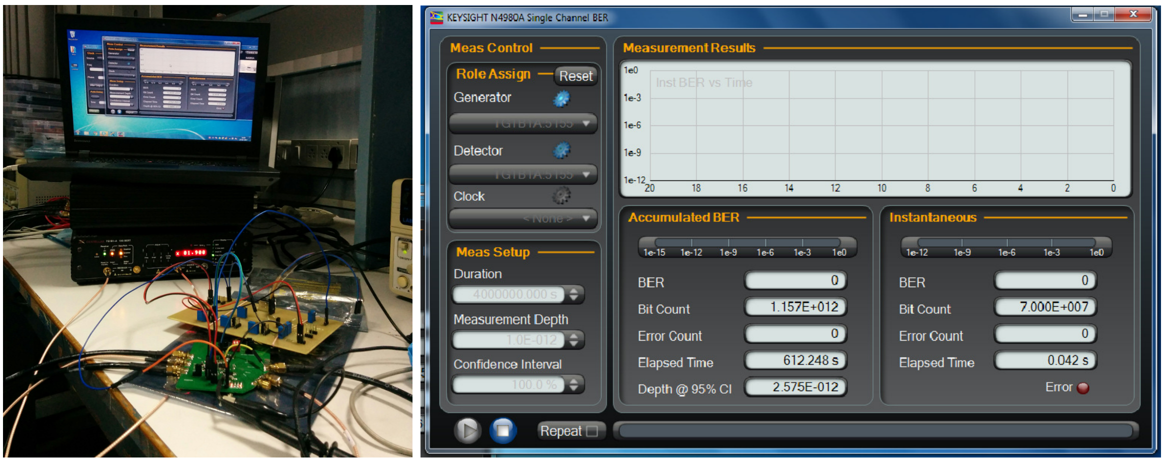


Figure 4.30: Photograph of the test setup for the BER measurement.

Control Loop hysteresis: As was discussed in the previous section, the fine control is overdesigned to have a range of 2 phase steps of the coarse control. Due to this, there can be two possible values of coarse and fine control state for lock. This also results in a hysteresis when the clock is swept in the forward and reverse directions. Figure 4.31 shows measured coarse control word as the phase is swept in the forward and reverse directions.

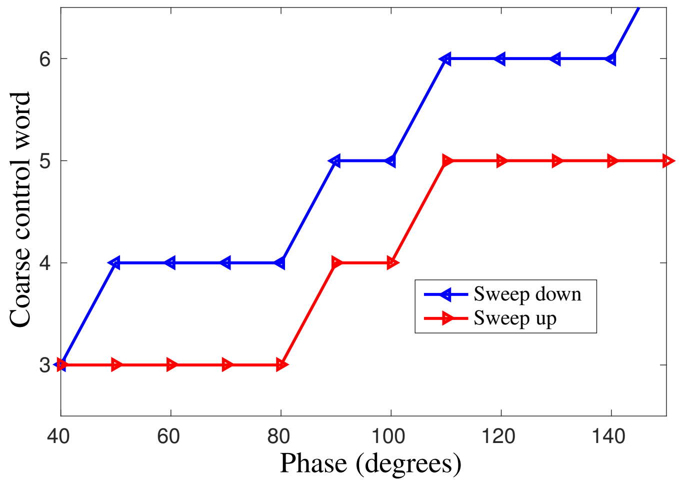


Figure 4.31: Measured coarse control word with phase sweeps in the forward and reverse directions showing the control loop's hysteresis.

In some cases the circuit can lock with the control voltage close to one of the thresholds of the window comparator. In such cases jitter in clock and noise on the control voltage can result in a coarse correction step. The hysteresis in the control loop helps in avoiding such coarse correction steps from occurring frequently, as subsequent lock will be away from the thresholds. Figure 4.32 shows the measured control voltage when the circuit locks close to the threshold. As seen in Figure 4.32, after about 1 μs, the circuit takes a coarse correction step and then locks with its control voltage around the center of the window.

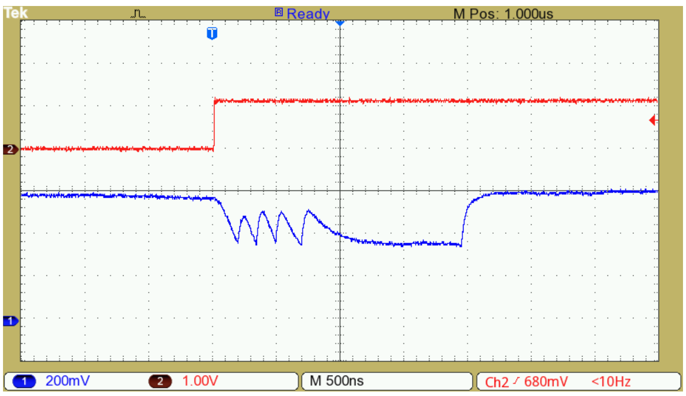


Figure 4.32: Measured control voltage when lock is achieved near one of the thresholds of the window comparator. Noise in the control voltage results in a coarse correction step and loop hysteresis ensures subsequent lock is around the center of the window. The upper waveform is the reset (active low) signal.

Correlated Jitter tolerance: As was discussed in Section 4.4, in on-chip interconnects the clock for the transmitter and receiver is derived from the same PLL. Hence, much of the jitter in the PLL output will be correlated between the transmitter and the receiver. We measured the circuits tolerance to correlated jitter for a BER $< 10^{-10}$. The clock synthesizer instrument (Centellax TG1C1-A) has 6 clock outputs out of which jitter can be added to three of the outputs and the other three are clean. For measuring correlated jitter tolerance, the transmitter clock and the receiver clock are derived from the clocks with jitter. Figure 4.33 shows the tolerance to correlated jitter.

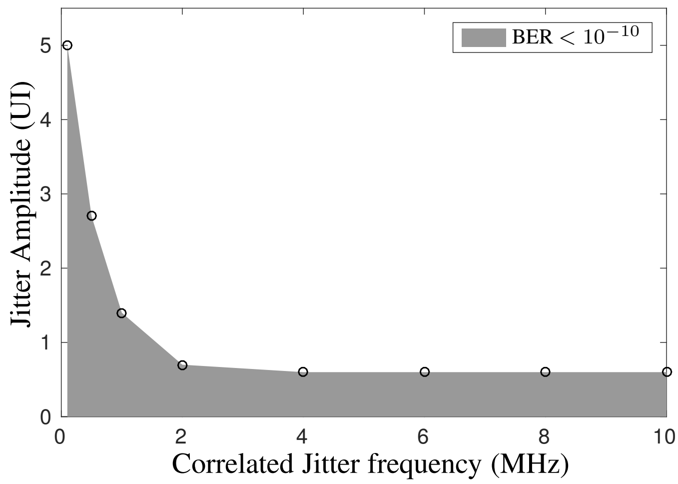


Figure 4.33: Measured jitter tolerance to correlated jitter for BER of $< 10^{-10}$.

Uncorrelated Jitter tolerance: Noise in the clock distribution networks can result in jitter which is uncorrelated between the transmitter and the receiver. Uncorrelated jitter is tracked by the control voltage. For measuring uncorrelated jitter tolerance, the transmitter clock is picked from a clean output of the synthesizer and the receiver clock is picked from the output with jitter. For the data generator and error detector (TG1B1-A), the data source is driven from a clean clock from the synthesizer and the error detector is driven from the recovered clock output of the test chip. Figure 4.34 shows the control voltage as the circuit tracks an uncorrelated jitter of 0.1 UI at a frequency of 100 KHz. The BER was $< 10^{-10}$ in this experiment. Since the jitter amplitude is small, no coarse correction steps occur in this case.

Higher amplitude jitter will result in the circuit taking coarse correction steps. Figure 4.35 shows the control voltage as the circuit tracks an uncorrelated jitter of 0.4 UI at a frequency of 1 MHz. The BER was $< 10^{-9}$ in this experiment.

Figure 4.36 shows the circuits tolerance to uncorrelated jitter of different amplitudes and frequencies for a BER of $< 10^{-9}$. It may however be noted that uncorrelated jitter of such high amplitudes, between two nodes of an on-chip clock distribution network, is not expected

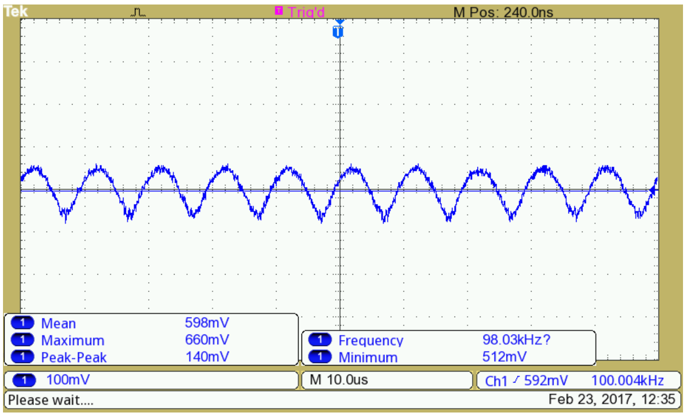


Figure 4.34: Measured control voltage when the circuit tracks an uncorrelated jitter of 0.1 UI at 100 KHz. The BER was $< 10^{-10}$. Since the jitter amplitude is small no coarse correction steps are required for tracking.

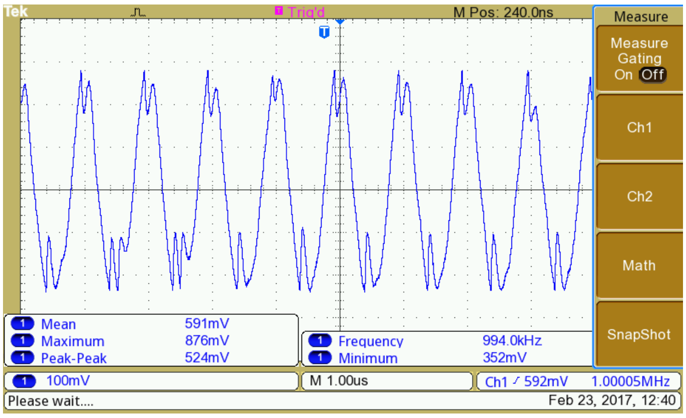


Figure 4.35: Measured control voltage when the circuit tracks an uncorrelated jitter of 0.4 UI at 1 MHz. The BER was $< 10^{-9}$. The large amplitude of the jitter requires coarse correction steps for tracking.

practically.

Recovered clock and data: Figure 4.37 shows the spectrum of the recovered clock. The Root Mean Squared (RMS) jitter in the recovered clock was measured using a spectrum analyzer and is found to be about 1.6 ps, as shown in Figure 4.37.

Figure 4.38 shows the eye diagram of the recovered clock and recovered data. For the measurement of eye diagrams, the data and clock were generated using the Kintex-7 FPGA based Genesys development board. The recovered clock and data signals from the chip were obtained after converting the resolved data to low swing by CML buffers for driving the off-chip interface.

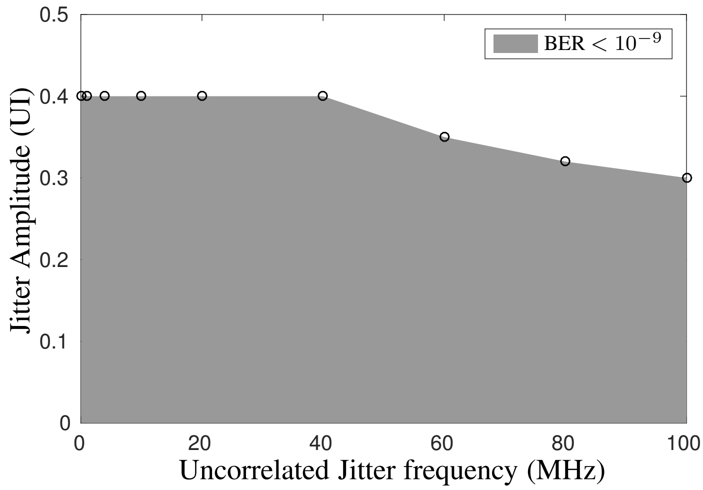


Figure 4.36: Measured jitter tolerance to uncorrelated jitter for BER of $< 10^{-9}$.

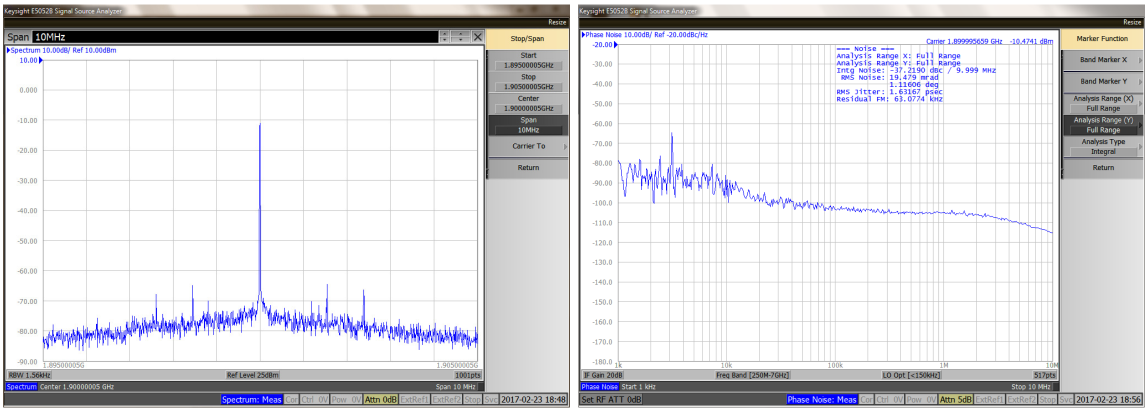


Figure 4.37: Measured spectrum of the recovered clock and the phase noise. The RMS jitter of the recovered clock is 1.6 ps.

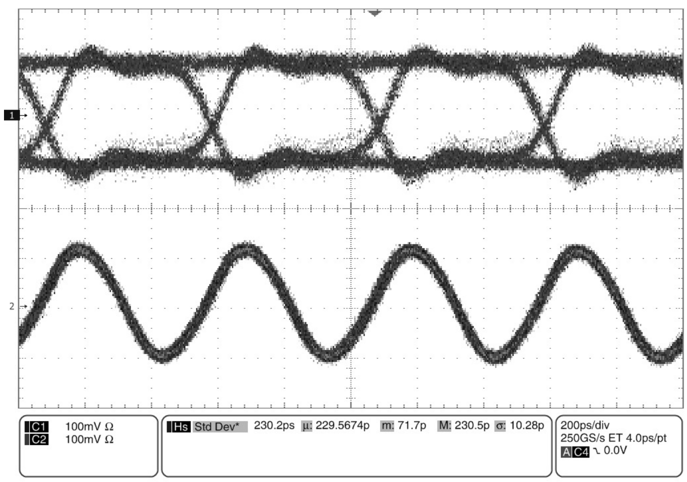


Figure 4.38: Measured recovered data and recovered clock of the receiver.

Summary of Results

Table 4.5 compares the performance of previously reported clock synchronizers for repeaterless interconnects with the presented design. As seen from Table 4.5, the proposed synchronizer for repeaterless low swing interconnects is the only one for on-chip interconnects that is adaptive and does not have phase quantization error. The proposed architecture includes clock domain transfer to the receiver's clock domain. Further, the architecture uses largely digital subcircuits, allowing it to scale well. The analog components are limited a) one VCDL which has a small range and no stringent linearity requirement b) static comparators, which which also do not impose any stringent requirements. Also, to the best of our knowledge, this is the only work which discusses clock domain transfer to receiver clock domain for low swing interconnects.

To confirm scalability of the architecture, the synchronizer was also designed and simulated in TSMC 65 nm CMOS technology, for a data rate of 4 Gbps. The clock division ratio (K) of 32 was used for deriving the clock for the coarse tuning loop. Rest of the implementation details are identical to the UMC 130 nm CMOS technology implementation. From layout extracted simulations, the power consumption was found to be 1.5 mW drawn from a 1 V supply. The area of the synchronizer is $48\mu m \times 50\mu m$.

4.6 Performance of the full link

In the previous chapter we had discussed implementation of repeaterless low swing interconnects with equalizers. Design of the receiver, i.e. the sampling clock retiming and clock domain transfer have been discussed in this chapter. Having designed a complete end-to-end link we shall now compare it to a standard repeater inserted link in UMC 130 nm CMOS technology with the aid of circuit simulations. For simulations, a 10 mm interconnect in M7 layer was used with a capacitively coupled transmitter discussed in Section 3.1 and the clock retiming circuit discussed in this chapter. The interconnect used has a width of 0.4 μm and spacing of 0.4 μm. The interconnect parasitics are R = 67.6 Ω/mm, C = 122 fF/mm and L = 1.1nH/mm. The link was designed for a data rate of 2.5 Gbps with a 1.2 V supply. This is assumed to be the logic speed of this process, which is determined by the frequency of oscillation of a 5 stage ring oscillator of NAND gates. In terms of FO4 delay, this is about 8FO4. Most microprocessors have cycle times in excess of 10FO4 [83].

The total power consumption of the transmitter is 725 μW and the total receiver power

Table 4.5: Comparison with other reported repeaterless synchronizers

	JSSC'14 [10]	JSSC'05 [57]	JSSC'03 [58]	This work	
Process	65 nm	110 nm	110 nm	130 nm (Measured)	65 nm (Simulated)
Interconnect type	On-chip	Off-chip	Off-chip	On-Chip	
Synchronizer type	Mesochronous	Plesiochronous	Plesiochronous	Mesochronous	
Controller type	Digital	Analog	Digital	Coarse digital + fine analog	
Adaptive control	No	Yes	Yes	Yes	
Clock domain transfer	No	No	No	Yes	
Data rate	3 Gb/s	10 Gb/s	10 Gb/s	1.9 Gb/s	4 Gb/s
Power Consumption	0.4 mW	220 mW	129 mW	2.9 mW	1.5 mW
Supply Voltage	0.9 V	1.5 V	1.2 V	1.2 V	1 V
Area ($\mu m \times \mu m$)	Not reported	250 × 1400	1600 × 2600	76 × 80	48 × 50

consumption is 3.4 mW at this frequency. Hence the total link consumes ~4.1 mW. The latency of the link is 450 ps. The worst case latency of the clock retiming circuit is 3 clock cycles. Hence, the worst case latency is about 4 clock cycles. For designing a repeater inserted link for driving the same interconnect, repeaters are needed every 200 μm. The buffer size used is W_N = 4 μm and W_P = 12 μm. Further, synchronizing flip-flops are needed after every 4 repeaters. The total power consumption of this repeater inserted link is 6.2 mW and the latency is 14 cycles. It must be noted that the power incurred in distributing a synchronous clock to the repeaters is not included for the repeater inserted case. Hence, in practice the power savings will be even more substantial.

It is worth noting that lower metal layers have higher parasitic resistance and capacitance as compared to top metal layers. Hence, the number of repeaters required is more when lower metal layers are used for long interconnects. This means that the benefits of using repeaterless interconnects are higher for lower metal layers.

4.6.1 A note on DVFS

Digital circuits generally use DVFS for power saving when the computing demand is low. It is hence of interest to investigate the performance of low swing interconnects with supply voltage scaling and frequency throttling.

Capacitively coupled transmitter

The low swing interconnect which was designed was simulated at a few supply voltage, frequency points to test its performance. Table 4.6 summarizes the results from this simulation. The link was functional across all these operating supply voltage, data rate combinations.

Table 4.6: Capacitively coupled link DVFS

Supply (V)	Frequency (GHz)	I_{static}	R_L
1.2	2.5	3.7 μA	16 kΩ
1	1.8	2.1 μA	23.8 kΩ
0.9	1.25	1.45 μA	30.8 kΩ

Since, the static current reduces and the load resistance increases with lowering of the supply voltage, the values of the coupling capacitors of the equalizer are still close to optimum (within 5%). Hence, the low swing interonnect circuit is compatible with DVFS.

Clock retiming circuit

The DLL in the clock retiming circuit was designed for a frequency of 2.5 GHz, with a tuning range of about 500 MHz. We first simulated the DLL to check if it can lock at lower supply voltages and correspondingly lower frequencies. Table 4.7 lists the supply and frequency combinations for which the DLL achieves lock.

Table 4.7: DLL with DVFS

Supply (V)	Frequency (GHz)	DLL locks?
1.2	2.5	✓
1	1.8	✓
0.9	1.25	✓

Correct operation of the circuit can be guaranteed as long as the range of the fine control loop exceeds a coarse correction step. As discussed earlier, for robust operation we design the fine control loop to have a range of two phase steps fo the DLL. The VCDL was simulated to test its range with dynamic voltage scaling. Table 4.8 summarizes these results.

Table 4.8: VCDL range with dynamic voltage scaling

Supply (V)	Frequency (GHz)	delay range (ps)	Two phase steps of DLL (ps)
1.2	2.5	80	80
1	1.8	192	110
0.9	1.25	350	160

We then simulated the complete clock retiming circuit at these frequency and supply conditions, and the circuit works correctly in these conditions.

4.7 Measuring clock skew and its variation with time

Techniques for measurement of skew between remote nodes, and its low frequency drifts, are needed to measure the frequency up to which the jitter is correlated. This information will enable appropriate design of the loop bandwidth of the synchronizers. A circuit for measuring clock skew between remote nodes, and its low frequency drift, using subsampling was reported in [67] which was discussed in Section 2.4.1 (refer to Figure 2.16). This technique achieves high accuracy, by performing block averaging of multiple measurements. As was discussed in Section 2.4.1, the skew introduced by the routing of the sampling clock introduces substantial measurement error and block averaging results in poor tracking bandwidth.

Figure 4.39 shows an improved skew measurement architecture, based on the subsampling technique reported in [67] (refer to Figure 2.16). In order to eliminate the error introduced by the routing delay of the sampling clock, two measurements are made by routing the sampling clock along two paths. These paths are shown in Figure 4.39, as path L_1 and L_2. These two paths feed the sampling clock to the two sampling flip-flops in opposite order along the same path. In the first measurement, the sampling clock is routed along Path L_1. The additional delay introduced by the routing of the sampling clock produces an error, t_{CM} in the measurement. Let

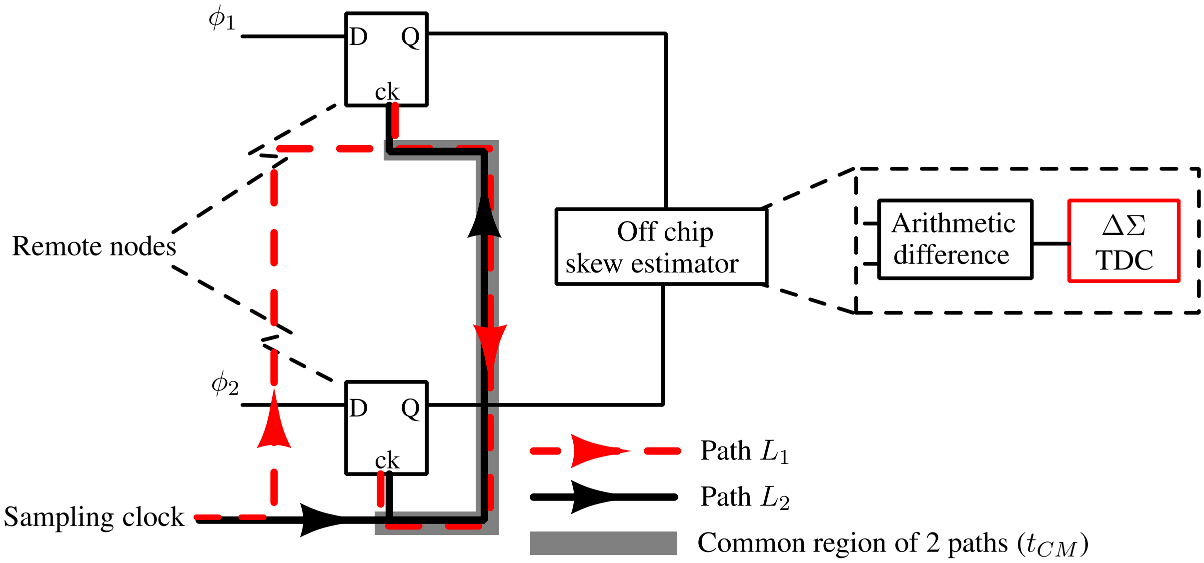


Figure 4.39: Subsampling based skew measurement circuit, with compensation for routing clock skew. The backend uses a $\Delta\Sigma$ modulator to achieve good tracking bandwidth with high accuracy. This is an improvement over the architecture reported in [67] (Figure 2.16), with the differences in color.

this measurement be

$$T_{skew}^{L_1} = T_{skew} + t_{CM}.$$

The second measurement cycle is performed by routing the clock along the Path L_2. In this measurement, the error in the skew measurement is identical but in opposite direction. This skew can be represented as

$$T_{skew}^{L_2} = T_{skew} - t_{CM}.$$

Now, by taking the average of the two measurements, the error introduced by the routing of the sampling clock can be eliminated.

The work reported in [67] used histogram analysis, which effectively averages multiple measurements. This allows improving the resolution by increasing the measurement time. This technique is good for measuring DC skew. However, a moving average scheme in the form of a sigma-delta Time to Digital Converter (TDC) is a better approach for measurement of skew with low frequency jitter. Hence, as shown in Figure 4.39, a $\Delta\Sigma$ modulator is used to convert the skew (amplified by the subsampling stage) to digital form. Despite the use of sub-sampling of the clock signals, the actual information signal, i.e. the skew, is oversampled. Typically, the clock signals are in the order of a few GHz and the skew variation is in the order of a few KHz. The sampling clock is off from the core clock by a small fraction, and hence, subsamples the core clock. However, the information of skew gets oversampled. This allows us to use $\Delta\Sigma$ based TDC, which shapes the quantizer noise to achieve high resolution.

Figure 4.40 shows the block diagram of the $\Delta\Sigma$ TDC used for skew measurement. The $\Delta\Sigma$ modulator is constructed using the following components, each described in analogy to the $\Delta\Sigma$ ADC.

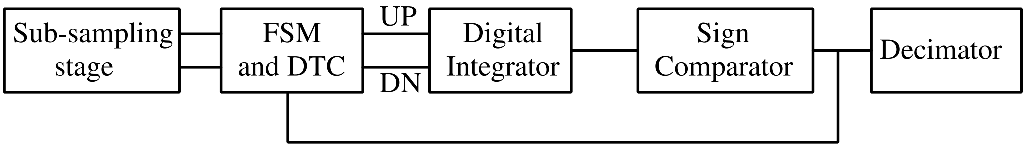


Figure 4.40: Block diagram of the $\Delta\Sigma$ TDC used for skew measurement.

1 bit DAC and subtractor: The 1 bit DAC in this case corresponds to a 1-bit Digital to Time Converter (DTC). This is implemented using multiplexers and fixed delay lines. The fixed delay is simply a long shift register, clocked by the fast clock. Figure 4.41 shows the circuit of the DTC. The subtraction is also performed by the same circuit. Here, depending on the 1-bit input, one of In_1 or In_2 is delayed with respect to the other.

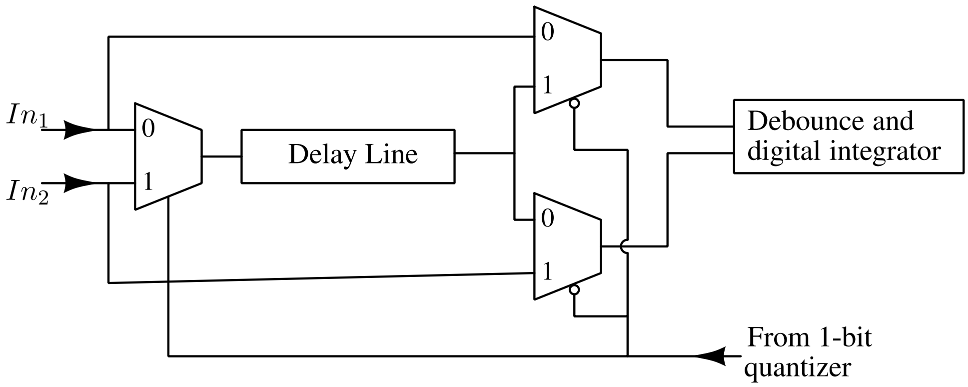


Figure 4.41: Circuit of the 1 bit digital to time delay converter.

Integrator: A digital counter is used as an integrator in this design. It is worth noting that the quantization noise of the digital integrator is not shaped by the modulator. However, the quantization noise of the digital integrator is much lower than the 1 bit quantizer that follows.

1 bit quantizer: The 1 bit quantizer is a simple sign comparator. The output of the comparator drives the multiplexer network, thus closing the modulator loop.

Decimator: The decimator of the $\Delta\Sigma$ TDC is identical to the one used in a conventional $\Delta\Sigma$ ADC.

The above approach culminates in an all-digital implementation, which can be ported to a Field Programmable Gate Array (FPGA), with only the sub-sampling stage on-chip. The sigma-delta TDC architecture has been designed and simulated using Simulink for digital integrators of resolutions N=8 and for a typical Over Sampling Ratio (OSR) = 64 with signals of 1 KHz frequency in the TDC. So, maximum input skew frequency is limited to 15.625 Hz. Input skew signal of frequency 3.76 Hz was given using phase modulated sinusoidal signal to one of two input clock signals. Hence, four samples per period of the input skew are obtained after decimation. Figure 4.42 shows the spectral plot of the output bit stream of the modulator. Since the input is generated by phase modulation, it inherently has infinite sidebands. These sidebands appear as harmonics in Figure 4.42 and are part of the input signal. The sigma-delta

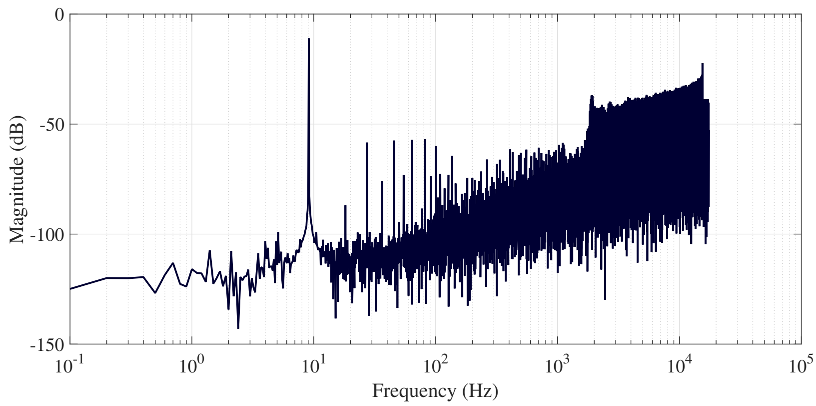


Figure 4.42: Spectral plot of output bit stream for $N = 8$.

TDC circuit design should ideally give an Effective Number Of Bits (ENOB) of 10 with an analog integrator for this setup. However, with a digital integrator, the ENOB is 9.47. Thus, an all-digital implementation is achieved in-lieu of a very small penalty in ENOB. The degradation in ENOB depends on the oversampling ratio and the digital integrator resolution. Figure 4.43 shows the ENOB with the resolution of the integrator for a few values of OSR. This skew measurement circuit has been implemented, with the sub-sampling stage in UMC 130 nm CMOS technology and the $\Delta\Sigma$ TDC in an FPGA and verified. A detailed report of these results can be found in [84].

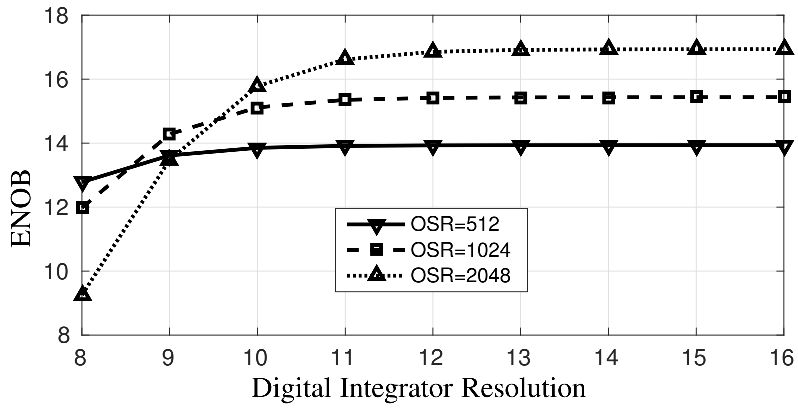


Figure 4.43: ENOB with digital integrator resolution N for a few values of OSR.

4.8 Summary

Repeaterless interconnects have to use active synchronizing circuits for achieving high performance. The synchronizer reported in [10] is a foreground synchronizer and cannot track environmental changes. It also suffers from quantization of the recovered clock phase. In this chapter, we report a background synchronizer, that allows the phase to be tracked while the link is in normal operation. Along with a coarse clock control, built around a multi-phase DLL, the circuit has a fine tuning loop using a VCDL. This allows a continuous phase control, eliminating the phase quantization error. Once the low swing input data is sampled and resolved to rail to rail levels, it must be transferred to the receivers clock domain. Multi flip-flop synchronizing circuits are generally used for such clock domain crossing. These synchronizers trade latency for MTBF [12]. We report a low latency deterministic synchronizer that makes use of the multiple clock phases available in the system.

The synchronizer has been fabricated in UMC 130 nm CMOS technology and tested. The scalability of the solution is confirmed by post layout simulations performed in TSMC 65 nm CMOS technology. The robustness of the synchronizer to correlated and uncorrelated jitter has been demonstrated experimentally with measurements of a test chip. Compared to other clock retiming circuits reported in the literature, the circuit reported by us does not have phase quantization error, can track delay variations in normal operation and includes clock domain transfer to the receiver clock domain.

A technique for measuring skew between remote nodes has also been reported in this chapter. The architecture features an all digital $\Delta\Sigma$ based implementation which achieves high resolution along with good tracking bandwidth.

Chapter 5

Design for testability of repeaterless interconnects

Testability of VLSI designs is very important for large scale and high volume applications. Design for testability for pure digital circuits has been well studied. Test techniques for digital systems have been developed which are now fairly standardized and automated tools are available for inserting scan chains and for generating test patterns. However, design for testability of analog circuits is more challenging. The circuit solutions reported for repeaterless interconnects use mixed signal circuits to achieve high performance at low power [10,11,19,21,31,40,46,49]. The testability of these schemes has however not been sufficiently explored. Replacement of a digital interconnect that uses repeaters with a repeaterless low swing interconnect, should ensure that the system is still testable. It is even better if these mixed signal circuits can tested with available test equipment that are designed for the test of digital systems. Design for testability of some mixed signal circuits like PLL's, DLL's, ADC's etc. has been reported in the literature [14–18]. These use the structural fault model for faults in the analog circuits and the test methodology is hand-crafted to the specific circuit under consideration.

The primary aim of the work reported in this chapter is to design a high speed low swing interconnect link that is testable and has high fault coverage in both the digital and analog sections of the circuit. Standard digital test interfaces like scan test, BIST with a digital output and DC tests with digital outputs are used so that no additional test equipment is needed. While the complete interconnect link (from transmitter to the recovered data at the receiver) has digital inputs and outputs, it must be noted that a scan test may not capture all defects. This is because, the interconnects are designed to operate at a few GHz, while scan tests are generally performed at less than 400MHz. At such low frequencies, links can work without equalization and sometimes even without clock recovery, and might fail at-speed due to timing faults. At-speed scan tests are generally used for capturing delay faults. For at-speed scan test, the test pattern is loaded at a low frequency and the capture event is triggered one application period (which is the clock period in normal operation) from the load of the last bit of the test pattern [85]. Clock recovery circuits are not guaranteed to operate correctly in such a sequence, and hence, these interconnects cannot be tested using at-speed scan tests. Hence, defect based testing is a better approach to qualify such designs. The stuck-at fault model is used for the digital components of the circuit. Structural fault model reported in [15] is used for the analog circuit components.

In this Chapter, we show that the low swing interconnect with equalizers and clock synchronizers that were discussed in Chapter 3 and Chapter 4 can be tested with high degree of fault coverage using some additional circuitry. The additional circuitry added only for the purpose of testing are shaded gray in all the figures in this chapter. These circuits will be turned off in normal operation.

5.1 DC and scan test of low swing interconnects

Scan test for digital circuits is done by using scan flip-flops instead of regular flip-flops. Scan

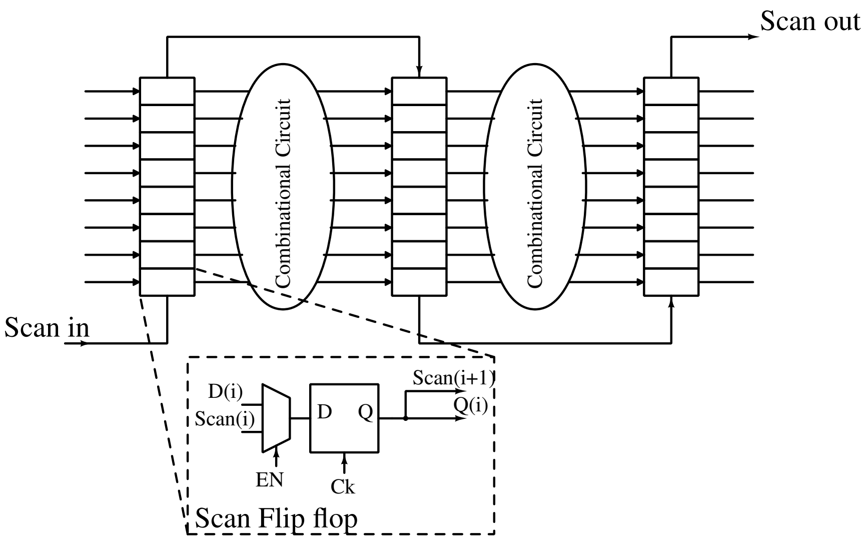


Figure 5.1: Concept of a scan chain.

flip-flops have a multiplexer at the input which can switch the entire system between normal mode and scan mode. Figure 5.1 illustrates a sequential circuit which can be scan tested. The scan flip-flop is shown in the inset figure. In scan test mode, all the flip-flops form a long shift register which can be preloaded to initialize the system to a desired state. After the preload step, the circuit is clocked once. The system is then changed to scan mode to read out the results. The low swing interconnect with a clock synchronizing receiver is a system that operates across three clock domains, which are the transmitter clock domain (ϕ_{Tx}), the sampling clock domain (ϕ_d), and the receiver clock domain (ϕ_{Rx}). The test of this system is done by dividing the system into two parts. The transmitter, interconnect and the phase detector at the receiver form the first half and the clock synchronizing circuit at the receiver excluding the phase detector forms the second half. Figure 5.2 shows the block diagram of the repeaterless interconnect system with the capacitively coupled transmitter described in Chapter 3 and the clock synchronizing receiver described in Chapter 4 with additional components included for making the design testable.

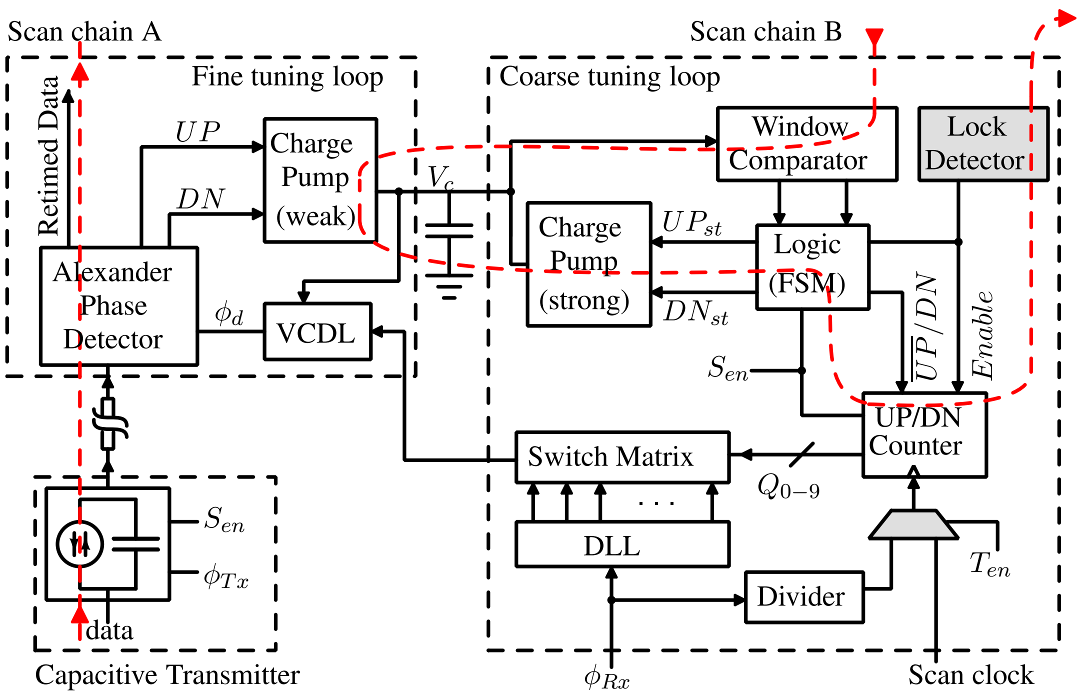


Figure 5.2: Block diagram clock synchronizer system, divided into fine tuning and coarse tuning loops. S_{en}: Scan enable signal, T_{en}: Test mode enable.

Since the receiver and the transmitter operate in different clock domains, the circuit is tested using two separate scan chains which are the data path scan chain (Scan chain A) and the clock control path scan chain (Scan chain B). The data path scan chain begins at the transmitter, goes through the interconnect and the phase detector at the receiver. The output of this scan chain is the retimed data output of the phase detector. The clock control path scan chain begins at the window comparator, goes through the strong and the weak charge pumps, the control FSM, UP DOWN counter and finally the lock detector block. When test is enabled, the coarse correction loop's clock input is driven from an external scan clock as shown in Figure 5.2. The divider in this circuit can be shared across multiple such receivers in the chip and tested separately.

5.1.1 The data path scan chain (Scan chain A)

The low swing transmitter (Figure 5.3) is the first component of the data path scan chain. This is the capacitively coupled transmitter with feed-forward equalization described in Chapter 3. A single ended version is shown for brevity, but actual implementation used a differential interconnect. Flip-flops are added to probe the driver side of the series capacitors (the shaded flip-flops in Figure 5.3) and thus enable the scan chain to cover all the nodes up to the series capacitors. The additional latch in the data path is added to optionally introduce a half cycle delay at the transmitter, which is required for testing the phase detector at the receiver. This latch is transparent during normal operation and it can be absorbed into the buffer that drives the line. Since the interconnect has high latency, it is not in the critical path. Hence, the delay added by the latch does not degrade the maximum operating frequency of the system.

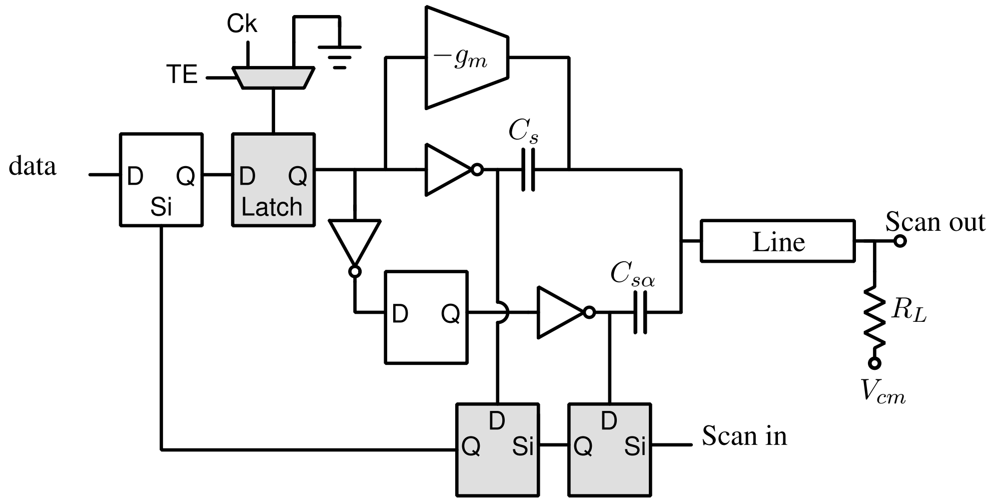


Figure 5.3: The capacitive feed-forward equalizer with the weak driver. All the flip-flops are clocked by the same transmitter clock, which is not shown.

Figure 5.4 shows the circuit diagram of the receiver termination with the test circuit. It uses comparators with programmed offset for the DC test. This circuit is a single stage opamp followed by an inverter (Figure 5.5). The input transistors of the comparator in Figure 5.5 are deliberately mismatched to have an offset of 15 mV. The interconnect is designed for a logic swing of 60 mV and when the circuit has no faults the comparator gets an input of 30

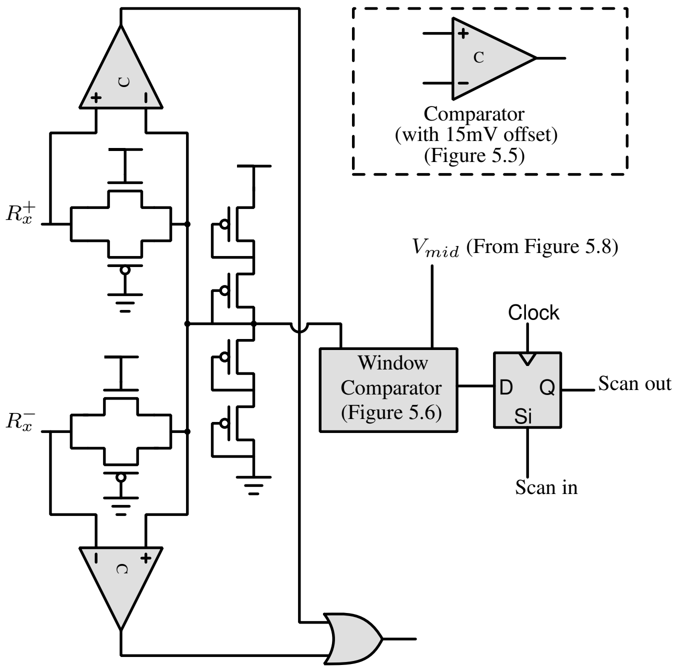


Figure 5.4: The termination of the interconnect at the receiver with test circuits.

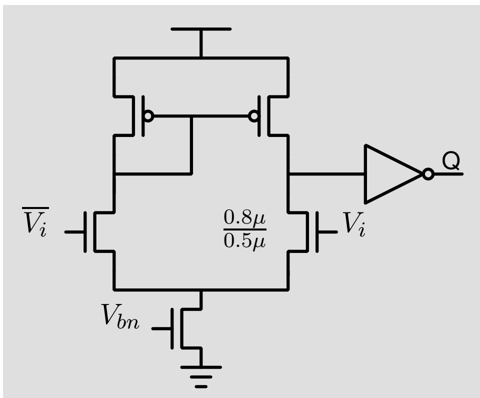


Figure 5.5: Comparator with programmed offset used in the receiver termination. All un-labelled transistors have W/L=$0.5\mu/0.5\mu$.

mV. The input transistor sizes are $0.5\mu/0.5\mu$ and $0.8\mu/0.5\mu$, which is sufficient to overcome any mismatch due to the manufacturing process. These comparators are used for two DC tests, with the interconnect input fixed at each of the logic levels, and can detect many faults in the transmitter and in the receiver termination.

The differential interconnect is a symmetric system under defect free conditions. Hence, in a good circuit, the common mode voltage does not show large voltage swings . However, any fault in the interconnect results in an asymmetry, which causes the common mode voltage at the receiver to swing. The window comparator in Figure 5.4, compares the receiver's common mode voltage with another voltage divider bias generator in the clock recovery circuit. The window comparator is constructed using two comparators with a programmed offset of +15 mV and -15 mV. Any fault results in an asymmetry in the interconnect and is detected by the comparators. Some faults, like drain open in one of the transistors of the transmission gate resistor, result in a dynamic mismatch. This is not detectable at DC. Hence, the window comparator is designed to operate at the scan frequency (which is assumed to be 100 MHz) and these faults are detected with a simple toggling data pattern during scan. Common centroid layout techniques were used to reduce the inherent offset in these comparators. Since these comparators are either used at DC or at scan frequencies, the additional parasitics of the common centroid layout are not a problem.

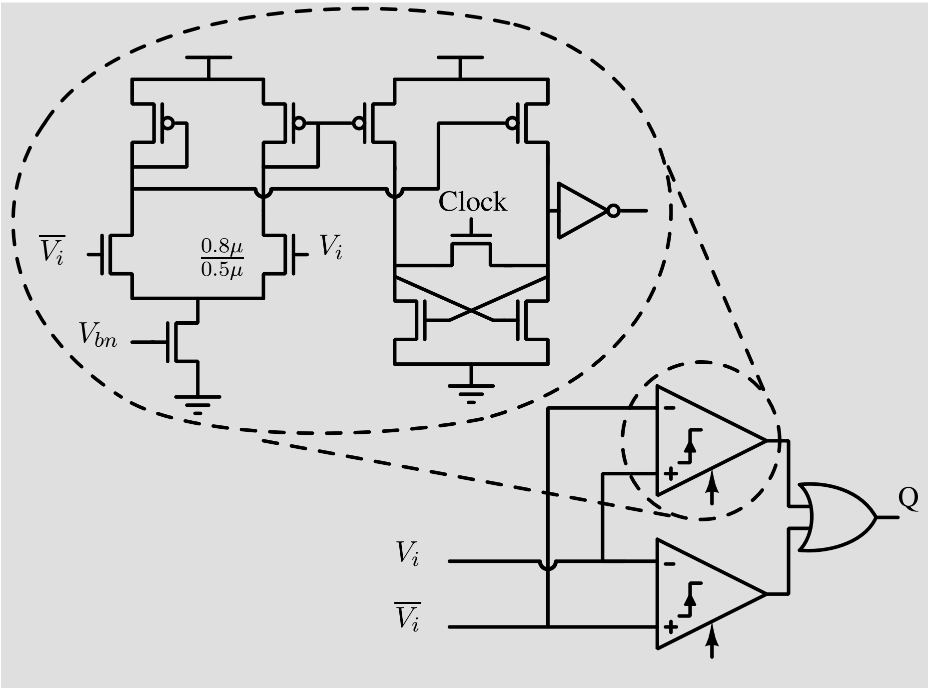


Figure 5.6: Window comparator used in the receiver termination circuit. All un-labelled transistors have W/L=$0.5\mu/0.5\mu$.

The last component in the data path scan chain is the Alexander phase detector at the receiver. The circuit diagram of the phase detector is shown in Figure 5.7. When the link is operated at the scan frequency, the phase detector always asserts the UP signal. This is because the interconnect latency is much less than half clock period of the scan frequency clock. To test the other signal path, the half cycle delay at the transmitter side is enabled, which makes the phase detector assert the DN signal. Thus, with two passes both these paths can be tested. The last flip-flop (which is driven by either ϕ_{Rx} or $\overline{\phi_{Rx}}$) in the data path is used to insert either a 1

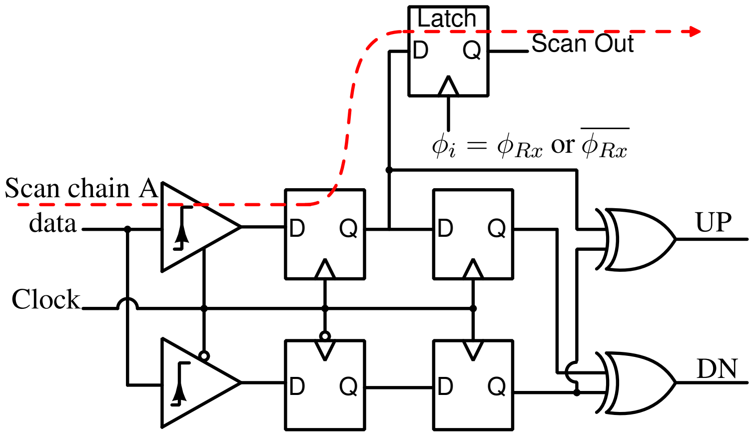


Figure 5.7: Scan testable Alexander phase detector.

clock or a half clock cycle delay. This is required for transferring the data to the receiver clock domain. Half cycle delay is chosen when the sampling clock is less than half a clock cycle away from the receiver clock and this is done by driving this last flip-flop with $\overline{\phi_{Rx}}$. For test purposes this can be controlled via the clock control path scan chain. When ϕ_{Rx} is chosen, it results in an increase in the length of Scan chain A register by 1 flip-flop.

5.1.2 Clock control path scan chain (Scan chain B)

The first circuit block in this scan chain comprises of the window comparator, the charge pumps and the control FSM. Figure 5.8 shows the circuit diagram of this part of the system.

The charge pump is an analog circuit and cannot be included in the scan chain as is. To work around this problem the charge pump is converted to a combinational circuit when scan test is enabled. This is done by connecting the bias voltages for the current sources in the charge pump to GND for the PMOS source and to V_{DD} for the NMOS sink. This essentially converts the analog charge pump into a combinational circuit with two inputs UP and DN and one output. Also when scan is enabled, the window comparator's input is connected to the middle of the thresholds thus forcing the output to be "00". Two flip-flops are used to capture the comparator's outputs, which are read through the clock control path scan chain.

Scan chain A is used to make the phase detector assert either UP or DN signal. This results in the control voltage V_c being driven to a logic '1' or '0' respectively. When the scan is disabled and the circuit is clocked before re-enabling scan mode, the control FSM resets the control voltage to within the window. Most of the faults in the charge pump result in the control voltage not being reset to within the window or in not being driven to the desired logic

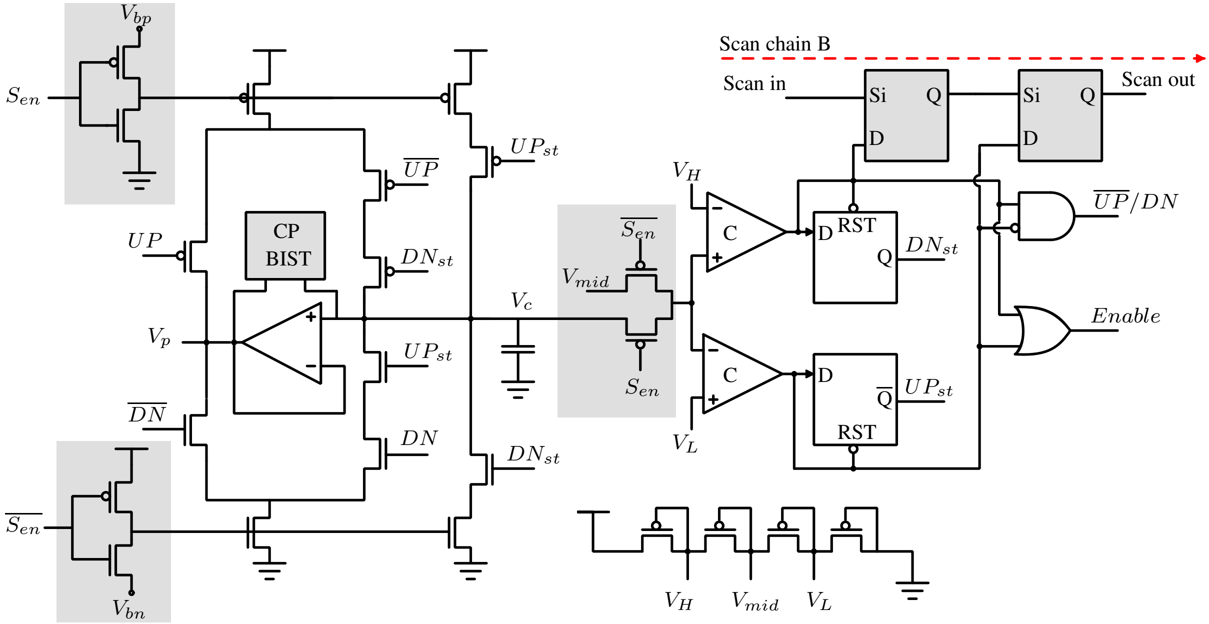


Figure 5.8: Control logic for generating $\overline{UP}/DN$ and $Enable$ signals for the ring counter and UP_{st} & DN_{st} signals for the strong charge pump. V_H, V_L are the upper and lower thresholds of window comparator respectively. All flip-flops are clocked with the divided clock of the coarse control loop.

level when scan is enabled. The comparators outputs can detect most faults in the charge pump circuit.

To test the ring counter, it is first preloaded with a 1 hot pattern. Scan chain A is used to drive the control voltage in the charge pump to logic '1' or '0'. The window comparators outputs enable the ring counter in this condition and the $UP/\overline{DN}$ signal is driven to '1' or '0' depending on the control voltage. This completes the pre-load step and by de-asserting scan enable (S_{en}) and clocking the circuit the ring counter can be made to count in the desired direction before enabling scan again and reading the results.

The last circuit block in the clock control path scan chain is the switch matrix. Defects in this block may lead to inability of selecting a desired phase for locking or inability to deselect a particular phase. This is tested by pre-loading the ring counter with all zeroes pattern. This causes none of the phases to be picked, resulting in Scan chain A not getting clocked. Simple continuity test of Scan Chain A can detect a permanently selected phase. Further pre-loading different one hot values into the ring counter and testing the continuity of Scan Chain A all the paths in the switch matrix can be tested.

5.2 BIST for improving fault coverage

The Lock Detector in Figure 5.2 is a simple saturating UP counter. It logs the number of times a coarse correction request has been issued. From any initial condition, the number of coarse corrections needed can be no more than half the number of DLL phases. The design used a 10 phase DLL and hence a 3 bit saturating UP counter is sufficient for the lock detector. For BIST the interconnect is run with random data at-speed. The receiver is expected to lock within 2 μs, which corresponds to less than 5000 cycles at 2.5 Gbps. Some of the faults in the charge pump, which are not detectable in the scan test, can be detected using this test. During scan test the charge pump's current sources were used as switches. This however masks a drain source short fault in the current source transistors. The BIST with the lock detector can detect such faults.

The scan test of the charge pump tested only the main charge pump path and the charge balancing path (that drives the node V_p in Figure 5.8) is not tested. Any faults in this second path or faults in the amplifier in the charge pump, result in the node V_p drifting towards V_{DD} or GND. This pushes one of the current sources to linear region and as a result causes increased jitter in the recovered clock, which can degrade the interconnect performance. The CP-BIST block in Figure 5.8 is a window comparator that is designed for a window of 150 mV. Figure 5.9 shows the circuit diagram of the window comparator that is built with two comparators with a programmed offset of 150 mV. Once lock is achieved, this comparator's output being high is an indicator of a fault in the charge pump that was not detected in the scan test.

The DLL in the receiver is not tested completely by this BIST. This DLL can be treated as a stand-alone unit and using the techniques reported in [16, 17], a complete test of the DLL can be integrated with the interconnect test.

5.3 Simulation Results: Fault coverage and area overhead

The interconnect system was designed in UMC 130 nm CMOS technology with a supply voltage of 1.2 V. The digital components are tested using scan test. Since the circuits are logically simple in nature, the stuck at fault coverage is 100%. The digital coarse correction is operated at a divided clock frequency which is in the range of scan test frequencies. Hence, the delay faults in this path are also tested with 100% coverage.

For the analog components, two DC tests with the interconnect input at logic 1 and logic 0 respectively can detect 50.4% of the structural faults in the circuit. Scan test of the analog

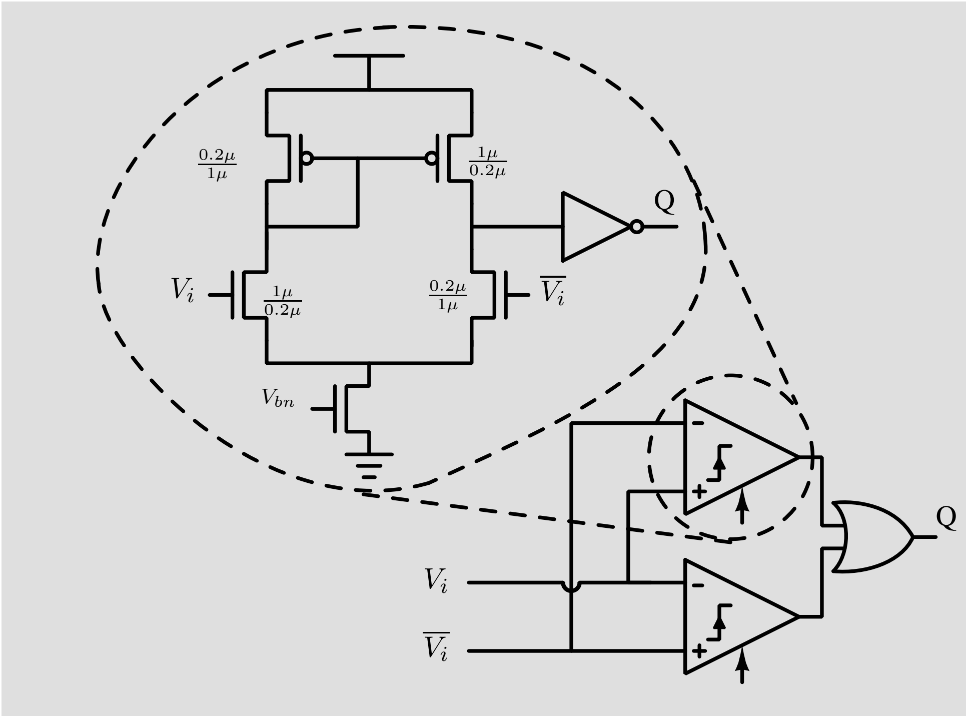


Figure 5.9: The window comparator used in the charge pump for BIST. All un-labelled transistors have W/L=$0.5\mu/0.5\mu$.

circuits in the receiver by converting the charge pump to a combinational circuit enhances the coverage to 74.3%. Most of the faults missed by the DC and scan test are detected in the BIST which improves the fault coverage to 94.8%. Table 5.1(a) tabulates the types of faults and their coverage statistics. The total additional circuits required are shown in Table 5.1(b).

5.4 Summary

Due to random manufacturing defects, the yield of VLSI chips is never 100%. High volume VLSI chips should be testable to screen and filter out faulty parts. Design for testability of digital circuits is well studied and test techniques have been standardized. However, to achieve best performance, long interconnects need to use repeaterless low swing architectures with equalization and clock synchronization. These circuits are mixed signal circuits and must be testable within the standard digital process flow. We have shown that, with a little additional circuitry, the low swing interconnects can be tested within the standard digital flow. Simple techniques are shown to integrate the analog components into the digital scan chain. With the aid of a simple BIST with small overhead a structural fault coverage of 95% can be achieved. This shows that

Table 5.1: Fault coverage statistics and circuit overhead of the testable low swing interconnect design.

(a) Coverage of different types of faults

Defect	Coverage
Gate open	87.8%
Drain open	93.9%
Source open	93.9%
Gate drain short	93.9%
Gate source short	100%
Drain source short	100%
Capacitor short	100%
Total	94.8%

(b) Circuit and control input overhead

Entity	Number
Flip-flop	7
Comparators (DC)	4
Comparators (100 MHz)	2
D-Latch	1
2×1 Multiplexer	2
3 bit saturating UP counter	1
Control signals	2
Logic gates	6
Area	29%

low swing interconnect circuits can potentially be used in large scale and high volume digital systems.

Chapter 6

Settling time of Mesochronous clock retiming circuits

6.1 Introduction

On-chip low swing interconnects are mesochronous systems, which means that the transmitter and receiver operate at the exact same frequency, with arbitrary phase relationship. As reported in the previous chapters, we had designed synchronizers for these low swing interconnect systems. The circuit recovers the phase offset between the incoming data and receiver clock and generates an appropriate sampling clock for sampling the data. The data is then transferred to the receiver clock domain with a clock domain transfer circuit. In this chapter we will investigate the settling time of this synchronizer circuit, particularly in the presence of timing jitter in the incoming data. It is known that timing jitter in the incoming data can degrade the BER of such links and this has been well studied [86]. However, timing jitter can also increase the settling time of clock recovery circuits indefinitely. This occurs when the circuit wakes up with its clock in the horizontally closed region of the data eye. The settling time of the circuit is analyzed by using a Markov chain model of the system and the model predictions are verified using behavioural simulations of the synchronizer and circuit level simulations as well.

Figure 6.1 shows the block diagram of a repeaterless interconnect system. The delay of the interconnect is expressed as $(n + \lambda)T$, where $n \in \mathbb{Z}^+$, $\lambda \in [0, 1)$ and T is the system clock period. In this Chapter, we will be using the example of mesochronous synchronizer for on-chip interconnects reported in Chapter 4 for all the analysis. However, the results also apply to off-chip interconnects that use forwarded clock from the transmitter to receiver, with a clock

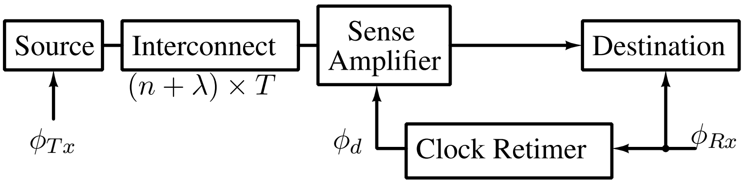


Figure 6.1: Block diagram of a repeaterless low swing interconnect system. ϕ_{Tx}: Transmitter clock, ϕ_{Rx}: Receiver clock, ϕ_d: retimed sampling clock, ($(n+\lambda)\times T$): Repeaterless interconnect delay, where $n \in \mathbb{Z}^+$, $\lambda \in [0, 1)$, T: system clock period.

phase recovery at the receiver, like those reported in [62].

6.2 Settling time of the clock retiming circuit

The settling time of the mesochronous clock retiming circuit depends on the initial phase error between the clock and the data. This initial phase error ($\Delta\phi$) is a continuous variable taking values in $[0, 2\pi)$. When the initial phase error is less than π, the circuit achieves lock by decreasing the phase to 0. On the other hand, when $\Delta\phi$ is greater that π, the circuit achieves lock by increasing the phase difference to 2π, which is the same as 0 by phase wrapping. The settling time of the clock recovery circuit depends on the gain of the system and the initial phase error. For mesochronous systems only the phase has to be corrected and the clock recovery circuit can be of the first order [20]. Hence, the loop filter is a single capacitor. When a binary phase detector is used, the capacitor voltage (V_c) is quantized to a step size given by

$$\delta V_c = \int_0^T \frac{I_{CP}}{C} dt = \frac{K_{CP}}{C}.$$

Here, K_{CP} is the gain of the charge pump, expressed in terms of the charge pump current I_{CP} and the clock period T. Hence, the step size of the phase corrections is

$$\delta\phi = \delta V_c K_{VC} = \frac{K_{CP} K_{VC}}{C}. \tag{6.1}$$

where K_{VC} is the gain of the phase modulator. The number of phase correction steps (M) needed for achieving lock can be written as

$$M = \begin{cases} \frac{\Delta\phi}{\delta\phi} & \text{when } \Delta\phi < \pi, \\ \frac{2\pi - \Delta\phi}{\delta\phi} & \text{when } \Delta\phi > \pi. \end{cases} \tag{6.2}$$

The settling time (t_s) can then be written as

$$t_s = \frac{MT}{\alpha}, \tag{6.3}$$

where α is the data activity factor. From equations (6.1), (6.2) and (6.3), t_s can be expressed as

$$t_s = \begin{cases} T\frac{C}{\alpha K_{VC} K_{CP}} \Delta\phi & \text{when } \Delta\phi < \pi, \\ T\frac{C}{\alpha K_{VC} K_{CP}} (2\pi - \Delta\phi) & \text{when } \Delta\phi > \pi. \end{cases} \tag{6.4}$$

6.2.1 The extension of the settling time

When the initial phase difference is equal to π, the circuit can settle to two discrete values of the phase difference, which are 0 and 2π (which is same as 0 by phase wrapping), and both the solutions are acceptable. Since the initial phase difference is a continuous variable, the above scenario is one of taking a discrete decision on a continuous input. Theoretically, such decisions can take infinite amount of time for certain initial conditions [87]. This happens when $\Delta\phi$ is exactly π radians and the system has no reason to choose one solution over another. This is akin to metastability in flip-flops. When $\Delta\phi$ is exactly π radians, the flip flops in the phase detector become metastable. In these conditions, the phase detector loop enters a state of indecision. However, for sustained indecision of the phase detector loop, the flip-flops in the phase detector must become metastable in every clock cycle. Practically however, given the narrow widths of the metastability windows, the fast recovery times and the inherent jitter present in the clocks, sustained loop indecision due to flip-flop metastability is highly unlikely. Hence, metastability of the flip-flops in the phase detector is *not* discussed in this work.

Effect of timing jitter: The window of susceptibility

When the data input has timing jitter, due to ISI in the data and/or random jitter in the clock, and the initial clock phase (ϕ_d^0) is in the horizontally closed region of the data eye, the expression of settling time in equation (6.4) is not valid. Figure 6.2 illustrates an eye diagram with timing jitter and when ϕ_d^0 is in the horizontally closed region of the data eye. This region is called the *window of susceptibility* ($\mathcal{W}$) henceforth. When the initial clock is in this window $\mathcal{W}$, the output of the phase detector is randomized by the jitter in the data, and hence, the phase error information is lost. This increases the settling time t_s indefinitely, till the system escapes this window $\mathcal{W}$. The width of the window ($T_{\mathcal{W}}$) depends on the amount of timing jitter present in the

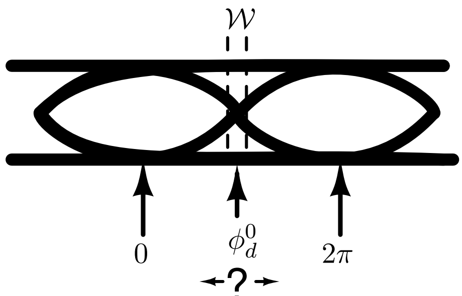


Figure 6.2: Illustration of an eye diagram with timing jitter and with initial clock position ϕ_d^0 in the horizontally closed region $\mathcal{W}$.

system and the threshold of the sampling comparators. The effect of offset on $T_{\mathcal{W}}$ is illustrated in Figure 6.3. For simplicity of analysis, we assume that the samplers have zero offset in the

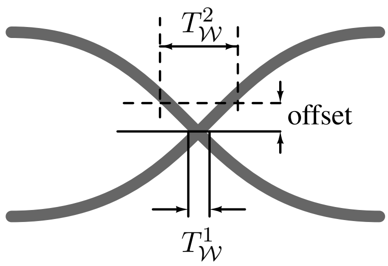
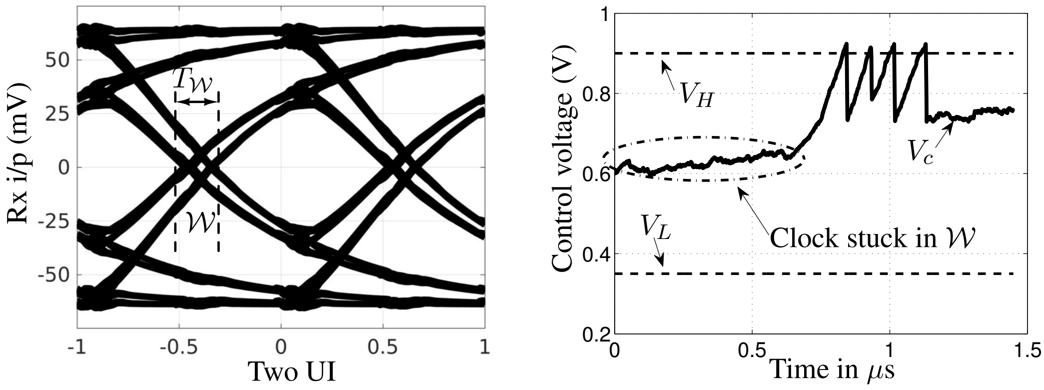


Figure 6.3: Illustration of the sampler offset increasing the size of the window $\mathcal{W}$ from $T_{\mathcal{W}}^1$ to $T_{\mathcal{W}}^2$.

rest of this work. The results can however be easily extended once an appropriate window $\mathcal{W}$ is defined after accounting for the offset.

Typical systems implemented on-chip can have hundreds of long interconnects [88, 89], and a horizontal eye opening of about 85% is not pessimistic [11, 31]. If the initial clock position is assumed to be uniformly distributed, there is a 15% chance that the circuit wakes up with its initial clock phase in the window $\mathcal{W}$, making it important to study and understand this problem. Circuit level simulations show that the system can remain stuck in the window for thousands of cycles. Figure 6.4(a) shows the eye diagram at the receiver input of a typical low swing interconnect. Here, $T_{\mathcal{W}}$ is the width of the window $\mathcal{W}$. Figure 6.4(b) shows the control voltage of a coarse-fine type clock retiming circuit [20] when the circuit is stuck with its clock in this window for an extended period of time. The simulation was done in UMC 130 nm CMOS technology, with a 10 mm interconnect and a low swing current mode transmitter with capacitive pre-emphasis [31].



(a) Eye diagram at receiver input.

(b) Control voltage trajectory.

Figure 6.4: (a) Eye diagram at receiver input indicating the window of susceptibility $\mathcal{W}$ and its width $T_{\mathcal{W}}$ and (b) Control voltage trajectory when initial sampling instant is within $\mathcal{W}$.

Practical demonstration of the extension of the settling time.

We demonstrated the extension of the settling time with the second prototype chip that was designed for the clock retiming circuit (as discussed in Chapter 4). This circuit was implemented with a low swing transmitter and a short interconnect. The short interconnect was loaded with a varactor to control the bandwidth of the interconnect, and hence the horizontal eye opening. The circuit had a reset input which reset the coarse control word as well as the analog control voltage to $V_{DD}/2$. Figure 6.5 shows the circuit diagram of the reset circuit in the test chip. The

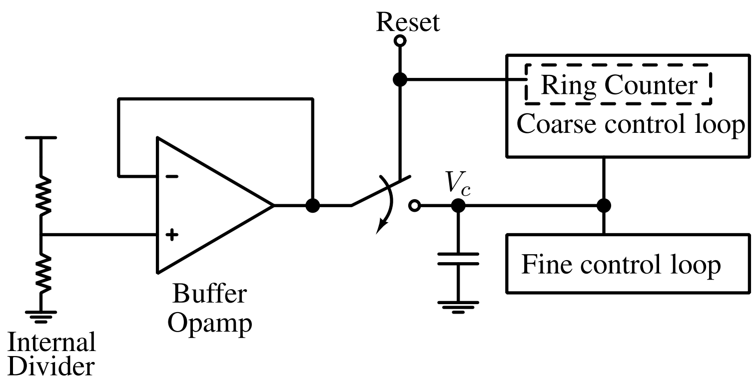


Figure 6.5: Circuit for resetting the receiver circuit and initializing the control voltage to $V_{DD}/2$.

chip was tested at a frequency of 2 GHz with Centellax TG1B1-A data generator and Centellax TG1C1-A clock synthesizer. In the clock synthesizer, one of the clock outputs can be phase shifted with a resolution 2°, which is used for programming the initial phase error. Figure 6.6 shows the simulated eye diagram at the receiver input.

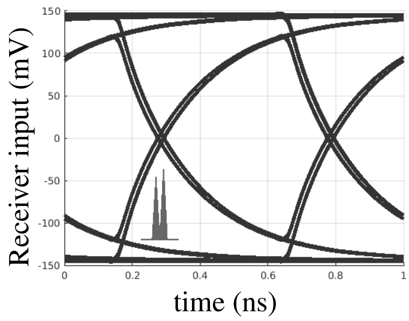


Figure 6.6: Simulated eye diagram of the data input. The inset plot shows the jitter induced zero crossing distribution.

With the clock initialized in the window $\mathcal{W}$, the settling of the control voltage was captured with an oscilloscope as shown in Figure 6.7. In this capture, the clock remains stuck in the

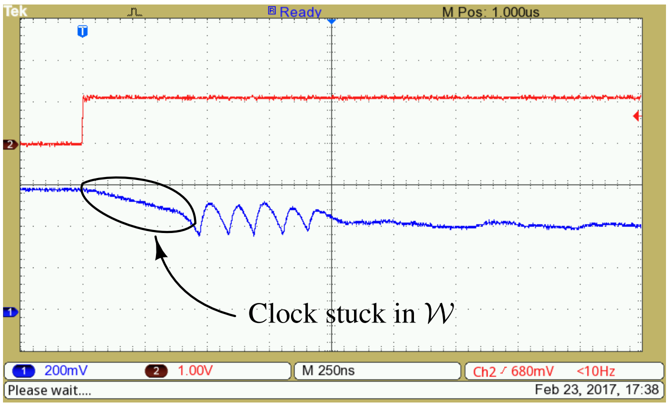


Figure 6.7: Measured extension (about 725 cycles) of the settling time, with the clock initialized within the window $\mathcal{W}$. The data is from a PRBS31 source with a Mark-to-space ratio of 0.5. X-scale: 250 ns/div. Upper waveform is the reset (active low) signal.

window for approximately 725 cycles. Notice the abrupt change in the slope of the signal when the circuit escapes the window $\mathcal{W}$, which happens after about 725 cycles in this example. When the circuit is initialized at the edge of this window, such extension of settling is not observed as shown in Figure 6.8.

The extension of the settling time was observed with different data patterns like PRBS7, PRBS15 and PRBS31. In addition, the TG1B1-A data generator produces outputs with different Mark-to-Space ratios. Mark-to-Space ratio is the ratio of the number of 1's to the number of 0's in the data stream. The Mark-to-Space ratios can be chosen between 0.5, 0.25 and 0.125. Even with these different patterns, the extension of settling time occurs when the clock is initialized in the closed region of the data eye. Figure 6.9 shows the control voltage evolution when the

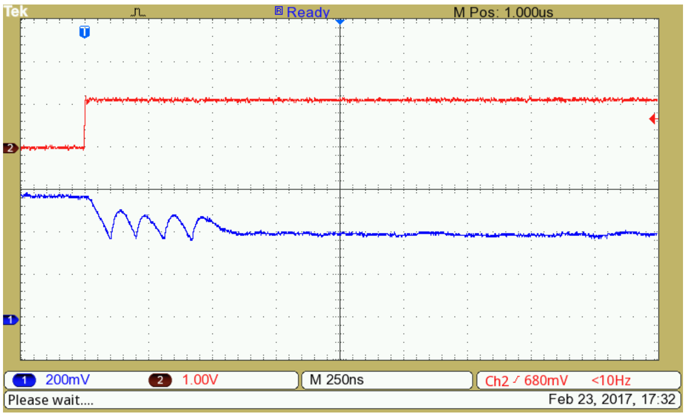


Figure 6.8: Measured settling behaviour with the clock initialized at the edge of the window $\mathcal{W}$. The data is from a PRBS31 source with a Mark-to-space ratio of 0.5. X-scale: 250 ns/div. Upper waveform is the reset (active low) signal.

circuit is stuck in the horizontally closed region of the eye with a data pattern with Mark-to-Space ratio of 0.125, in which case the circuit remains stuck with its clock in the window $\mathcal{W}$ for about 1200 cycles. As expected the total settling time is higher for a lower data activity

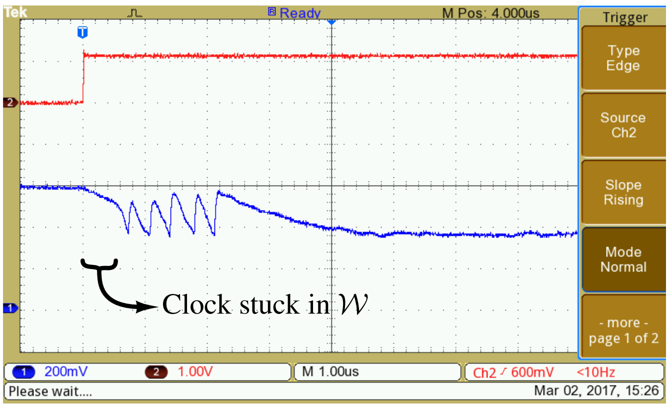
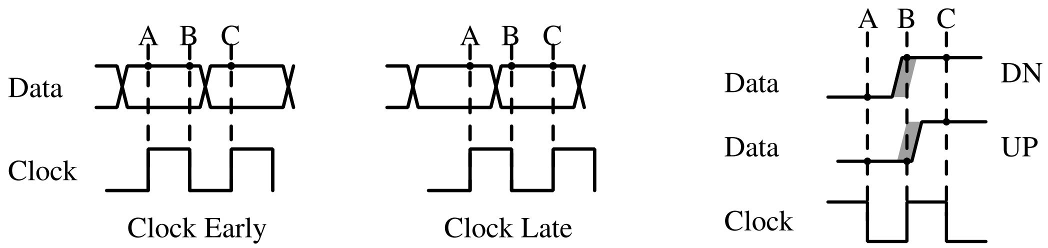


Figure 6.9: Measured extension of the settling time (about 1200 cycles), with the clock initialized within the window $\mathcal{W}$. The data is from a PRBS31 source with a Mark-to-space ratio of 0.125. X-scale: 1 μs/div. Upper waveform is the reset (active low) signal.

factor. It is worth noting that changing the Mark-to-Space ratio only changes the data activity factor. The minimum run length for logic 1 and logic 0 is still 1-bit. We will later show that by changing the minimum run length for one of the logic levels, the settling time can be reduced considerably.

Working of phase detectors in the window of susceptibility

The timing diagram of the Alexander phase detector [77] is shown in Figure 6.10(a). The phase



(a) Sampling instants of the phase detector. **(b)** Sampling in the presence of jitter.

Figure 6.10: (a) The sampling instants of the Alexander phase detector under normal conditions and (b) The sampling instants of the Alexander phase detector when $\Delta\phi \sim \pi$ and the data is corrupt with jitter.

detector takes 2 samples per bit period and takes a binary decision of shifting the clock to the right or to the left, based on the last three samples. When there is no ISI, the phase detector consistently produces either UP or DN pulses. Thus, for a given data activity factor α, the settling time can be calculated using the expression in equation (6.4).

Figure 6.10(b) shows the timing diagram of the Alexander phase detector when the data is corrupt with timing jitter and the initial phase error is close to π radians, i.e. the initial clock position is in the window $\mathcal{W}$. Due to jitter, the sample of the data signal taken close to the data transition, the value of which determines the phase detectors decision, causes the phase detector to generate UP or DN signals depending only on the current data transition time and not on the average data arrival time. Hence, the phase detector produces UP and DN pulses randomly and the clock recovery circuit can remain stuck, with its clock in this region, indefinitely.

Linear phase detectors, like the Hogge phase detector [78], behave similar to binary phase detectors in the window of susceptibility. The ideal characteristics of the Alexander and Hogge phase detectors are shown in Figure 6.11. The highlighted region shows the window $\mathcal{W}$, and within this window the variation in the gain of the phase detector is negligible. Hence, the analysis of the system behaviour in the window $\mathcal{W}$ is applicable to both these types of phase detectors.

In the rest of this chapter we will analyze the settling time of mesochronous clock retiming circuits in detail. We will model the system as a Markov chain for the purpose of analysis and analyze the settling time in presence of different types of jitter. We will also present some techniques of reducing the settling time.

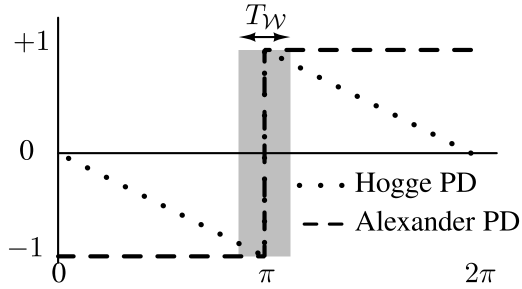


Figure 6.11: The characteristics of linear and binary Phase Detectors (PD). Highlighted region indicates the window $\mathcal{W}$. The phase detector gains are normalized.

6.3 The Markov chain model of the clock recovery circuit

In order to analyze the settling time, when the initial clock is in the window $\mathcal{W}$, the circuit is modeled as a Markov chain with absorbing states. The binary phase detector produces UP or DN pulses on every data transition, which are converted into an analog control voltage using a charge pump. The control voltage is used to delay the clock (either using a phase interpolator or a delay line). Since the input is binary, the output phase is quantized. Hence, the clock position can be discretized to the step size (τ) of the phase detector loop update. In order to keep the jitter in the recovered clock low, controllers typically use step sizes of less than 0.1% of the clock period [20, 57]. The region where the input data eye is closed, i.e. the window $\mathcal{W}$, is of particular interest. A Markov chain model of the system is constructed, in which the states designate the clock positions and the phase corrections performed by the clock retiming circuit form its state transitions. The edges of the window $\mathcal{W}$ are modeled as absorbing states.

The sources of timing jitter are data dependent (ISI induced) jitter in the data and noise induced random jitter in the clocks of the transmitter and receiver. The analysis of each type of jitter is done independently, followed by an analysis for data dependent and random jitter together. For analyzing the effect of data dependent jitter, an interconnect link with different bandwidth's is considered. The interconnect is modeled as a 20 section RC network. For simulating different amounts of ISI, different values of RC time constants are chosen and the eye diagrams and timing jitter histograms for a few considered channel bandwidths are listed as cases 1 through 3 as follows:

Case 1: For benign channels the horizontal eye opening approaches 100% and the jitter histogram has a narrow distribution as shown in Figure 6.12.

Case 2: As the bandwidth of the channel decreases, the jitter histogram splits and produces two distinct peaks [90]. This shows that the ISI due to the immediate previous bit is dominant (Figure 6.13).

Case 3: Further reduction in the bandwidth shows that the jitter histogram splits into 4 regions as shown in Figure 6.14. This shows that the ISI due to previous two bits is dominant.

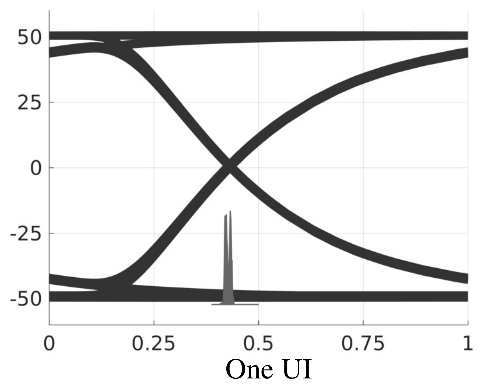


Figure 6.12: Eye diagram and zero crossing histogram for a benign channel.

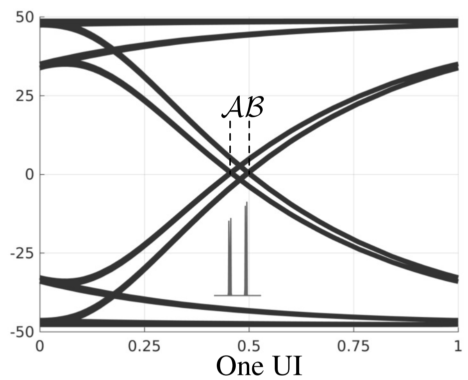


Figure 6.13: Eye diagram and zero crossing histogram for a channel for which the ISI due to the immediate previous bit is dominant.

For modeling effect of random jitter, the random jitter is assumed to have a gaussian distribution. The system is modeled for the cases when a) the jitter is induced by ISI due to 1 previous bit, b) the jitter is induced by ISI due to 2 previous bits, c) the jitter is random with a Gaussian distribution and d) the jitter is induced by random jitter and ISI due to 1 previous bit.

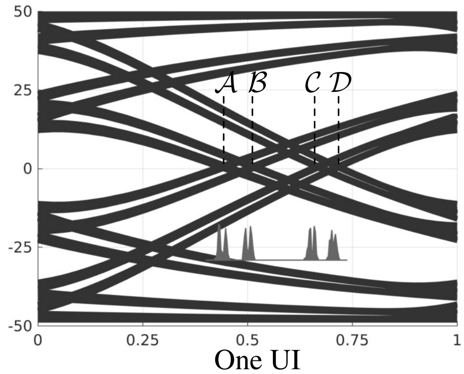


Figure 6.14: Eye diagram and zero crossing histogram for a channel for which the ISI due to previous 2 bits is dominant.

6.3.1 Markov chain model for jitter induced by 1 bit ISI

When the ISI due to the immediate previous bit is dominant, the zero crossings of the data are bunched into two narrow distributions as shown in Figure 6.13. This is approximated to the data signal following one of two distinct traces. Figure 6.15 illustrates an eye diagram with ISI due to exactly 1 previous bit. Here, t_i is the initial sampling instant of the clock, γ is the

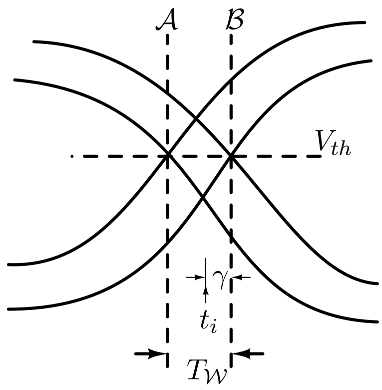


Figure 6.15: Cartoon of an eye diagram with ISI limited to 1 previous bit.

distance to the right edge of the window and V_{th} is the threshold of the samplers in the phase detector. $\mathcal{A}$ and $\mathcal{B}$ represent the two distinct zero crossing times of the data signal. For the eye diagram shown in Figure 6.13, the size of the window $\mathcal{W}$ is about 4% of the bit period. This means $T_{\mathcal{W}}$ is about 40τ. Assuming that the source outputs 1 and 0 with equal probability, the bit combinations that result in traces with zero crossing at $\mathcal{A}$ and $\mathcal{B}$, respectively, are listed in Table 6.1. Here, b_0 is the current bit corrupt with ISI due to b_{-1}. A data transition occurs when $b_0 \neq b_1$. Table 6.1 lists all 3 bit combinations which cover all possible data traces.

Here, LT and RT indicate that the clock is shifted to the left and to the right respectively,

Table 6.1: Possible sequences and data traces for 1 bit ISI.

b_{-1}	b_0	b_1	Zero crossing time	Action
0	0	0	-	NA
0	0	1	$\mathcal{B}$	LT
0	1	0	$\mathcal{A}$	RT
0	1	1	-	NA
1	0	0	-	NA
1	0	1	$\mathcal{A}$	RT
1	1	0	$\mathcal{B}$	LT
1	1	1	-	NA

by a step of size τ. NA indicates no corrective action in that cycle. When all sequences are equally likely, the probabilities $P(RT) = 0.25 = P(LT)$ and $P(NA) = 0.5$. Note that the system escapes the window of susceptibility $\mathcal{W}$ when, for the first time, either $(n_R - n_L)\tau \geq \gamma$ or $(n_L - n_R)\tau \geq (T_{\mathcal{W}} - \gamma)$. Here, n_R and n_L are the total number of times the clock position has been shifted to the right and to the left, respectively, from start-up.

This system is modeled as a one dimensional Markov chain. The states of the Markov chain are the positions of the sampling clock in the window $\mathcal{W}$. Once the clock escapes this window, the time taken to lock to the center of the eye can be calculated using equation (6.4). Thus, the edge positions of the window are modeled as absorbing states. This makes the model a Markov chain with absorbing states [91]. Figure 6.16 shows the state diagram of the Markov chain for this system. By knowing the state space and the transition probabilities of a Markov

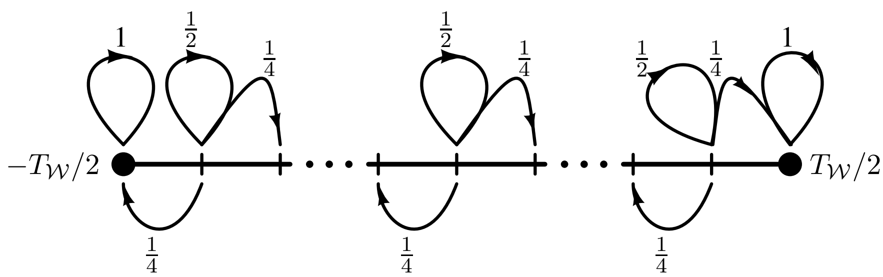
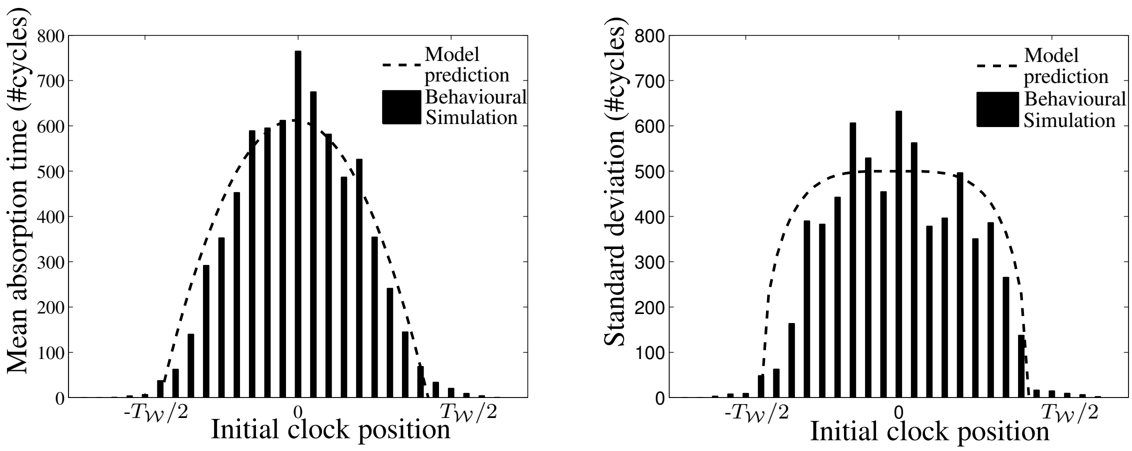


Figure 6.16: Model of the clock recovery circuit, when the data has ISI limited to 1 previous bit, as a Markov chain with absorbing states.

chain, one can calculate the mean time to absorption from any initial state [91]. The calculation of the mean time to absorption (and its variance) are presented in Appendix B.

The combined plots of the mean time to absorption for the 1 bit ISI case obtained from behavioural simulations and Markov chain model predictions are shown in Figure 6.17(a). The model predictions were computed by solving for the mean absorption time of this Markov chain using a linear equation solver, while the behavioural simulations were done using a 20 section RC interconnect and a VerilogA behavioural description of the clock retiming circuit. As one would expect, the settling time is maximum when the initial sampling phase is at the center of the window $\mathcal{W}$.



(a) Mean absorption time. **(b)** Standard deviation of absorption time.

Figure 6.17: Mean and standard deviation of the absorption time as predicted from the Markov chain model and from behavioural simulations. Each data point in the behavioural simulation is an average of 100 runs.

It is worth noting that the variance in the absorption time is quite high. Figure 6.17(b) shows the standard deviation of the absorption time as predicted by the Markov chain model and as observed in the data obtained from behavioural simulations.

6.3.2 Markov chain model for 2 bits ISI

When the ISI due to previous two bits is significant, the data transitions histogram has 4 peaks as shown in Figure 6.14. This can be approximated to ISI of two bits which results in 4 distinct data transition times. Figure 6.18 illustrates an eye diagram when ISI is limited to two bits and the four data transition times are labeled as $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$ and $\mathcal{D}$. Thus, the window of susceptibility $\mathcal{W}$ can be divided in to three sub-windows which are $\mathcal{W}_{\mathcal{A}-\mathcal{B}}$, $\mathcal{W}_{\mathcal{B}-\mathcal{C}}$ and $\mathcal{W}_{\mathcal{C}-\mathcal{D}}$. Unlike the 1 bit ISI case, the occurrences of the traces with zero crossing time at $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$ and $\mathcal{D}$ are not independent of each other. Thus, the system is no longer memoryless, and hence, requires a

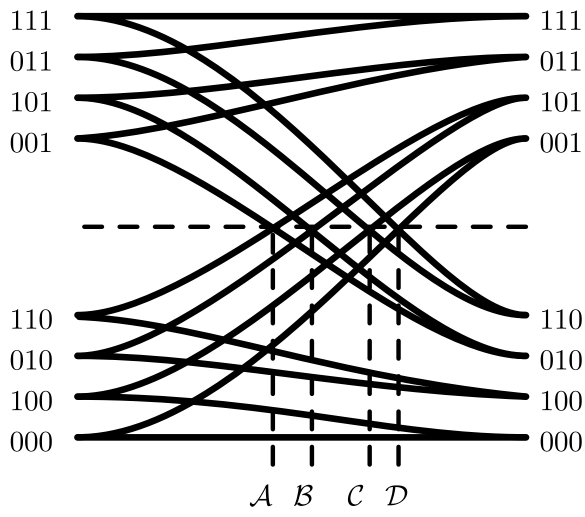


Figure 6.18: Cartoon of an eye diagram with ISI due to previous 2 bits, showing the four discrete data transition times.

second order Markov chain model. A second order Markov chain model is, in general, difficult to analyze. A technique to convert this second order Markov chain to a first order Markov chain (with a slightly bigger state space) is described in the following. This greatly simplifies the analysis, as known results for first order Markov chains can be directly applied.

To construct the Markov chain model of the system, one needs to first consider the order of occurrences of the traces with zero crossing times at $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$ and $\mathcal{D}$. Figure 6.19 shows the state diagram of the source data. The FSM is limited to 3 bit combinations as the ISI for every bit is limited to previous two bits. The state transitions are labeled with the zero crossing locations of the data signal $\mathcal{A}$ through $\mathcal{D}$. $\mathcal{X}^{(i)}$,$i = 1, 2$ indicates a state transition without a data transition.

The second order Markov chain model for this system is shown in Figure 6.20. To capture the memory of the system, six states are used for each clock position. Four of these states correspond to the previous data transition (T_{n-1}) occurring at $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$ and $\mathcal{D}$. The remaining two states are i) $\mathcal{X}^{(1)}$ which indicates no data transition in the previous cycle and ii) $\mathcal{X}^{(2)}$ which indicates no data transition in the previous two cycles.

For example, when $T_{n-1} = \mathcal{A}$, the source can be in either state S_2 or S_5 (refer Figure 6.19), and the next cycle will either not have a data transition, or have a data transition at position $\mathcal{B}$. When the initial clock is in the sub-window $\mathcal{W}_{\mathcal{B}-\mathcal{C}}$, a data transition at $\mathcal{B}$ results in a clock shift to the right by 1δ. Thus, the transitions from state at $T_{n-1} = \mathcal{A}$ are either to state $\mathcal{X}^{(1)}$ at the same clock position or to state $T_{n-1} = \mathcal{B}$ of the next clock position to the right. Similarly, by analyzing the FSM in Figure 6.19, the rest of the state transitions can be determined. In

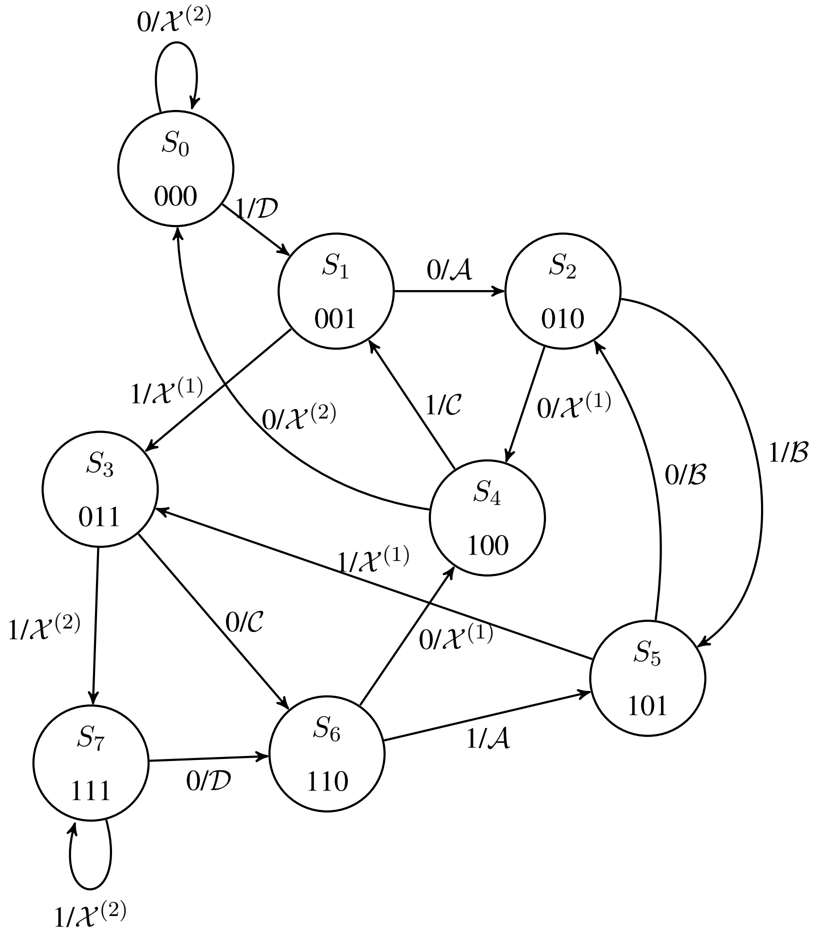


Figure 6.19: State diagram indicating the possible transitions and data zero crossing positions, for data corrupted with ISI from previous 2 bits.

Figure 6.20, the transitions are shown for only one clock position for brevity, for the clock positions in the sub-window $\mathcal{W}_{\mathcal{B}-\mathcal{C}}$. All the state transitions have a probability of 0.5 when the data source outputs 0 and 1 with equal probability.

Similar to the 1 bit ISI case, the mean time to absorption for the 2 bit ISI case was determined using a Markov chain model. Also, behavioural simulations were performed. For the eye diagram shown in Figure 6.14, the window size is about 30% of the clock period. Here, $\mathcal{W}_{\mathcal{A}-\mathcal{B}} \sim 8\%$, $\mathcal{W}_{\mathcal{B}-\mathcal{C}} \sim 15\%$, and $\mathcal{W}_{\mathcal{C}-\mathcal{D}} \sim 7\%$. Accordingly, the Markov chain was constructed and analyzed using a linear equation solver. The mean time to absorption for each clock position was taken as the average of the mean times to absorption from each of the six states in the Markov chain that correspond to that position. For the behavioural simulations, a 20 section RC interconnect was used with a VerilogA behavioural description of the clock retiming circuit. In order to reduce the simulation time and the required sample size, a step size

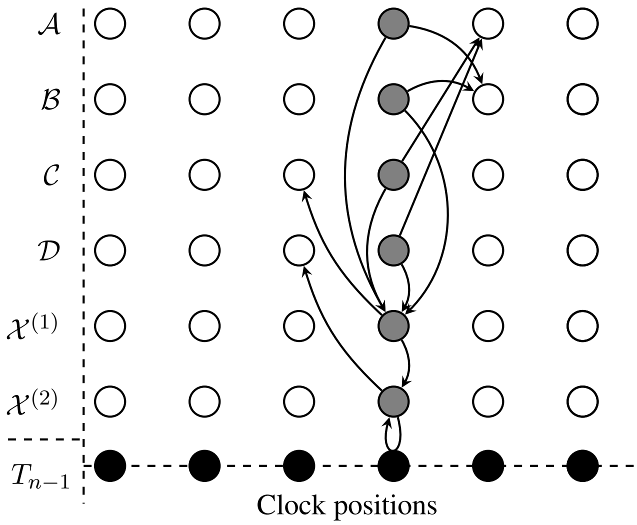
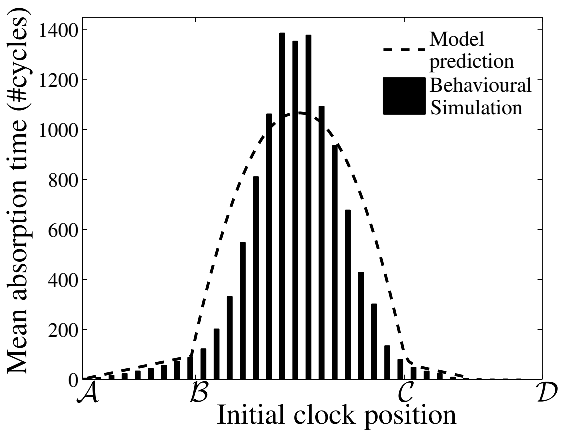
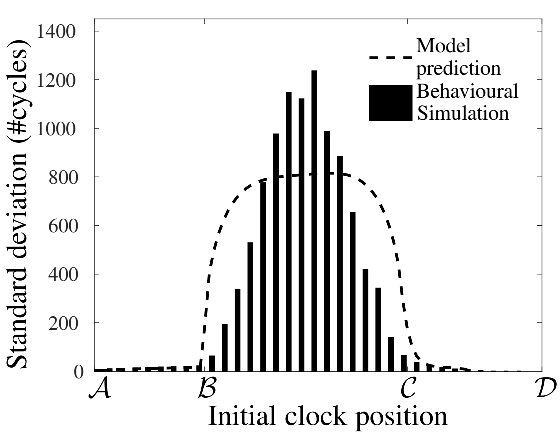


Figure 6.20: First order Markov chain model (with extended state space) for two bit ISI case. Indicated clock positions and transitions correspond to the sub-window $\mathcal{W}_{\mathcal{B}-\mathcal{C}}$. Transitions are indicated for only one clock position for brevity. T_{n-1} indicates location of immediate previous transition. $\mathcal{X}^{(i)}$, $i = 1, 2$ indicates no data transition in the previous cycle and in the previous two cycles respectively. All transitions out of a state have equal probability.

of 3.5τ was used for this analysis. The combined plots of the mean time to absorption for 2 bit ISI case obtained from behavioural simulations and Markov chain model predictions are shown in Figure 6.21(a). The standard deviation of the absorption time is shown in Figure 6.21(b). The behavioural simulations used an RC circuit for the interconnect, whereas the Markov chain



(a) Mean absorption time.



(b) Standard deviation.

Figure 6.21: Mean absorption time and standard deviation of absorption time as predicted from the Markov chain model and from behavioural simulations. Each data point in the behavioural simulation is the average of 350 runs.

model truncates the ISI to 2 bits. Note that the peak of the mean time to absorption, as predicted by the model, is about 1100 cycles when the step size is 3.5τ. This can be as high as 12000 cycles when the step size is 1τ.

6.3.3 Markov chain model for Gaussian distributed random jitter

This section discusses the increase in the settling time due to random jitter in the clock. There are two sources of clock jitter in digital systems: inherent jitter in the clock generating oscillator and jitter introduced by the clock distribution network. The former can be modelled as a Gaussian distribution [92]. The latter, however, depends on several factors and can be random with arbitrary distribution [93]. The analysis in this section assumes that the jitter has a Gaussian distribution.

For ease of analysis, it is assumed that the receiver clock is noise free and all the jitter is in the transmitter clock. This assumption is valid as long as the channel response is constant over the spectrum of the noisy clock. The jitter histogram is assumed to be a Gaussian distribution, with a standard deviation of σ_{ck}. The sampling clock position (t_{ck}), with respect to the mean data transition position, can be represented as as $m\tau$, where m is an integer taking values from $-\infty$ to ∞. Hence, with respect to the sampling clock position, the mean of the data transition can be written as $-m\tau$. Unlike the analysis for ISI induced jitter, wherein the window of susceptibility $\mathcal{W}$ was bounded, random jitter is unbounded and a window $\mathcal{W}$ of finite size cannot be defined. For tractable analysis, a window size of $\pm 3\sigma$, i.e. $|m\tau| < 3\sigma_{ck}$, is used. This covers $> 99.999\%$ of the possible data transition positions.

When the clock transition is to the right of the mean data transition position ($m > 0$), the circuit shifts the clock to the right towards final lock (m is incremented). However, even when $m > 0$, the random jitter in the data can cause the instantaneous data transition to occur to the right of the sampling clock position. This results in a clock shift in the wrong direction. Figure 6.22 shows the probability distribution of the clock transition time and the data transition position. The shaded area represents the probability that the data transition occurs to the right of the sampling clock position, when the mean of the data transition is to the left of the sampling clock transition ($m > 0$). This is the probability of a phase update in the wrong direction. This

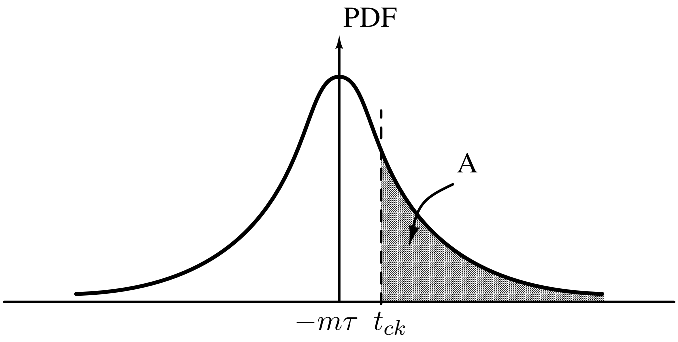


Figure 6.22: Probability distribution of clock transition time and the position of data transition for Gaussian random jitter.

probability can be calculated as follows.

$$\begin{aligned}
P(update: wrong) &= \int_{t_{ck}}^{\infty} \frac{e^{\frac{-(x+m\tau)^2}{2\sigma_{ck}^2}}}{\sqrt{2\pi}\sigma_{ck}} dx \\
&= 1 - \int_{-\infty}^{m\tau} \frac{e^{\frac{-(x+m\tau)^2}{2\sigma_{ck}^2}}}{\sqrt{2\pi}\sigma_{ck}} dx \\
&= 1 - \frac{1}{2}\left[1 + erf\left(\frac{2m\tau}{\sqrt{2}\sigma_{ck}}\right)\right] \\
&= \frac{1}{2} - \frac{1}{2} erf\left(\frac{2m\tau}{\sqrt{2}\sigma_{ck}}\right).
\end{aligned}$$

The probability of an update in the correct direction can then be calculated as $P(update : right) = 1 - P(update : wrong)$. Note that these probabilities will be scaled by the probability of transitions. For a source which outputs 1 and 0 with equal probability, the probability of transition is also $1/2$. Since the data transitions are independent of the clock jitter, the probabilities can be multiplied. This equation is then used to construct the probability transition matrix of the Markov chain, to predict the mean time to absorption and the standard deviation of the time to absorption. As noted earlier, absorption in this case is defined as the phase difference between the mean of data position and sampling clock position being $> 3\sigma_{ck}$. Figure 6.23 shows the plots of the mean time to absorption and the standard deviation of the time to absorption with the initial clock position as estimated by the Markov chain model, when $\sigma_{ck} = 20\tau$, i.e. standard deviation of the clock jitter is 2% of the clock period.

6.3.4 Modeling the effect of ISI and random jitter

The Markov chain model of the system can be easily extended to analyze the combined effect of data dependent and random jitter. An example case of jitter induced by 1 bit ISI and Gaussian

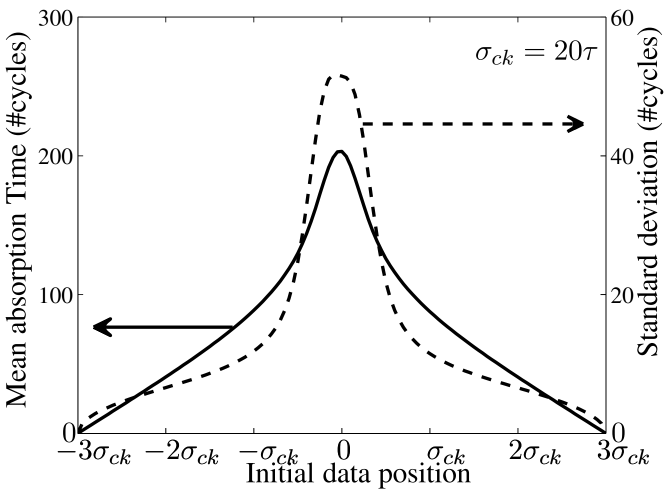


Figure 6.23: Mean absorption time and standard deviation of the absorption time with Gaussian distributed random jitter. The standard deviation of the clock jitter is assumed to be 20τ (i.e. 0.02UI).

distributed random jitter is shown in this section. When the jitter is solely due to 1 bit ISI, the data follows one of two distinct traces as discussed in Section 6.3.1. When random jitter is also present, the zero crossing of the data spreads around these two zero crossing positions. Since the random jitter is independent of the data, the probability of zero crossing at any position is equal to the product of the contributions of each of the sources of jitter. For the purpose of analysis it assumed that the receiver clock is jitter free and the transmitter clock has all the random jitter and the channel introduces the data dependent jitter.

Figure 6.24 shows the distribution of the data zero crossing positions and the sampling clock position when the data signal has both random jitter and 1 bit ISI induced jitter. The window of susceptibility is now widened to $3\sigma_{ck} + \mathcal{W}_{\mathcal{A}-\mathcal{B}} + 3\sigma_{ck}$.

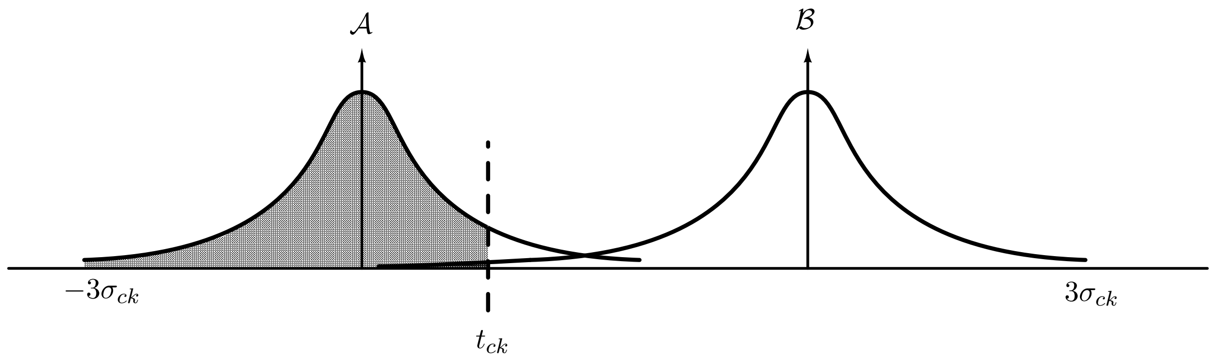


Figure 6.24: Probability distribution of clock transition time and the position of data transition for Gaussian random jitter along with 1 bit ISI.

The effect of the data jitter can be modeled as a shift in the mean of the gaussian random jitter. Hence, for random jitter along with jitter due to 1 bit ISI, the jitter at any time can be considered to be Gaussian distributed, with a standard deviation of σ_{ck} and mean at either $\mathcal{A}$ or $\mathcal{B}$. Consider an example clock position (t_{ck}) as shown in Figure 6.24. Choosing $\mathcal{A}$ as the reference position for calculations, the probability of a data transition occurring to the left of the clock is given by

$$P(left) = P(\mathcal{A}) \times \Phi\left(\frac{t_{ck} - 0}{\sigma_{ck}}\right) + P(\mathcal{B}) \times \Phi\left(\frac{t_{ck} - \mathcal{W}_{\mathcal{A}-\mathcal{B}}}{\sigma_{ck}}\right),$$

where $\Phi\left(\frac{x-\mu}{\sigma}\right)$ is the Cumulative Distribution Function (CDF) of a general Gaussian distribution with mean μ and standard deviation σ. The probability of a data transition to the right of the clock can then be calculated as

$$P(right) = 1 - P(NT) - P(left),$$

where, $P(NT)$ is the probability of not having a data transition in that clock cycle.

The Markov chain and its transition probabilities were computed in this manner and the mean settling and the standard deviation of the settling time was analyzed. Figure 6.25 shows the mean absorption time and the standard deviation of the absorption time obtained for data with timing jitter induced by 1 bit ISI in addition to Gaussian distributed random jitter with standard deviation of 1% of the clock period.

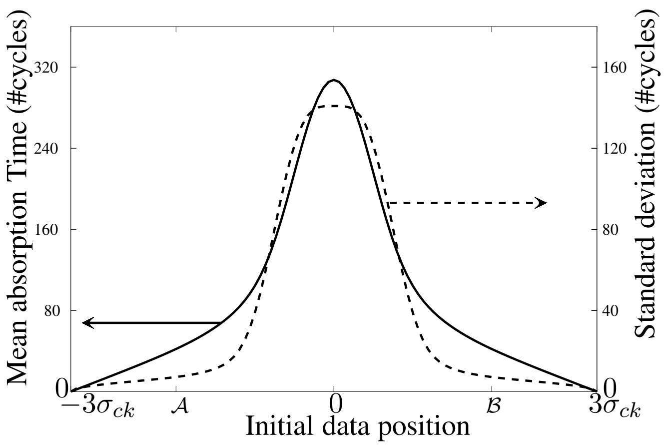


Figure 6.25: Mean absorption time and standard deviation of the absorption time when data has 1 bit data dependent jitter and Gaussian distributed random jitter. The standard deviation of the random clock jitter is assumed to be 10τ (i.e. 0.01UI).

6.4 Quantification of settling time for given confidence level

Since the settling time is not deterministic, it is not possible to quantify the settling time absolutely. However, a bound on the settling time can be assured for a given confidence level. The settling time is maximum when the system starts with its initial clock in the center of the window of susceptibility. We shall estimate the settling time from this position as this is the worst case settling time.

Let $P_{(0)}$ be a row matrix representing the Markov chain's initial condition. In this case this matrix will contain a 1 at the entry corresponding to the center position and the rest of the entries as zeroes. The probability of the Markov chain being in any state after a transition can be calculated by multiplying this row matrix with the PTM of the Markov chain. Similarly, after n transitions the probability of the Markov chain being in any of the states is given by

$$P_{(n)} = P_{(0)}.PTM^n$$

In the presence of absorbing states, the probability of absorption after n iterations is simply the sum of all the entries of $P_{(n)}$ that correspond to the absorbing states. Hence, one can iteratively find the number of transitions required for absorption with a given confidence level.

For the Markov chain of the 1 bit ISI case, with a window of 40 steps, the probability of absorption with the number of transitions is calculated as described above and shown in Figure 6.26 This is actually the cumulative distribution function of the probability of absorption

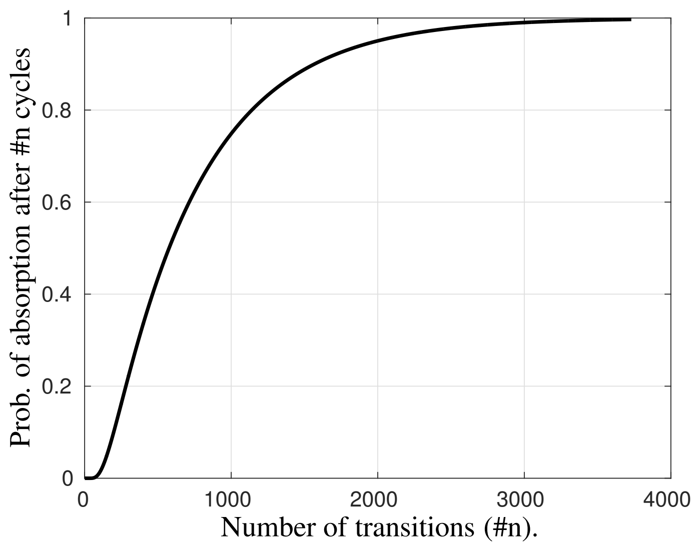


Figure 6.26: Probability of absorption as a function of the number of transitions.

in a given number of transitions. Hence, the probability of absorption in exactly $\#n$ cycles can be found by taking the derivative of the above graph. This is shown in Figure 6.27. This shows

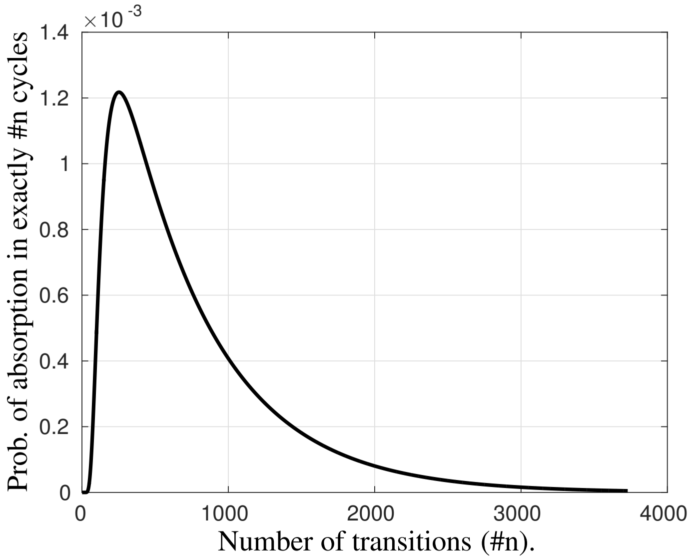


Figure 6.27: Probability of absorption in exactly #n transitions as a function of #n.

that the distribution of probability of absorption is skewed as one would expect. The probability of absorption is zero for #n less than the distance to the nearest absorbing state. For very large values of #n the probability approaches zero asymptotically.

As seen from Figure 6.26, for absorption with 99% surety, the Markov chain needs about 3000 state transitions. An analysis of the number of transitions needed for absorption with 99% confidence with the size of the window was done. Figure 6.28 shows the plot obtained from this analysis.

6.5 Reducing the settling time of the circuit

6.5.1 The coarse first synchronizer

The settling time of the circuit depends on the loop gain or the step size of the phase correction circuit. A large step size will mean that the size of the window of susceptibility (in terms of number of steps) is small. This results in quick settling, as was discussed in the previous section. However, a large step size results in high jitter in the clock after lock is achieved. The synchronizer proposed in Chapter 4, uses a coarse and a fine correction loop for accurate synchronization. In the designed circuit, the circuit first tries to achieve lock using the fine tuning loop. If lock cannot be achieved in the limited range of the fine tuning loop, a coarse

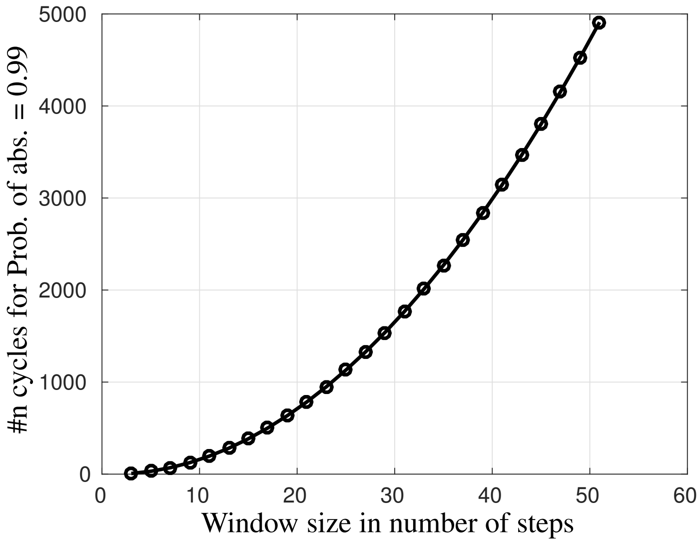


Figure 6.28: Probability of absorption in $\leq n$ transitions, with 99% confidence, as a function of window size for 1 bit ISI example.

correction is initiated. If the fine tuning loop is disabled during the initial settling period, it enables us to have a much larger step size, and hence a much faster settling. When a 10 phase DLL is used, and even if a horizontal eye opening of 50% is available, the size of $\mathcal{W}$ is only 5 steps.

Figure 6.29 shows the block diagram of a retiming circuit that allows a coarse first mode. This is a small modification over the circuit reported in Figure 4.6 of Chapter 4. Figure 6.30 shows the charge pump circuit for this coarse first synchronizer. Here, during the initial period after startup, a switch disconnects the loop filter capacitor and another set of switches convert the charge pump to a combinational circuit, as was done for the scan test in Chapter 5, (i.e. connecting the gate of the PMOS current source to GND and gate of the NMOS current sink to VDD). This essentially disables the fine tuning loop and ensures a coarse tuning step in every cycle of the divided clock that drives the coarse tuning loop. The amount of time for which the coarse tuning loop should be run before enabling the fine tuning loop depends on the size of $\mathcal{W}$ and on the desired confidence for achieving lock within this period. In this example circuit, the coarse synchronizer used a 10 phase DLL. Assuming an extreme condition that the horizontal eye opening is only 50%, it means that the window size is 5 steps. From the analysis presented in the previous section, an absorption with 99% confidence needs 32 cycles. This corresponds to 256 ns in absolute time, if the coarse controller operates on a divide by 16 clock for a 2 GHz system clock.

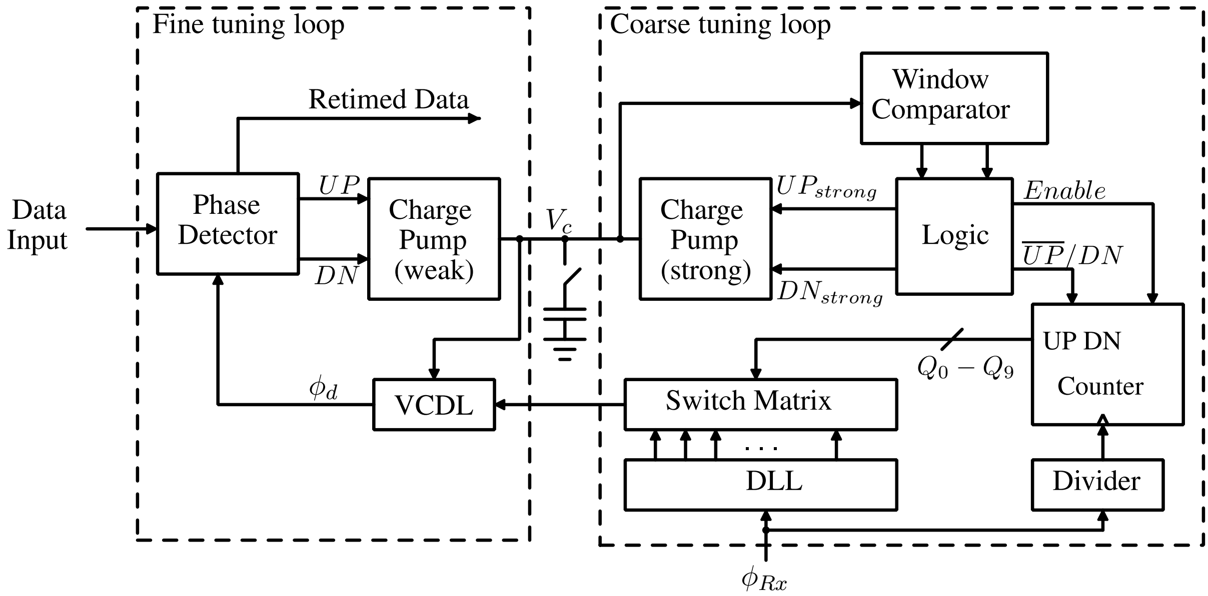


Figure 6.29: Block diagram of the clock synchronizer that used coarse and fine correction for achieving lock, with coarse tuning preceding the fine synchronizer.

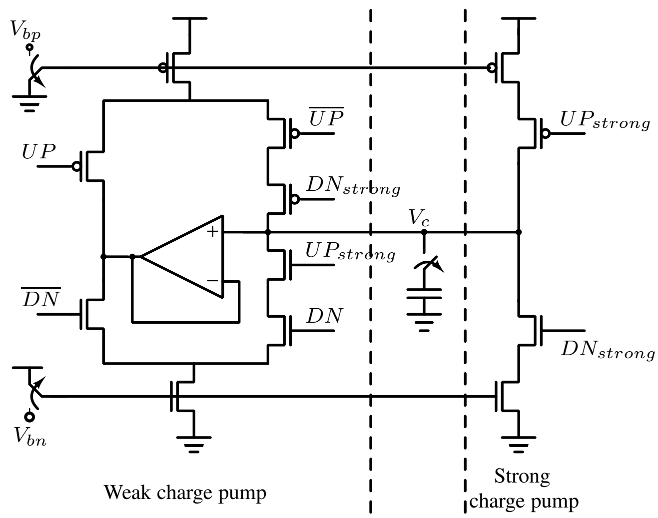


Figure 6.30: Charge pump circuit for enabling coarse first operation of the synchronizer. UP, $\overline{UP}$, DN and $\overline{DN}$ are driven by the fine tuning loop. UP_{strong} and DN_{strong} are driven by the coarse tuning loop. V_{bn} and V_{bp} are the bias voltages for the NMOS and PMOS current sink and source respectively. The switches are used to enable the coarse-first mode.

The coarse first synchronizer was designed and simulated for a benign channel. The receiver input eye diagram for this test is shown in Figure 6.31. The settling behaviour of this synchronizer is shown in Figure 6.32, from an initial clock phase in the center of $\mathcal{W}$. As seen, the circuit escapes the window $\mathcal{W}$ very fast and settles accurately once the fine tuning loop is enabled. The lower waveform is the select signal for switching from coarse first mode to normal

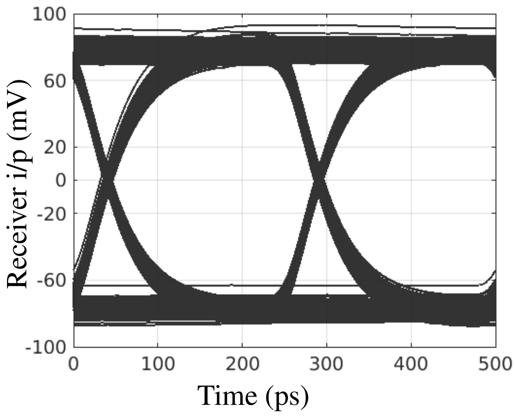


Figure 6.31: Eye diagram and zero crossing histogram for a benign channel.

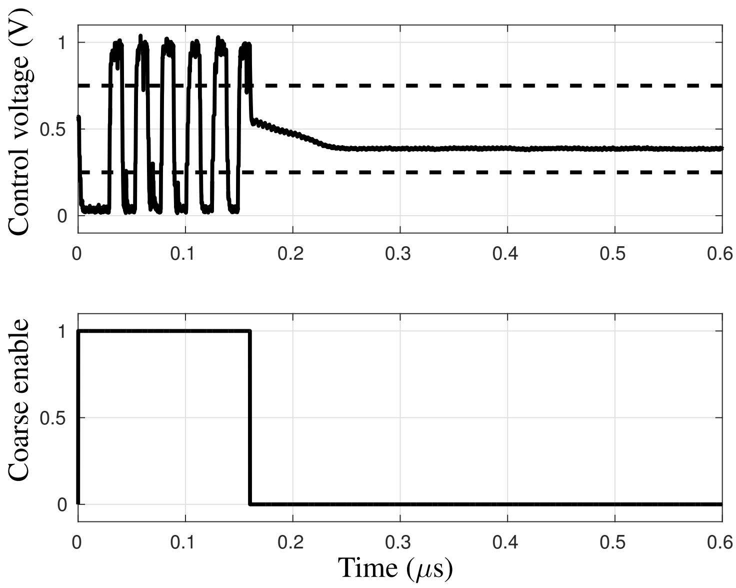


Figure 6.32: Settling of the coarse first synchronizer with input having jitter from a benign channel. The lower waveform is the control signal for enabling the coarse-only mode.

mode, which is run for 160 ns in coarse mode. This signal can be generated using a power on reset circuit.

The same synchronizer circuit was simulated with an input that had very high ISI. The receiver input eye diagram for this test is shown in Figure 6.33. With this data input, the settling of the synchronizer is shown in Figure 6.34. The clock was initialized to the center of the window $\mathcal{W}$. Even for this input, the circuit escapes the window $\mathcal{W}$ within the coarse first operating period. The noise in the control voltage is primarily due to the jitter in the incoming data.

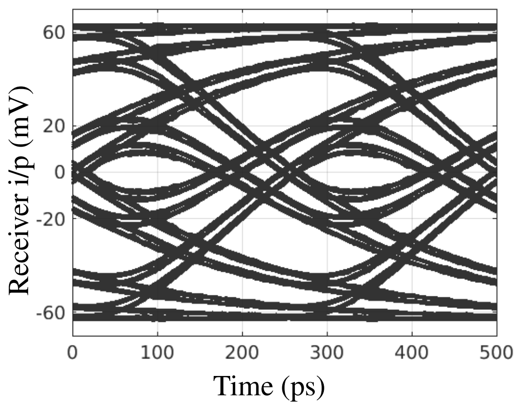


Figure 6.33: Eye diagram and zero crossing histogram for a channel with high ISI.

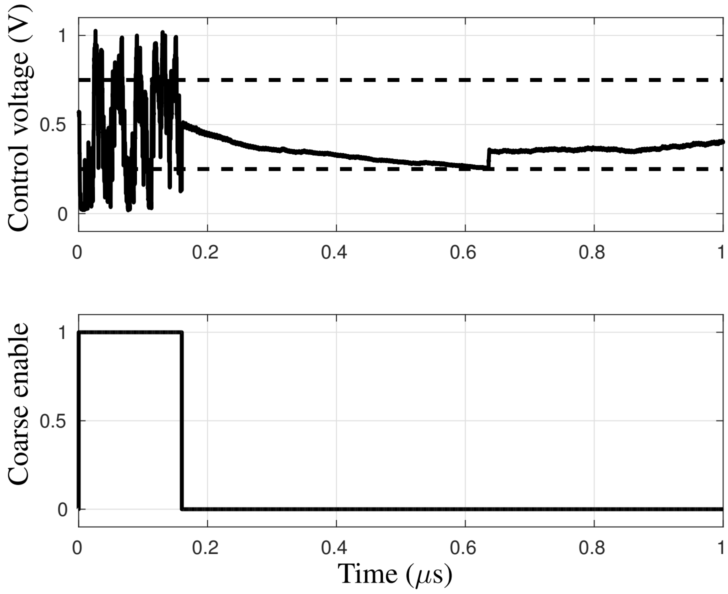


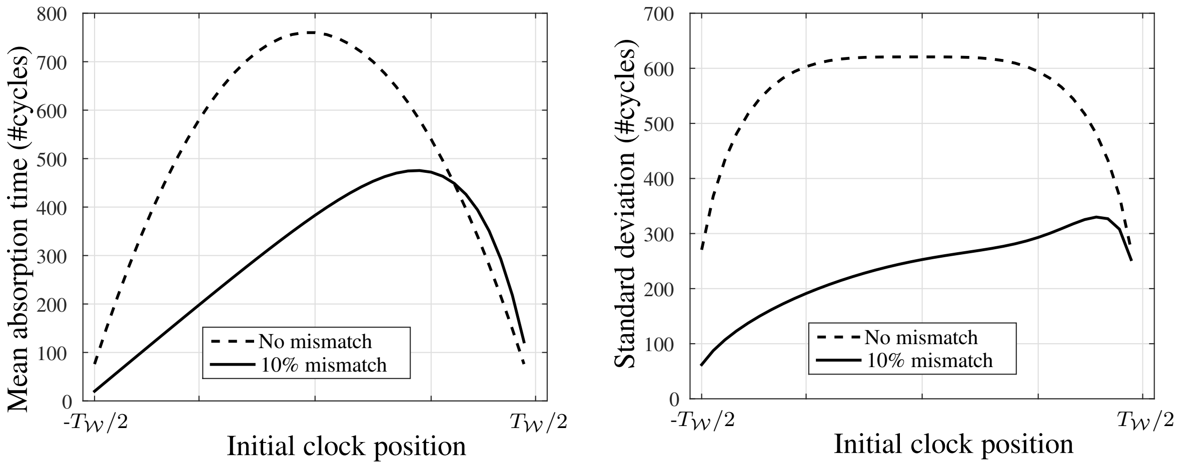
Figure 6.34: Settling of the coarse first synchronizer when the input signal has high ISI, with only 50% horizontal eye opening. The lower waveform is the control signal for enabling the fine correction.

6.5.2 Fine first synchronizer with bias to one of the absorbing states

As discussed in the previous section, the settling time of mesochronous clock retiming circuits can be quite high along with a large standard deviation. In this section, a technique to reduce the settling time by introducing a mismatch in the relative strengths of the UP and DN updates is discussed. The 1 bit ISI model is used for illustration purposes.

Figure 6.35(a) shows the mean time to absorption as a function of initial sampling phase, when the step sizes for the left and the right shifts are (i) equal and (ii) mismatched by 10%.

Note that a 10% mismatch in the step size of the left/right shift reduces the mean ab-



(a) Mean absorption time.

(b) Standard deviation of the absorption time.

Figure 6.35: (a) Mean absorption time and (b) standard deviation of the absorption time, for a symmetric Markov chain and for a chain with a 10% bias to left shift.

sorption time by up to 40%. The effect of this mismatch is more pronounced in the standard deviation of the absorption time as a small asymmetry is shown to result in a large reduction in the standard deviation. Figure 6.35(b) shows the plot of the standard deviation of the absorption time with and without the UP/DN mismatch. Hence, to reduce the settling time of the circuit, one can deliberately introduce a mismatch in the UP/DN strengths of the charge pump in the circuit. Alternately, a training sequence that generates a similar mismatch, by producing ISI biased towards one side of the window $\mathcal{W}$, can reduce the settling time. The settling time, with 99% confidence, reduces from 3000 cycles to 1000 cycles with a 10% mismatch in the left right jump probabilities.

6.6 Results and Discussions

As discussed before, typical systems on chip can have hundreds of long interconnects and a horizontal eye opening of about 85% results in a 15% chance that a system wakes up with its initial clock in the window of susceptibility. This section discusses the results of the suggested techniques for reducing the settling time. An implementation of a 10 mm interconnect with a capacitively coupled transmitter and a coarse-fine type clock recovery circuit [20], in UMC 130 nm CMOS technology, operating at 2.5 GHz, is used for the simulation experiments.

6.6.1 Reduction in settling time with charge pump mismatch

Figure 6.36 shows the reduction in the settling time of the circuit with the introduction of a 10% mismatch in the UP and the DN currents of the charge pump. Under identical initial conditions, a simulation with 10% mismatch showed >80% reduction in the time taken to exit the window $\mathcal{W}$. Charge pump mismatch can result in increased jitter once the circuit has locked. Hence, if

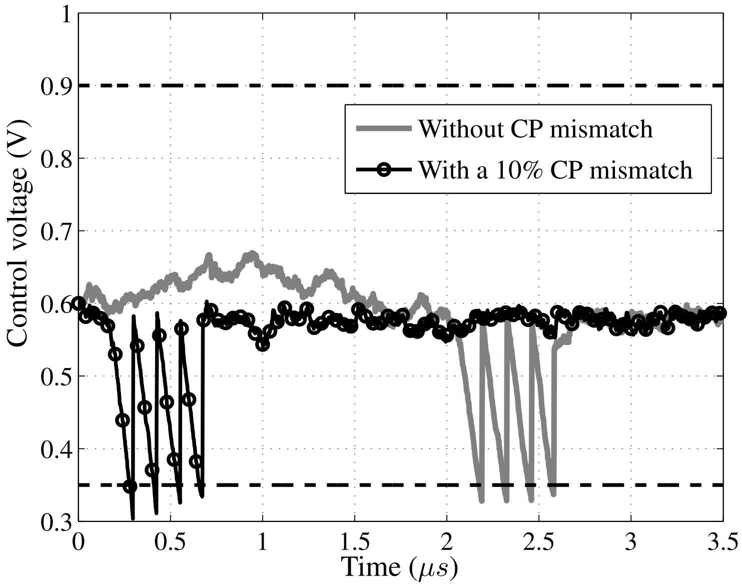


Figure 6.36: Control voltage of the clock recovery circuit when $\Delta\phi \sim \pi$ radians, showing the quick escape from the window $\mathcal{W}$ when the charge pump has a mismatch in the UP and DN currents.

good jitter performance is desired, the introduced mismatch can be switched off after lock has been achieved.

6.6.2 Reduction in settling time with training sequence

A training sequence that biases the system to either one of the directions can be used to reduce the settling time. One such training sequence is "...0010011100100111...". This sequence has a minimum run length of 1 bit for a logic '1' and 2 bits for logic '0'. Hence, the ISI for logic '1' is always more than that for logic '0' and the phase detector's output in the window $\mathcal{W}$ is biased towards the right. Figure 6.37 shows the control voltage as the circuit settles, when the above training sequence is used as well as with random equiprobable binary sequence. Notice the considerable reduction in the settling time when the deliberately biased training sequence is used. (The final value of the control voltage is not the same because of the loop hysteresis.

Refer to Figure 4.31 and associated discussion). One could also use an alternating 1 and 0 training sequence which reduces the jitter due to ISI to zero. However, random uncorrelated jitter between the transmitter and receiver clocks will still result in a non-zero size of the window $\mathcal{W}$.

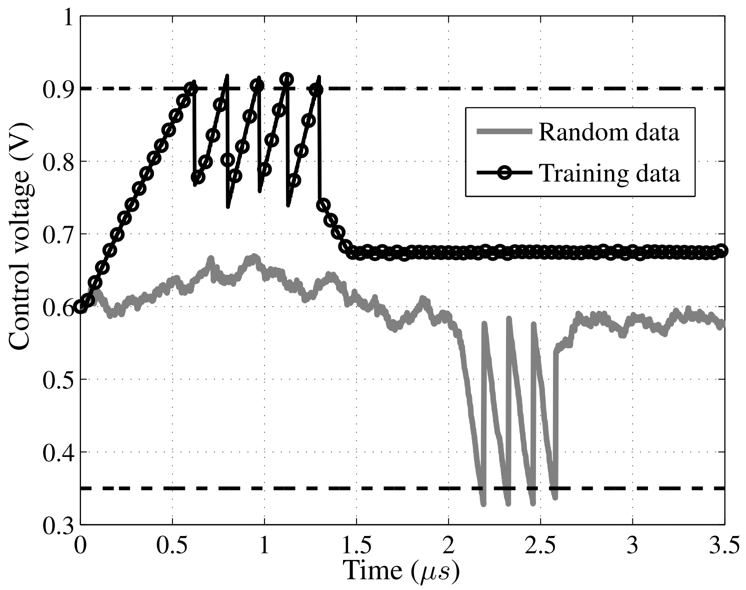


Figure 6.37: Control voltage of the clock recovery circuit when $\Delta\phi \sim \pi$ radians, showing the quick escape from the window $\mathcal{W}$ when a biased training sequence is used.

6.7 Summary

The effect of jitter on the settling time of mesochronous clock retiming circuits is discussed in this chapter. It is shown how ISI induced jitter and random jitter can increase the settling time of clock recovery circuits indefinitely, using simulations as well as with a practical demonstration in a fabricated chip. A model of the system as a Markov chain with absorbing states is developed. Using Markov chain models, the effect of different types of jitter is analyzed. The include, data dependent jitter introduced by 1 previous bit, by 2 previous bits, random jitter, and simultaneous data dependent and random jitter. The model predictions of the settling time in terms of the mean absorption time of the Markov chain match well with behavioural simulations.

Techniques for reducing the settling time are reported, which originate from the insights gained from the model. A coarse first synchronizer, a variant of the clock retiming circuit reported in Chapter 4, that uses only coarse correction steps initially, is reported. This architecture achieves quick settling, in the presence of substantial ISI. Another technique of reducing the settling time by introducing a mismatch between the phase updates in either direction of the clock retiming circuit is also discussed. This is applicable to phase interpolator based clock retiming circuits and also to the clock retiming circuit discussed in Chapter 4. This mismatch in the is achieved either by introducing a mismatch in the charge pump or by using appropriately designed training data. All the suggested fast settling synchronizers are verified with circuit simulations.

Chapter 7

Conclusions and Future scope

7.1 Conclusions

The work reported in this thesis started when low swing repeaterless interconnects, were being suggested as a power efficient alternative to the power hungry repeater inserted long interconnects. Many works had reported the significant power and performance gains that can be achieved with low swing interconnects that use equalizers to achieve high speed data transmission. These propositions raised a lot of interest, but also some skepticism about their practicality due to the following facts.

1. The low swing interconnects still had high latency. Hence, clock synchronizing circuits would be required at the receiver to sample the data correctly. The sampled data has to then be transferred to the receiver clock domain reliably.

2. The use of mixed signal circuits meant that the testability of the system would be affected.

3. Digital designs have to keep pace with technology nodes which change pretty quickly. While it is easy to design digital standard cells in new technologies, the same cannot be said about the design of low swing interconnects.

The work reported in this thesis attempts to address these challenges.

Chapter 3 discusses the design of equalizers for low swing interconnects. A capacitively coupled two tap feed-forward equalizer is presented. The circuit uses a weak driver, that enables arbitrarily low data activity factor or frequency of operation. A design optimization method, that uses worst case sequences, is also shown. This design method allows fast design of this type of transmitter. Chapter 3 also reports a low latency DFE circuit, using the sense amplifier comparator. The circuit uses a switched capacitor implementation of the feedback path, which allows minimum latency. Further, it also allows the offset of the comparator to be compensated using the same feedback transistors.

A clock synchronizing circuit for repeaterless low swing interconnects is discussed in Chapter 4. The synchronizer uses a coarse tuning loop along with a fine tuning loop to achieve accurate synchronization. A clock domain transfer to the receiver clock domain is included in the circuit. The circuit is shown to have a good BER performance and be tolerant to both correlated and uncorrelated jitter. This has been verified with measurements on a fabricated test chip in UMC 130 nm CMOS technology.

Testability of digital systems is very important for high volume applications. We show that, with a little additional circuitry, repeaterless low swing interconnects that use equalizers and clock synchronizers can be tested with a high degree of fault coverage. The testability of repeaterless interconnects is discussed in Chapter 5.

Timing jitter in the data can indefinitely increase the settling time of mesochronous clock retiming circuits. The effect of timing jitter on the settling time is discussed in Chapter 6. We have demonstrated this extension experimentally by measuring the settling time of the circuit fabricated in UMC 130 nm CMOS technology. Using a Markov chain model of the clock retiming circuit, the settling time in presence of different types of timing jitter is analyzed. Chapter 6 then reports techniques for building fast synchronizers, which are tested using circuit simulations.

To summarize, the work in this thesis has contributed in taking repeaterless interconnects a step closer to realization in practical systems.

7.2 Future directions of research in this area

There are many directions of research which stem from the work in this thesis. Some of them can be listed as follows

Design of bus interconnects: The work in this thesis considered optimization of single lane low swing interconnects. Extending these schemes to design a bus interconnect with crosstalk compensation and including clock synchronization for these bus interconnects forms one of the dimensions along which this work can be carried forward. Worst case based design methods for crosstalk compensation will be a useful contribution. For clock synchronization, the proposed circuits can be replicated to form an interconnect bus. At the receiver, some of the retiming circuits components like the DLL can be shared for the bus, thus reducing the overhead. Figure 7.1 shows a suggested architecture for this building over the single lane receiver proposed in this thesis. The line at the center of the

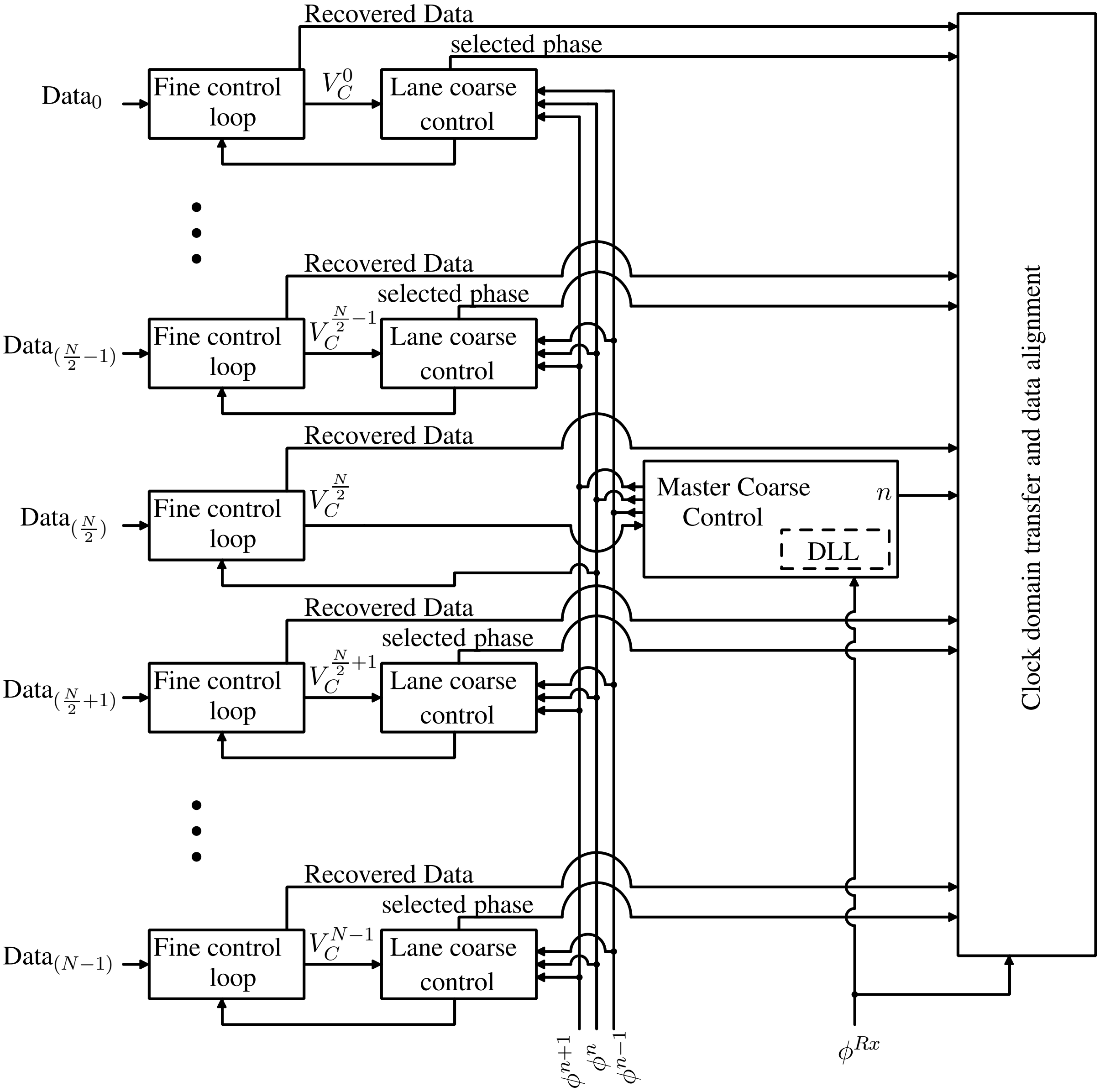


Figure 7.1: A suggested architecture for a mesochronous clock retiming circuit for a bus interconnect.

bus can use a clock retiming circuit that is identical to the single lane design. Using this lane's tapped DLL phase and its adjacent phases, a lane specific coarse control that picks from one of these three said phases of the DLL along with an individual fine control can retime the sampling clocks for all the lanes. This arrangement ensures differential delays of both polarities, with respect to the central lane, can be corrected for.

Dynamic voltage and frequency scaling: All of modern high performance digital designs use DVFS for power saving. The mesochronous clock synchronizers in this thesis were designed for a single frequency of operation. Design of a mesochronous synchronizer that allows DVFS reliably is a useful extension of this work. One of the ways of designing such a synchronizer could be by using a multi-band DLL in the synchronizer proposed in this work.

Automatic synthesis of chips using low swing interconnect: Automatic synthesis of digital chips using low swing interconnects is a direct extension of this work. Synthesis of digital circuits using a library of standard interconnect lengths can be investigated. Ultimately using the optimization methods for designing equalizers proposed in this work, automatic synthesis of interconnects can be developed as a useful extension of this work.

Automatic test pattern generation for mixed signal test: This work has investigated the test of mixed signal circuits using techniques used for testing pure digital circuits. Automatic test pattern generation for such mixed signal circuits for efficient design and evaluation of such circuits is another useful extension of the work in this thesis. Automatic fault simulation for evaluating coverage of structural faults will also form a very useful contribution.

Appendix A

Analysis and design of the Sense amplifier based clocked comparator

In the receiver of the low swing interconnect system, the most critical component is the clocked comparator. It is the resolution, speed, offset and power consumption of the receiver comparator which determine the performance and reliability of the receiver. Decision feedback equalization can be integrated with the comparator at the receiver as well [31]. Thus, an in-depth understanding of the comparator circuit and careful design is necessary. This chapter analyzes the sense amplifier based comparator in some detail. We shall first explain the basic working of this comparator and construct an analytical model of the first stage and point out some design guidelines. Design of some slave latch architectures is discussed next.

Analysis of the sense amplifier based comparator

Figure A.1 shows the circuit diagram of the first stage of the sense amplifier based comparator. The operation of any clocked comparator can be divided into four phases, namely precharge phase, sampling (or evaluation) phase, regeneration phase and latching phase. A similar phase decomposition and analysis is discussed in [94]. We will look at these phases one at a time.

Precharge phase

In the precharge phase, the comparator is reset and the outputs $\overline{S}$ and $\overline{R}$ are precharged to V_{DD}. The comparator is not sensitive to the inputs at this time. Figure A.2 shows the equivalent circuit diagram of the circuit in this phase.

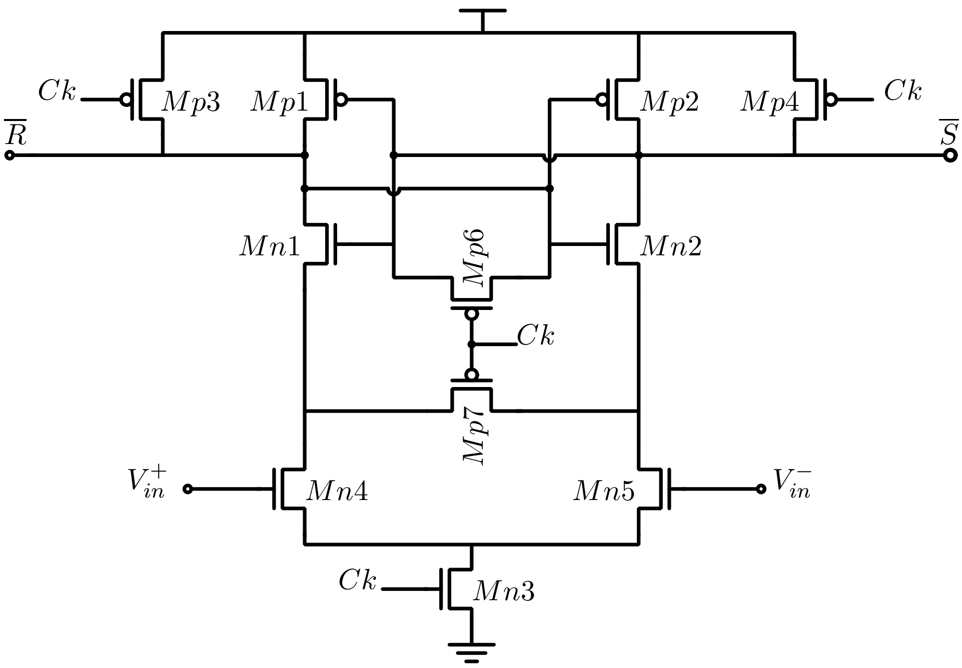


Figure A.1: First stage of the sense amplifier based comparator.

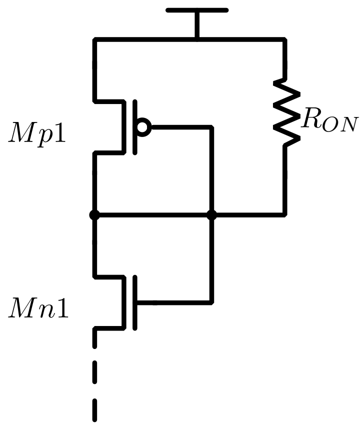


Figure A.2: Sense amplifier based comparator: Equivalent circuit for the precharge phase.

The precharge phase lasts for half the clock cycle, i.e. as long as the clock is low for the implementation shown in Figure A.1. The precharge transistors $Mp3$ and $Mp4$ should be sized such that the outputs $\overline{S}$ and $\overline{R}$ are precharged to at least $0.99 \times V_{DD}$ in this time. From this constraint we can write

$$R_{onM_{3,4}} = \frac{T_B}{9.2 \times C_l}$$

where C_l is the total load capacitance on $\overline{S}$ and $\overline{R}$ nodes.

Sampling phase

When the clock transitions from low to high, it triggers the sampling phase. This is when the comparator evaluates the inputs and establishes the initial voltage in the direction of the input for the regeneration phase. Figure A.3 shows the equivalent circuit diagram during the evaluation phase. The inputs bias the transistors $Mn4$ and $Mn5$ to act as current sources, I and $I + \Delta I$ depending on the input voltage. The common mode determines the absolute current I and the difference determines the differential current ΔI.

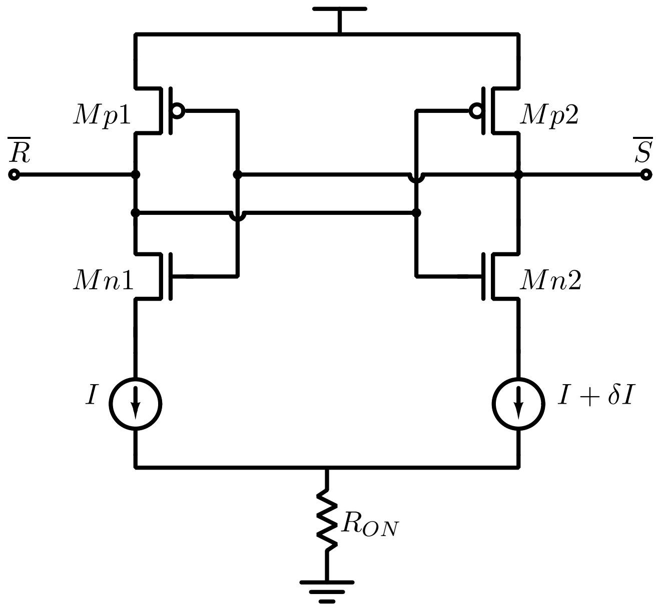


Figure A.3: Sense amplifier based comparator: Equivalent circuit during the Evaluation/Sampling phase.

Figure A.4 shows the characteristics of the inverters which are connected in positive feedback. When the evaluation phase just begins, the output nodes $\overline{S}$ and $\overline{R}$ discharge through the transistors $Mn1$ and $Mn2$ at rates of I and $I + \Delta I$. The operating point of the two inverters fall from V_{DD} as shown by the dashed lines till they reach close to the trip point of the inverters. Until this time the inverters are not active and in a very low gain region. When the operating point reaches sufficiently close to the trip point, due to the difference in the two branch currents a small difference between $\overline{S}$ and $\overline{R}$ is established. This marks the beginning of the regeneration phase. The transition between the sampling and regeneration phases is gray and cannot be well defined. The initial voltage established at the nodes $\overline{S}$ and $\overline{R}$ at the end of the sampling phase can be then written as

$$\Delta V_{in0} = \frac{\Delta I \times (V_{DD} - V_M)}{I}$$

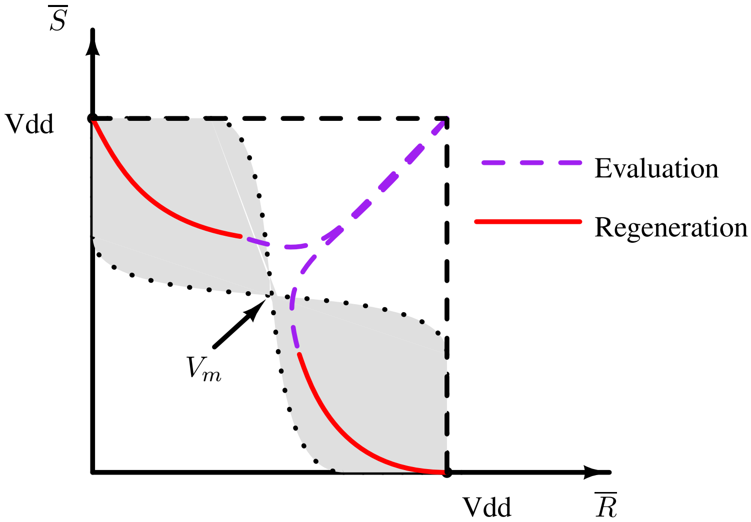


Figure A.4: Sense amplifier based comparator: Evaluation phase signals trajectory.

Hence, the sensitivity of the sampling phase increases with decrease in input common mode, which reduces the absolute discharge current I. However, this also means that the evaluation phase lasts longer, leaving less time for the regeneration phase.

Regeneration phase

Figure A.5 shows the equivalent circuit diagram during the regeneration phase. When the operating point of the inverters reaches close to the trip point at the end of the evaluation phase, the inverters are in high gain region. The high gain and the positive feedback pull the nodes $\overline{S}$ and $\overline{R}$ apart in the direction established by the evaluation phase.

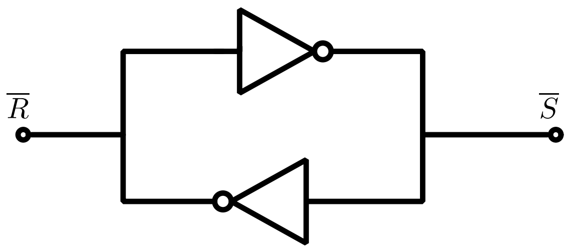


Figure A.5: Sense amplifier based comparator: Equivalent circuit during the regeneration phase.

Figure A.4 shows the characteristics of the two inverters and the trajectory of the operating points during the evaluation and regeneration phases. We can write the differential output voltage at the end of the regeneration phase as

$$\Delta V = \Delta V_{in0} \exp \frac{g_m \times t_2}{C_l}$$

Where t_2 is the time spent in the regeneration phase.

Latching phase

The second stage of the comparator is the slave latch, which is required to provide valid data while the first stage is reset for the next sample. Performance of the slave latch is important when comparators are designed for high speed and high resolution. The slave latch may be static or dynamic and may or may not use the clock signal. The next section discusses some slave latch architectures.

Slave Latch for the sense amplifier

The NAND SR Latch

A simple SR NAND latch (Figure A.6) may be used for latching the data sampled by the first stage of the comparator.

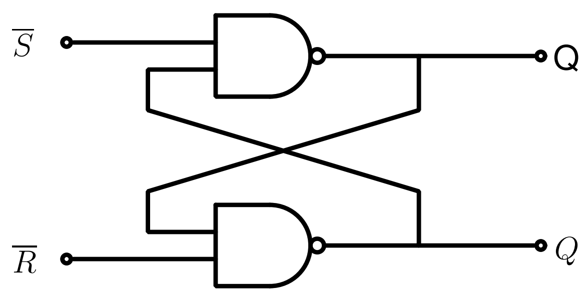


Figure A.6: Sense amplifier based comparator: Simple SR NAND latch.

During the reset phase $\overline{S}$ and $\overline{R}$ are both reset to V_{DD}. At this time the output is held to the previous bit. Depending on which of $\overline{S}$ or $\overline{R}$ goes low, the output Q is either set or reset. The operation of the NAND latch is such, that one of the outputs Q or $\overline{Q}$ is always delayed by one gate delay more than the other. If $\overline{R}$ resets $\overline{Q}$ to zero, Q is driven to one by $\overline{Q}$ going to zero. Similarly, when $\overline{S}$ goes low, it resets Q to zero which in-turn sets $\overline{Q}$ to one. Additionally, any load on one of the outputs affects the delay of the other, making the rise and fall delays asymmetric. These limitations were identified by the authors of [74] and an improved symmetric latch was proposed.

Symmetric CMOS Latch from [74]

The modified slave latch from [74] is shown in Figure A.7.

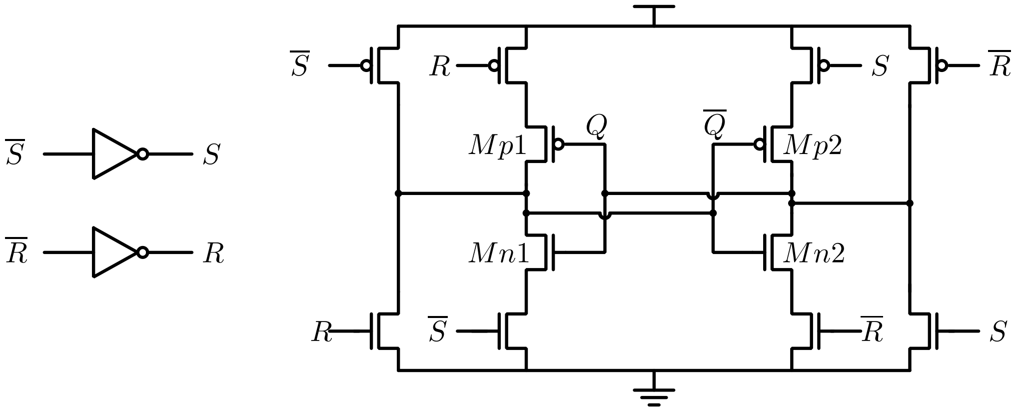


Figure A.7: Sense amplifier based comparator: Symmetric CMOS Latch. Adapted from [74].

The transistors $Mn1$,$Mp1$ and $Mn2$,$Mp2$ form two inverters, which are connected back-to-back in positive feedback. When one of $\overline{S}$ or $\overline{R}$ goes low, the positive feedback is disabled, and the first stage drives the output to the resolved value. Both the outputs are driven simultaneously. When $\overline{S}, \overline{R}$ return to 1 during the reset, the positive feedback is enabled to latch the output.

The modified slave latch does not have any of the problems of the NAND latch, and hence operates faster. However, one major drawback of the modified slave latch is that it needs $\overline{S}, \overline{R}$ as well as S, R. Inverters are used to generate S, R from $\overline{S}$ and $\overline{R}$. However, when operating close to its resolution, $\overline{S}$ and $\overline{R}$ are close to each other as shown in Figure A.8. For reliable operation the inverters that generate S, R from $\overline{S}$, $\overline{R}$ should have their trip point positioned at the center of the two voltages as shown by the dotted line. However, this point changes with process and input common mode. This slave latch treats $\overline{S}$ and $\overline{R}$ as single ended signals. A latch that has a differential input and a differential output can also be used with the same sense amplifier, which will not have this requirement.

Static and dynamic low swing latches

Figure A.9 shows a different slave latch which treats $\overline{S}$ and $\overline{R}$ as one differential pair and latches the output. The first stage of the slave latch (Stage A) is a buffer amplifier, which isolates the latch from the first stage of the comparator. It also provides a small gain. The second stage (Stage B) is the positive feedback stage. Stage B is enabled on the falling edge of the clock, i.e.

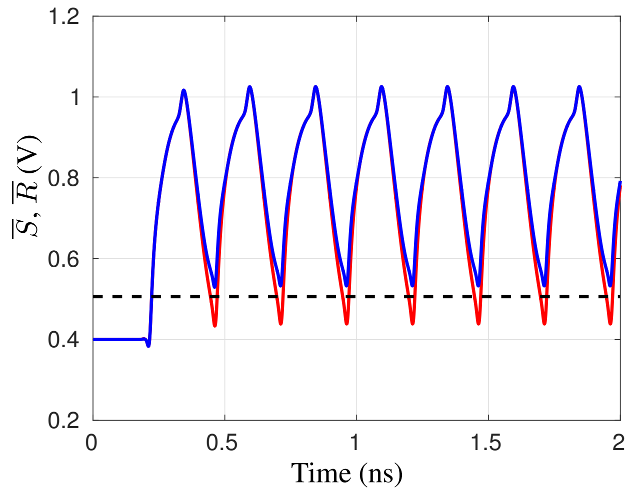


Figure A.8: $\overline{S}, \overline{R}$ and inverter trip point for optimum design.

at the end of the regeneration of the first stage. During this phase the output is regenerated if necessary and held by the positive feedback. Stage C converts the low swing resolved data to CMOS levels.

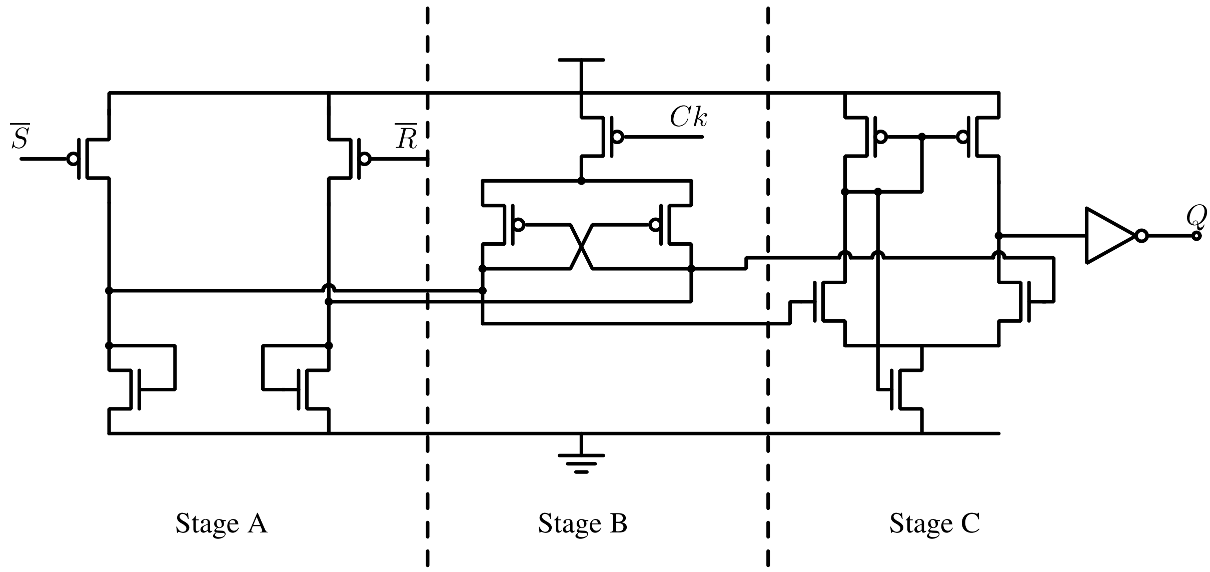


Figure A.9: Sense amplifier based comparator: Low swing static slave Latch.

Due to its differential nature, the performance of the slave latch is not sensitive to process variation or input common mode. However this comes with increased power consumption.

Figure A.10 shows an alternate implementation which holds the outputs dynamically in contrast to the above design. This is a derivative of standard static CMOS comparator [95]. Since the positive feedback is not disabled, the first stage of the comparator has to overcome the residual previous bit. This makes the dynamic latch slower than the static latch. The delay is of the dynamic latch is further affected by the poor common mode rejection ratio of the differential to single ended converter. The low common mode in the hold phase disallows the use of current source biased differential to single ended converter. Attempts to increase the output common mode of the first stage using diode connected transistors to ground reduces the bandwidth considerably. A very similar latch is reported in [96].

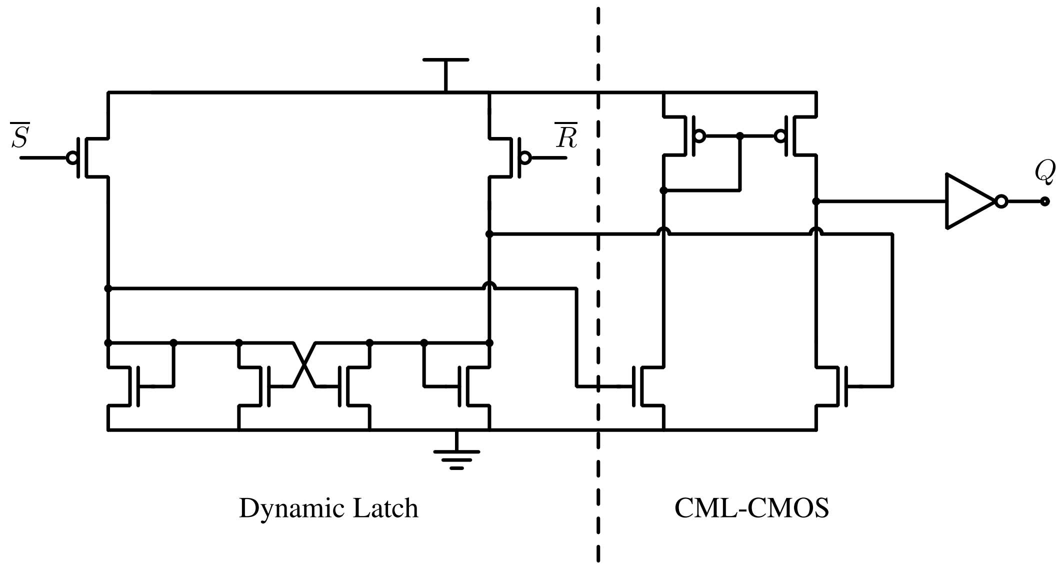


Figure A.10: Sense amplifier based comparator: Low swing dynamic slave Latch.

Apart from higher power consumption, another limitation of the low swing latches is the difference in rise and fall delays. This manifests due to the differential to single ended converter, wherein one of the input-to-output path is through the current mirror load.

Summary and simulation results

Figure A.11 shows the phase diagram of the comparator with various phases labeled. The total gain of the comparator depends on how much time is spent in regeneration phase. High gain trades with less time for evaluation. Thus, there is a trade off between the sensitivity and gain. As can be seen the comparator with low swing latch has the highest regeneration time. In order to test the comparators, we simulated the comparators with a input terminated with a differential 16 kΩ resistance. A 4 section RC interconnect with R = 100 Ω and C = 100 fF per section was used which produced an eye diagram shown in Figure A.12. The eye was sampled at the point with maximum opening. Table A.1 shows the performance of the comparator with different slave latches. Because of the ISI in the data, the time taken by the comparators to resolve is not

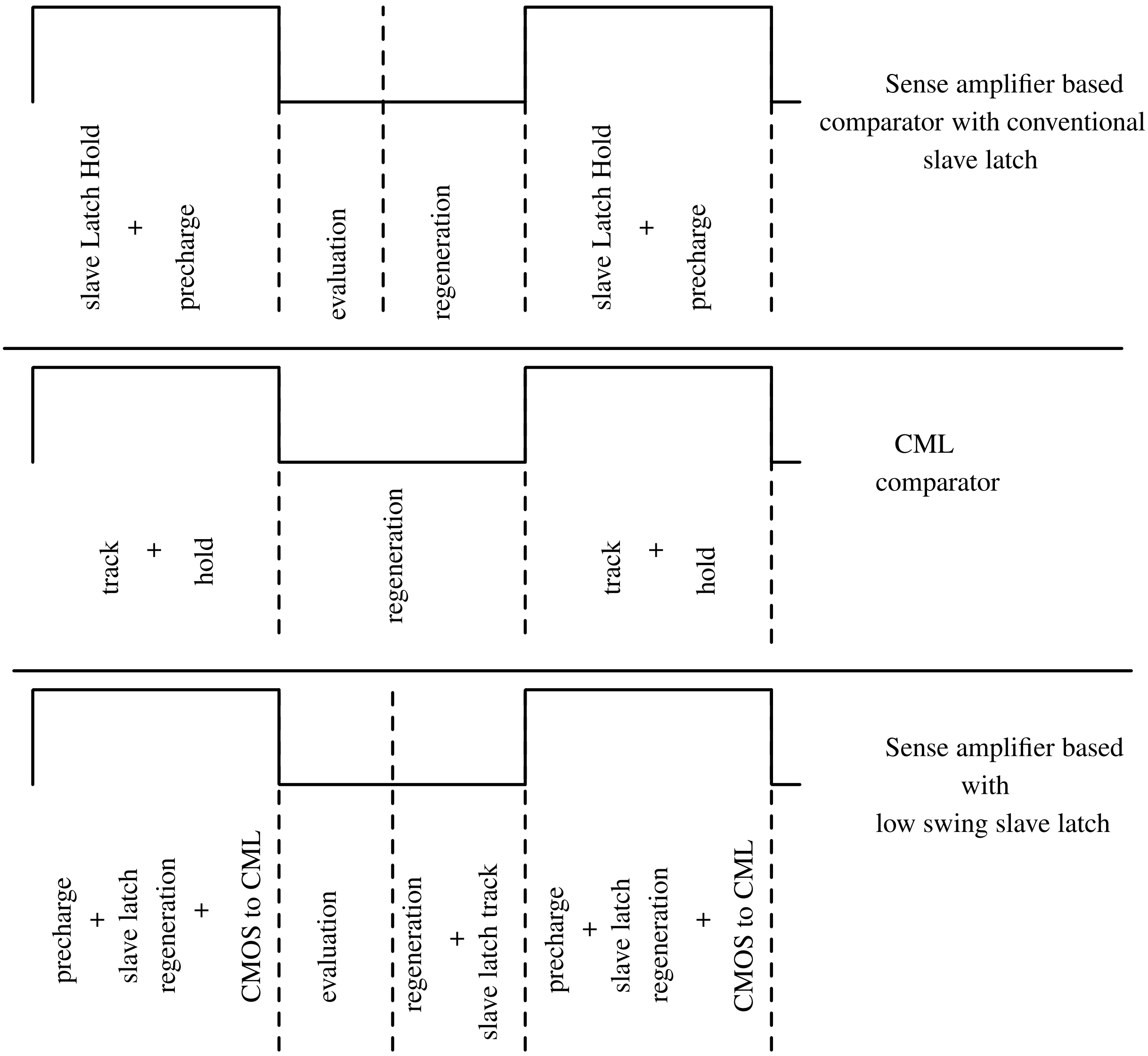


Figure A.11: Comparison of operation of CML comparator, Sense amplifier based comparator with conventional slave latch and Sense amplifier based comparator with low swing slave latch.

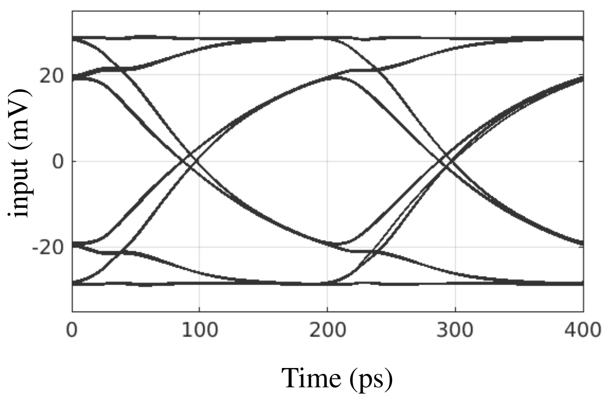


Figure A.12: Eye diagram of the input signal, of 5 GHz, used to test the comparators with different slave latches.

Table A.1: Performance of the Sense amplifier comparator and the different slave latches. The half-rate comparator was designed to operate at 2 x 2.5 GHz

Parameter	Dynamic latch	Static latch	CMOS latch [74]	Sense amplifier
Power consumption	2 x 115 μWatts	2 x 113 μWatts	2 x 50 μWatts	150 μWatts
Jitter (peak-peak)	120 ps	80 ps	50 ps	-NA-
Speed in SS corner	2 x 1.9 GHz	2 x 2.1 GHz	2 x 1.9 GHz	-NA-

constant. This results in jitter in the recovered data, which is shown in Table A.1. Both the low swing latches were designed so that they present the same load to previous stage. From Table A.1 we can see that the dynamic latch has the highest jitter. This can be attributed to the residual positive feedback of the previous bit. The comparators were designed in UMC 90 nm standard CMOS process. All results are prelayout results, with estimated layout parasitics added to all nodes.

Appendix B

Mean time to absorption of an absorbing Markov chain

The state diagram of the Markov chain representation of the clock recovery circuit, for data with 1 bit ISI, is shown in Figure 6.16. The states corresponding to $-T_{\mathcal{W}}/2$ and $T_{\mathcal{W}}/2$, which are at the edges of the window of susceptibility, are absorbing states. The *probability transition matrix* P for this Markov chain can be written as

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 & \cdots & & & & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 & \cdots & & & & 0 \\ 0 & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & \cdots & & & & 0 \\ & \vdots & & & \ddots & & & \vdots & \\ 0 & \cdots & & & & 0 & \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & \cdots & & & & 0 & 0 & 0 & 1 \end{bmatrix}.$$

To calculate the mean time to absorption (and the variance of the time to absorption), P is first written in the canonical form [91]. This is obtained by reordering the entries in P to separately aggregate all the transient and absorbing states respectively. Hence, P can be written as

$$P = \left[\begin{array}{c|c} Q & R \\ \hline 0 & I \end{array}\right] = \left[\begin{array}{cccccccc|cc} \frac{1}{2} & \frac{1}{4} & 0 & 0 & \cdots & & & 0 & \frac{1}{4} & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 & \cdots & & & 0 & 0 & 0 \\ & \vdots & & & \ddots & & & \vdots & \vdots & \vdots \\ 0 & \cdots & & & & & \frac{1}{4} & \frac{1}{2} & 0 & \frac{1}{4} \\ \hline 0 & \cdots & & & & & 0 & 0 & 1 & 0 \\ 0 & \cdots & & & & & 0 & 0 & 0 & 1 \end{array}\right].$$

Here, Q is a square matrix which captures the transitions from one transient state to another transient state, R captures the transitions from transient states to absorbing states and I is an identity matrix. The fundamental matrix N can be computed using the equation $N = (I_t - Q)^{-1}$, where I_t is an identity matrix of the same dimensions as Q. Conditioned on the starting position, the mean time to absorption (T_{mean}), in terms of the number of steps, is given by

$$T_{mean} = NC,$$

where, C is a column vector with number of rows equal to the number of transient states, and with all entries as 1. The variance of the time to absorption (in terms of number of steps), conditioned on the starting position, is calculated as

$$T_{var} = (2N - I_t) \times T_{mean} - T^{sq}_{mean}.$$

Here, T^{sq}_{mean} is obtained by squaring the elements of T_{mean}.

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